

Local Time Acceleration:

A Phenomenological Solution to the Hubble Tension Driven by Kinematic Anomalies in the Local Void

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Abstract

The tension between local ($H_0 \approx 73$ km/s/Mpc) and early-universe ($H_0 \approx 67$ km/s/Mpc) measurements of the Hubble constant remains the defining crisis of modern cosmology. In this work, we investigate whether this tension can be ameliorated by a phenomenological “Local Time Acceleration” (LTA) parametrization, which modifies the observed redshift via a bounded, observer-anchored decay function. We fit this model to the Pantheon+SH0ES dataset using the full released covariance matrix and a ladder-anchored likelihood. We report a statistically significant preference for the LTA model over standard Λ CDM ($\Delta\chi^2 \approx 13.65$, corresponding to $\sim 3.7\sigma$), with a Monte Carlo calibrated tail probability $p_{\text{MC}} \approx 8 \times 10^{-4}$ under $N = 10000$ baseline null injections. Tomographic analysis reveals that this signal is driven entirely by supernovae in the local volume ($z < 0.03$), effectively correcting a known “dip” in the Hubble residuals often attributed to peculiar velocities in the Local Void. Crucially, we perform a rigorous Jackknife resampling of all major constituent surveys (CfA, CSP, PS1, etc.) and find the signal is robust to the exclusion of any single telescope group, ruling out instrument-specific calibration systematics. While the effect vanishes at higher redshifts, its persistence and coherence in the local volume suggest that the local Hubble flow exhibits structured kinematic deviations that are better described by a smooth temporal decay than by standard peculiar velocity corrections alone.

Keywords: cosmology, Type Ia supernovae, BAO, local time acceleration, redshift mapping, negentropy

1 Introduction

The persistent mismatch between late-time (local) determinations of the Hubble constant ($H_0 \simeq 73$ km s $^{-1}$ Mpc $^{-1}$) and early-universe inferences under minimal flat Λ CDM ($H_0 \simeq 67$ km s $^{-1}$ Mpc $^{-1}$) has become the central anomaly in precision cosmology [1, 2]. What makes the “Hubble tension” unusually stubborn is that it lives at the intersection of multiple tightly constrained datasets: attempts to modify early-time physics must preserve the CMB acoustic peak structure, while late-time modifications to the expansion history are strongly constrained by the inverse distance ladder (BAO + SNe Ia) once the sound horizon scale is fixed.

A less explored possibility is that part of the discrepancy is not best described as a global change in the background expansion $H(z)$, but rather as a small, observer-linked distortion in the mapping between observed redshift and the cosmological redshift that enters background distances. This is not a semantic distinction: the SH0ES-style distance-ladder inference is dominated by nearby Type Ia supernovae and is therefore maximally sensitive to the detailed kinematics of the local Hubble flow. The observer resides in a structured environment (including a local underdensity often discussed under the umbrella of “Hubble bubble” / “Local Void” phenomenology), and the lowest-redshift portion of the Hubble diagram exhibits coherent residual structure that is not naturally captured by purely stochastic peculiar-velocity scatter alone. If the local Hubble flow contains an additional smooth, causal, line-of-sight effect that is not modeled in the standard pipeline, it should manifest precisely in the redshift range where the distance ladder is anchored and where local-flow systematics are most acute.

In this paper we introduce and test a phenomenological extension of Λ CDM which we call *Local Time Acceleration* (LTA). Operationally, LTA is implemented as an accumulated fractional frequency drift along the photon trajectory, producing an additive contribution to $\ln(1+z)$ and therefore a multiplicative correction to $(1+z)$:

$$1 + z_{\text{obs}} = (1 + z_{\text{cos}}) \exp[I(\chi)]. \quad (1)$$

The defining modeling choice is that the drift strength is evaluated using an *Earth-retarded lookback time* argument, so that the effect is explicitly causal and tied to the observer’s lightcone rather than to a global function of the scale factor. We further impose boundedness (the drift saturates and becomes negligible beyond a finite activation epoch) and monotonicity of the mapping (so that $z_{\text{cos}}(z_{\text{obs}})$ remains single-valued), making the model both numerically well-posed and falsifiable.

Our goal in this work is deliberately specific: *does a causal, bounded, observer-anchored redshift mapping of the LTA type improve the joint fit to late-time distance–redshift data when those data are analyzed using their released full covariances?* We emphasize what this statement does and does not entail. A detection of a preference for LTA would be an empirical statement about the adequacy of a particular mapping for describing the Hubble diagram and BAO observables; it would not, by itself, uniquely identify a microphysical origin. At the same time, this is not an unconstrained curve-fitting exercise: the mapping is nested ($s_{\text{anchor}} \rightarrow 0$ recovers ΛCDM exactly), it is restricted by causality and boundedness, and it is tested jointly against BAO, whose radial observable depends directly on the redshift Jacobian implied by the mapping.

Our analysis is intentionally conservative in data handling. For SNe Ia we use the Pantheon+SH0ES public release [1, 2], restricting to the `shoes_global` selection and adopting the released *full* covariance matrix, including the nontrivial cross-correlations between calibrators and Hubble-flow supernovae. Because calibrator/Hubble-flow correlations are substantial, we construct a ladder-anchored SN likelihood in which calibrators provide an absolute-distance anchor and the cosmology-sensitive Hubble-flow likelihood is obtained by conditioning on calibrators via a Schur complement (Sec. 10.2). We combine this with a BAO consensus likelihood (Sec. 10.3)[3], treating the sound horizon through a single scaling parameter α_{rd} . Crucially, BAO provides an independent lever arm on the mapping: the LTA redshift Jacobian modifies the radial BAO prediction in a way that a purely SN-driven distortion generally cannot evade.

For inference we compare baseline flat ΛCDM against $\Lambda\text{CDM}+\text{LTA}$ under the same data treatment and numerical safeguards. Because the LTA extension is nested and we impose physical bounds on the LTA amplitude, the null is a boundary point and standard asymptotic χ^2 -difference heuristics are not guaranteed to be reliable.

Because the LTA amplitude is constrained (e.g. $s_{\text{anchor}} \geq 0$), the null is a boundary point and standard Wilks-theorem χ^2 asymptotics need not hold; in simple one-sided cases the likelihood-ratio statistic approaches a nonstandard mixture distribution rather than χ^2_1 [30, 31]. In astronomy/cosmology contexts this issue is well known in the detection of additional model components and motivates Monte Carlo calibration when regularity conditions fail [32]. We therefore treat the parametric bootstrap as the primary calibration for $\Delta\chi^2$ in this paper (Sec. 12.2).

We therefore calibrate the detection significance using a Monte Carlo parametric bootstrap (null injections) under the baseline model, generating mock realizations with the *same* released covariances and refitting both models end-to-end (Sec. 10.6 and Sec. 12). In addition, we perform falsification-oriented robustness tests, including leave-one-group-out (jackknife) refits by telescope/survey grouping to check whether the preference is driven by any single subset, and supplementary cross-validation diagnostics designed for strongly correlated data (App. G).

Roadmap. Section 3 explains the LTA ansatz as a causal frequency-drift mechanism and makes explicit the model’s falsifiability requirements. Section 7 defines the $\Lambda\text{CDM}+\text{LTA}$ mapping, the Earth-retarded time construction, and the Jacobian needed for BAO. Sections 9–10 describe the datasets and the ladder-anchored likelihood used to respect the released covariance structure. Section 11 presents the primary fits and residual diagnostics, including the localization of the improvement in redshift. Section 12 summarizes jackknife robustness and Monte Carlo null calibration. Finally, Section 13 discusses interpretation and independent tests that can confirm or decisively falsify an LTA-like observer-local redshift-mapping effect.

2 Context: Relation to existing approaches to the Hubble tension

The discrepancy between late-time determinations of the Hubble constant and early-universe inferences under minimal flat ΛCDM has generated an exceptionally broad literature. Comprehensive reviews emphasize that proposed resolutions fall into three overlapping categories: (i) unrecognized systematics in one or more datasets and inference pipelines; (ii) new physics that modifies early-universe calibration, typically by

reducing the sound horizon r_d and thereby raising the CMB-inferred H_0 ; and (iii) new physics (or unmodeled astrophysics) that modifies the late-time distance–redshift relation or the interpretation of low- z anchors (e.g. dynamical dark energy, modified gravity, or local-structure effects) [5, 6]. Because the observational inputs are heterogeneous (SNe Ia, Cepheids/TRGB, BAO, CMB anisotropies, lensing, time delays, masers, and standard sirens), it is useful to state explicitly where the LTA framework tested here sits within this landscape, and how it relates to the failure modes that have been most discussed.

At a high level, the LTA hypothesis explored in this paper is *not* an attempt to modify the global expansion history $H(z)$ as a free function, nor is it designed to alter recombination-era physics to shift r_d . Instead, it is a phenomenological proposal that the mapping from observed redshift to the background cosmological redshift entering homogeneous FLRW distances may acquire an additional *observer-anchored, causal, bounded* contribution in the local Universe. This places LTA closest in spirit to the class of explanations in which part (or all) of the SH0ES–Planck discrepancy is driven by effects that are specific to the *local Hubble-flow anchor regime*, including local kinematics, local structure, or other observer-linked biases that are not captured by standard peculiar-velocity modeling.

2.1 Early-universe resolutions: shifting the sound horizon

A large fraction of proposed “new physics” solutions aim to reduce the sound horizon at the drag epoch, r_d , while preserving the excellent fit of Λ CDM to the CMB angular power spectra and other early-time observables. If r_d is smaller than in minimal Λ CDM, then the inverse distance ladder (BAO + SNe Ia), when anchored by CMB physics, can accommodate a larger inferred H_0 . Representative ideas include early dark energy (EDE) episodes that briefly contribute non-negligible energy density near matter–radiation equality [7], additional relativistic degrees of freedom (larger N_{eff}), modifications to recombination, and scenarios in which neutrino physics is altered (e.g. self-interactions) in ways that reshape the CMB damping tail and inferred parameters [8]. The review literature emphasizes that these models are tightly constrained: they must preserve the CMB peak structure, lensing, and BBN consistency, and they are increasingly challenged by the combined CMB+BAO+large-scale-structure dataset hierarchy [5].

The LTA analysis in this paper is largely orthogonal to this program. Operationally, we introduce an explicit sound-horizon scaling parameter α_{rd} in the BAO likelihood, and we test whether an *observer-anchored redshift mapping* can improve the fit in the regime that dominates local distance-ladder anchoring, without requiring a shift of early-time physics. Consequently, if future work finds a residual tension after accounting for observer-local effects, LTA-like mappings could in principle be combined with early-time modifications rather than being in direct competition with them. Conversely, if an early-time solution fully resolves the discrepancy across all late-time probes, it would predict that no additional observer-local mapping distortion is required; this provides a clear empirical discriminant.

2.2 Late-universe modifications: dynamical dark energy and modified gravity

A second broad class of proposals modifies late-time cosmology by changing the expansion history or the growth of structure relative to minimal Λ CDM. These include dynamical dark energy parameterizations ($w \neq -1$, evolving $w(z)$), interacting dark sector models, and modified-gravity theories in which the relation between geometry and stress–energy differs from GR on cosmological scales [5, 6]. Such models are constrained not only by the SNe Ia Hubble diagram but also by BAO (both transverse and radial), weak lensing, redshift-space distortions, cluster abundances, and CMB lensing.

The empirical signature isolated in this paper differs from what these models typically target. Our best-fit LTA profiles are deliberately *bounded and saturating*, and the improvement is tomographically localized to $z \lesssim 0.03$, leaving the best-fit background (H_0, Ω_m) and the higher- z Hubble diagram close to baseline. This behavior is more naturally interpreted as a correction to the *local* mapping between z_{obs} and the effective redshift argument used in distances, rather than as a genuine modification of $H(z)$ over a broad redshift lever arm. Accordingly, LTA does not aim to replace late-time dark energy phenomenology globally; rather, it tests whether the low- z ladder anchor regime exhibits a structured deviation that is better captured by a smooth, causal, observer-local mapping than by standard local-flow scatter models.

2.3 Systematics and independent late-time anchors

A persistent theme in the literature is that the SH0ES–Planck discrepancy could reflect residual systematics in one or more components of the distance ladder (Cepheid calibration, SN Ia standardization, selection effects, dust, host correlations) or in early-universe inference. For this reason, independent determinations of H_0 have been heavily emphasized. These include TRGB-calibrated ladders [11], strong-lensing time-delay cosmography [12, 13], geometric megamaser distances [14], and gravitational-wave standard sirens [15]. A key point from review work is that the most informative cross-checks are those that change both the astrophysical systematics and the inference architecture, rather than merely reprocessing the same ladder ingredients [5, 6].

The contribution of this paper is not to adjudicate between all of these external H_0 anchors. Instead, we take the Pantheon+SH0ES public release at face value and analyze it using the published *full* covariance structure (including calibrator–Hubble-flow cross-correlations), and we ask a narrower question: within that released statistical model, is there evidence for a local, smoothly varying mapping distortion that improves the Hubble-flow residual structure, and does that distortion survive falsification-oriented robustness tests (jackknife, null injections) and remain consistent with BAO once the mapping Jacobian is propagated? In that sense, LTA should be read as an *internal-consistency hypothesis test* on a specific late-time dataset combination, rather than as an attempt to supersede the full cross-probe literature on H_0 .

2.4 Local structure, cosmic variance, and redshift interpretation

A third major theme, directly relevant to the present work, is whether the local environment of the observer can bias low- z determinations of H_0 . This includes (i) the possibility that the observer resides in a local underdensity or otherwise atypical region, affecting inferred distances and velocities (“Hubble bubble” / Local Void phenomenology), and (ii) irreducible cosmic variance in the locally measured Hubble flow even in a perfect experiment [9]. Detailed analyses in the SN context have generally concluded that a simple spherically symmetric local void of realistic amplitude is insufficient, by itself, to reconcile SH0ES with Planck [10], though local-flow modeling and sample selection remain active topics.

LTA is deliberately aligned with the *spirit* of this class of explanations—namely, that the anchor regime of the distance ladder may encode structured local effects—but it differs in its implementation. Rather than modeling a particular density profile or velocity field, we introduce a nested, bounded, monotonic mapping in which photons acquire an accumulated contribution to $\ln(1+z)$ parameterized as a line-of-sight integral and evaluated with an Earth-retarded time argument. This choice yields a concrete, testable prediction beyond a generic “local effect” statement: the mapping implies a redshift Jacobian $dz_{\text{obs}}/dz_{\text{cos}}$ that directly affects radial BAO. In other words, LTA is not merely an empirical correction to the SN Hubble diagram; it is a mapping whose consistency can be independently checked in observables that depend on redshift derivatives.

2.5 What is novel in the present analysis

Within the broader literature, the main methodological distinctions of this work are:

1. **A specific observer-anchored redshift-mapping ansatz.** We implement the hypothesis as a multiplicative correction to $(1+z)$ (additive in $\ln(1+z)$), accumulated along the photon trajectory, with explicit bounded activation and an Earth-retarded time construction to enforce causal bookkeeping.
2. **Full released-covariance treatment with calibrator cross-correlations respected.** The Pantheon+SH0ES release contains nontrivial calibrator–Hubble-flow covariance coupling; we construct a ladder-anchored likelihood via a Schur complement so that calibrators act as an absolute-distance anchor without discarding the published correlation structure.
3. **Significance calibration under a boundary-nested null.** Because the LTA amplitude is bounded ($s_{\text{anchor}} \geq 0$), the null is a boundary point; we therefore calibrate $\Delta\chi^2$ using parametric bootstrap null injections that preserve the released covariance and rerun the full refitting pipeline.
4. **A joint SN+BAO test that propagates the mapping Jacobian.** BAO is not used only as a late-time distance lever arm; it provides a falsification channel because radial BAO depends explicitly on the redshift Jacobian implied by the mapping.

5. **Tomographic localization of the effect.** The preference is driven by a narrow low- z shell ($z \lesssim 0.03$), consistent with a local kinematic/mapping effect and inconsistent with a broad modification of the late-time expansion law.

Taken together, these points place LTA in a well-defined position relative to the existing solution space reviewed in [5]: it is a *phenomenological, observer-local mapping extension* designed to address structured low- z Hubble-flow residuals while remaining explicitly testable against BAO through the Jacobian constraint. If the preference survives further external scrutiny (alternative SN compilations, alternative flow models, sky-split tests, independent anchors), then it motivates a broader program of checking whether part of the Hubble tension is best understood as a local redshift-interpretation bias rather than as a single global modification of early- or late-time cosmology.

3 Conceptual Framework: Local Time Acceleration

The motivating hypothesis behind this work is unconventional, so it behoves us to be explicit about what is being proposed at the observational level, what requirements we impose to keep it well-posed, and what claims the present analysis does and does not make. Our aim here is not to argue for a specific microphysical origin, but to define a concrete and falsifiable phenomenological mapping between cosmological redshift and observed redshift, and then to test whether that mapping is preferred by late-time distance–redshift data analyzed using their released covariances.

3.1 LTA as an accumulated, observer-local contribution to $\ln(1+z)$

In standard FLRW cosmology, the observed redshift is determined by the ratio of scale factors, $1+z = a(t_{\text{obs}})/a(t_{\text{emit}})$, up to conventional local corrections (peculiar velocities, gravitational potentials, etc.) that are treated as separate effects. LTA modifies this assumption by introducing an additional *line-of-sight accumulated* fractional frequency drift. The observational statement is that photons acquire a small multiplicative factor in $(1+z)$ beyond the cosmological contribution:

$$1+z_{\text{obs}} = (1+z_{\text{cos}}) \exp[I(\chi)]. \quad (2)$$

Writing the effect in $\ln(1+z)$ is not a stylistic choice: it is the natural representation for a process that composes multiplicatively and accumulates additively along a trajectory. The quantity $I(\chi)$ is constructed as the line-of-sight integral of an effective drift-rate field $s(\chi)$ (with units of a Hubble rate), so that s/c acts as a fractional frequency-drift per unit comoving distance (Sec. 7.2).

This is not a coordinate relabeling. A pure coordinate transformation cannot change dimensionless local observables without additional physical content. In contrast, LTA is defined precisely as an additional physical contribution to the observed frequency ratio relative to the baseline Λ CDM prediction, parameterized in a way that is nested, bounded, and testable.

3.2 Why an Earth-retarded time argument appears

The distinctive structural element of the LTA ansatz is that the drift-rate profile is evaluated as a function of an *observer-retarded* lookback time t_{ret} , rather than as a generic function of cosmic time or scale factor. The purpose of this construction is causal bookkeeping: if the underlying chronometric coupling is a *universal* law but the dominant contribution in the present epoch is sourced by the observer’s nearby environment, then any influence on the observed past light cone must be consistent with propagation at or below the speed of light. We therefore tie the effective drift strength at a point along the photon path to a retarded-time assignment based on the observer worldline and the background geometry (Sec. 7.3).

Importantly, “Earth-sourced” here should be read as *observer-anchored* rather than as a commitment to a particular terrestrial mechanism. The present analysis does not require (and does not assume) that the relevant state variable is literally biological. Instead, the Earth-retarded construction is a compact proxy for a wide class of observer-local causal influences whose effective strength depends on the observer’s local history or environment. The empirical question tested below is therefore: does an *observer-anchored, causal, bounded* mapping of this type improve the description of the data?

3.3 Bounded activation and controlled degrees of freedom

To remain physically viable and empirically falsifiable, the LTA contribution must not grow without limit. We therefore impose a bounded activation through a function $g(t) \in [0, 1]$ such that the effect decays to zero at a finite cutoff epoch t_{life} (Sec. 7.4). This ensures that the additional drift saturates and becomes negligible for sufficiently large lookback times, preventing the model from contaminating high-redshift cosmology where early-universe constraints are exceptionally tight.

A second requirement is that the redshift mapping remain monotonic:

$$\frac{dz_{\text{obs}}}{dz_{\text{cos}}} > 0, \quad (3)$$

so that the inverse relation $z_{\text{cos}}(z_{\text{obs}})$ is single-valued and the Hubble diagram remains well-defined (Sec. 7.5). This is not merely a numerical convenience: non-monotonic mappings would imply multi-valued distance predictions at fixed observed redshift and would be directly incompatible with the interpretation of both SNe Ia and BAO as functions of redshift.

Finally, the model is explicitly *nested*. In the zero-amplitude limit ($s_{\text{anchor}} \rightarrow 0$), the integral $I(\chi) \rightarrow 0$ and the mapping reduces exactly to baseline Λ CDM. This nesting allows us to compare Λ CDM and Λ CDM+LTA at fixed data treatment using standard likelihood-ratio-style test statistics, while still calibrating their null distribution via Monte Carlo because the null lies on a boundary (Sec. 10.6).

3.4 Why BAO matters for falsification

Although the primary low-redshift leverage comes from SNe Ia, BAO plays a crucial conceptual role in this test. The LTA mapping implies not only a shift in the distance–redshift relation but also a redshift Jacobian $dz_{\text{obs}}/dz_{\text{cos}}$. Radial BAO observables depend explicitly on $d\chi/dz$ and therefore directly probe the Jacobian induced by the mapping (Sec. 10.3). This provides an independent check against mappings that might otherwise fit SNe Ia by distorting redshift in a way that would be inconsistent with a standard ruler measured in the radial direction. For this reason, joint SN+BAO fitting is an essential part of rendering the LTA hypothesis falsifiable with late-time data.

3.5 What this paper claims and what it does not

The restricted empirical claim supported by this paper (if the preference persists under the full set of robustness checks) is: *within the adopted data treatment and model family, an observer-anchored, causal, bounded LTA mapping provides an improved fit to late-time distance–redshift data relative to baseline Λ CDM.*

This is not a claim of unique microphysical identification. The same phenomenology could in principle arise from unmodeled local kinematics, residual calibration structure, or other observer-local effects not captured in the baseline pipeline. Accordingly, our strategy is to (i) implement a precise mapping with clear causal and boundedness requirements, (ii) fit it using the released full covariance structure (including calibrator/HF cross-covariance), and (iii) report falsification-oriented diagnostics that test whether the preference is distributed across the data, robust to leave-out refits, and unlikely under baseline null injections.

In this sense, LTA is treated here as a *testable phenomenological hypothesis*: the cosmology data are used to evaluate whether the mapping is statistically preferred, while the interpretation of its ultimate origin is left to independent constraints and future targeted tests (Sec. 13).

4 Photon-frequency transport and the physical meaning of an integrated $\Delta \ln(1+z)$

This paper’s data analysis is built around an *integrated* redshift modification, $\Delta \ln(1+z) \equiv I(\chi)$, and a corresponding local drift field $s(\chi)$. Because this structure is central (and is used directly in the BAO Jacobian), we make explicit here how an integrated correction arises from standard photon-frequency transport once one posits a universal chronometric coupling between *organization* and *realized clock time*.

4.1 Redshift as a frequency ratio measured by matter clocks (Jordan frame)

In geometric optics, a photon is described by a null wavevector k^μ tangent to a null geodesic of the *physical* (matter) metric. In the Jordan-first completion adopted in this revision, that physical metric is $g_{\mu\nu}^J$. Thus

$$k^\mu k_\mu = 0, \quad k^\nu \nabla_\nu^J k^\mu = 0, \quad (4)$$

and the frequency measured by an observer with Jordan-frame 4-velocity u^μ is

$$\nu_J \propto -k_\mu u^\mu \quad (\text{contraction with } g_{\mu\nu}^J). \quad (5)$$

Operationally, redshift is a *ratio of measured frequencies*:

$$1 + z_{\text{obs}} \equiv \frac{\nu_{\text{emit},J}}{\nu_{\text{obs},J}} = \frac{(-k_\mu u^\mu)_{\text{emit}}}{(-k_\mu u^\mu)_{\text{obs}}}. \quad (6)$$

For comoving observers in an unperturbed Jordan-frame FLRW background, $g_{\mu\nu}^J dx^\mu dx^\nu = -dt_J^2 + a_J^2(t_J) d\mathbf{x}^2$, Eq. (6) yields the familiar scale-factor law

$$1 + z_{\text{cos}} = \frac{a_J(t_{J\text{obs}})}{a_J(t_{J\text{emit}})}. \quad (7)$$

(See, e.g., standard GR/cosmology treatments [23].)

4.2 Derived chronometric factor from Jordan-frame scalar–tensor structure

Earlier versions of this work introduced a universal chronometric factor $A(\psi)$ as a direct postulate relating “realized clock time” to a background proper time. In the Jordan-frame-first scalar–tensor completion adopted here, the chronometric factor is *derived* by the standard conformal transformation between Jordan and Einstein frames.

Jordan vs. Einstein frames and a derived A . Recall from Sec. 4.5 that the Einstein-frame metric is defined by $g_{\mu\nu}^E = F(\psi) g_{\mu\nu}^J$, so that matter becomes conformally coupled in Einstein variables. The conformal factor that appears in the matter coupling is

$$A(\psi) \equiv F(\psi)^{-1/2}, \quad C(\psi) \equiv \ln A(\psi) = -\frac{1}{2} \ln F(\psi), \quad (8)$$

which is not an independent ingredient; it is fixed once $F(\psi)$ is specified.

Clock-time relation (frame conversion). Jordan proper time is the physical clock time, while Einstein proper time is a computational convenience. Along any worldline,

$$d\tau_J = A(\psi) d\tau_E. \quad (9)$$

Frequency conversion and redshift bookkeeping. A detector counts cycles N per unit proper time. Therefore the frequency measured by matter clocks in the physical frame is

$$\nu_J \equiv \frac{dN}{d\tau_J} = \frac{1}{A(\psi)} \frac{dN}{d\tau_E} = \frac{1}{A(\psi)} \nu_E. \quad (10)$$

Using Eq. (6), the observed redshift may be written in Einstein variables as

$$1 + z_{\text{obs}} = \frac{\nu_{\text{emit},J}}{\nu_{\text{obs},J}} = \frac{\nu_{\text{emit},E}/A(\psi)_{\text{emit}}}{\nu_{\text{obs},E}/A(\psi)_{\text{obs}}} = (1 + z_E) \frac{A(\psi)_{\text{obs}}}{A(\psi)_{\text{emit}}}, \quad (11)$$

where $1 + z_E \equiv \nu_{\text{emit},E}/\nu_{\text{obs},E}$ is the redshift defined in the Einstein variables.

Homogeneous cancellation (why the background ladder is standard in Jordan). In homogeneous FLRW, the Jordan and Einstein scale factors are related by $a_E = \sqrt{F(\psi)} a_J = a_J/A(\psi)$. Therefore

$$(1 + z_E) \frac{A(\psi)_{\text{obs}}}{A(\psi)_{\text{emit}}} = \frac{a_E(t_{E\text{obs}})}{a_E(t_{E\text{emit}})} \frac{A(\psi)_{\text{obs}}}{A(\psi)_{\text{emit}}} = \frac{a_J(t_{J\text{obs}})}{a_J(t_{J\text{emit}})} \equiv 1 + z_{\text{cos}}, \quad (12)$$

so that in the strictly homogeneous limit $z_{\text{obs}} = z_{\text{cos}}$ and the standard ladder relations are recovered in the physical Jordan frame (Sec. ??). In other words, the *background* effect of $A(\psi)$ is already encoded in $H_J(z)$ once one solves the Jordan-frame scalar–tensor background.

Where an additional $\Delta \ln(1 + z)$ imprint can arise. Any extra mapping between an “underlying” cosmological redshift and the observed redshift must come from *inhomogeneous* structure (endpoint and integrated effects) or from the backreaction of the homogeneous mode on $H_J(z)$. Writing $\psi = \bar{\psi}(t_J) + \delta\psi(x)$ and $\mathcal{C}(\psi) = \bar{\mathcal{C}}(t_J) + \delta\mathcal{C}(x)$, the observer-local imprint tested by the LTA mapping corresponds to packaging the net perturbative contribution as

$$\ln\left(\frac{1 + z_{\text{obs}}}{1 + z_{\text{cos}}}\right) \equiv I \simeq [\delta]_{\text{obs}} - [\delta]_{\text{emit}} + \int_{\gamma} k^{\mu} \nabla_{\mu}^J(\delta) d\lambda, \quad (13)$$

which motivates the purely phenomenological $I(\chi)$ parameterization used in the SN+BAO pipeline as a controlled observer-local limit (Sec. 7.2).

4.3 Why the correction is naturally written as an *integral* in $\ln(1 + z)$

Even though Eq. (12) may be written as an endpoint difference plus an integrated term, it is mathematically (and physically) natural to represent the *perturbative* chronometric imprint as an *accumulated* contribution to $\ln(1 + z)$.

Along a photon path γ , the difference $\mathcal{C}_{\text{obs}} - \mathcal{C}_{\text{emit}}$ may be written as a line integral of the local gradient:

$$\mathcal{C}_{\text{obs}} - \mathcal{C}_{\text{emit}} = \int_{\gamma} d\mathcal{C} = \int_{\gamma} (\nabla_{\mu} \mathcal{C}) dx^{\mu} = \int_{\gamma} (k^{\mu} \nabla_{\mu} \mathcal{C}) d\lambda, \quad (14)$$

where λ is any affine parameter along the null geodesic.

This motivates defining an effective *local* fractional frequency-drift contribution along the line of sight and integrating it in $\ln(1 + z)$, exactly as done in the LTA pipeline. In the isotropic observer-local approximation used for SN+BAO fitting, we parameterize this imprint as a function of comoving distance alone:

$$I(\chi) \equiv \mathcal{C}_{\text{obs}} - \mathcal{C}_{\text{emit}}(\chi), \quad I(\chi) = \frac{1}{c} \int_0^{\chi} s(\chi') d\chi', \quad s(\chi) \equiv c \frac{dI}{d\chi}. \quad (15)$$

Combining Eq. (7) with Eq. (14) yields the LTA mapping used in the likelihood:

$$1 + z_{\text{obs}} = (1 + z_{\text{cos}}) \exp[I(\chi)]. \quad (16)$$

A useful differential form makes explicit the “sum along the path” intuition:

$$\frac{d}{d\chi} \ln(1 + z_{\text{obs}}) = \frac{d}{d\chi} \ln(1 + z_{\text{cos}}) + \frac{s(\chi)}{c} = \frac{H(z_{\text{cos}})}{c} + \frac{s(\chi)}{c}, \quad (17)$$

where we used $dz_{\text{cos}}/d\chi = H(z_{\text{cos}})/c$ in FLRW. Thus $s(\chi)$ is precisely the additional local contribution to the *log-redshift accumulation* per unit comoving distance.

4.4 Universal law vs. the observer-local limit tested in this paper

Conceptually, Eq. (8) is intended as a *universal* law: wherever sustained organization increases ψ , the realized accumulation rate of clock time is biased by $A(\psi)$. In a fully general cosmological setting, $\psi(x)$ would be sourced by *all* organized systems (and potentially by organized eras) and may vary across space and time.

The net imprint on an incoming photon would then be described by the same local transport statement (Eq. (13)), with possible direction dependence:

$$I[\gamma] = \int_{\gamma} (k^{\mu} \nabla_{\mu} \mathcal{C}) d\lambda. \quad (18)$$

However, the empirical anomaly we isolate in the SN Hubble diagram is sharply localized to the nearby volume. For the present paper we therefore test a *controlled observer-local dominant* limit: we model the redshift-dependent part of $I(\chi)$ as arising primarily from the evolution and causal reach of the chronometric field sourced in the observer’s local environment. This produces an approximately isotropic mapping at low redshift while remaining falsifiable through the BAO Jacobian (Sec. 10.3). Generalizing $I[\gamma]$ beyond the observer-local limit (i.e. including distributed cosmological sources of ψ) is an important next step, but it is not required to test whether the *local* Hubble-flow residual structure prefers an integrated $\Delta \ln(1+z)$ mapping.

4.5 Jordan-frame scalar–tensor completion and a derived chronometric factor

To make the chronometric hypothesis mathematically well-posed *and* to ensure a self-consistent late-time cosmology capable of mimicking dark-energy-like expansion histories, we adopt a standard scalar–tensor (ST) structure and take the *Jordan frame* to be the physical (matter) frame. Operational observables—atomic clock rates, Cepheid periods, supernova light-curve timescales, ruler lengths, and the redshift measured by spectroscopy—are defined with respect to the Jordan metric $g_{\mu\nu}^J$.

Jordan-frame action (physical matter frame). We work with signature $(-, +, +, +)$ and define $M_{\text{Pl}}^2 \equiv (8\pi G)^{-1}$. A minimal ST completion with an additional nonequilibrium “order” sector is

$$\mathcal{S} = \int d^4x \sqrt{-g_J} \left[\frac{M_{\text{Pl}}^2}{2} F(\psi) R_J - \frac{f^2}{2} (\nabla_J \psi)^2 - f^4 U(\psi) \right] + \mathcal{S}_m[g^J, \Phi] + \mathcal{S}_{\text{neq}}[g^J, \psi, \Xi], \quad (19)$$

where F and U are dimensionless functions, f is a stiffness scale (mass dimension 1), $(\nabla_J \psi)^2 \equiv g^{J\mu\nu} \nabla_{\mu} \psi \nabla_{\nu} \psi$, and $\mathcal{S}_{\text{neq}}$ is the covariant insertion point for sustained nonequilibrium activity / constraint-maintenance that sources ψ .

Einstein frame and the *derived* chronometric factor. Define the Einstein-frame metric by the standard conformal transformation

$$g_{\mu\nu}^E \equiv F(\psi) g_{\mu\nu}^J. \quad (20)$$

In this frame the gravitational sector is canonical, but the matter action becomes conformally coupled:

$$\mathcal{S}_m[g^J, \Phi] = \mathcal{S}_m[A(\psi)^2 g^E, \Phi], \quad A(\psi) \equiv F(\psi)^{-1/2}, \quad (21)$$

so the conformal/chronometric factor $A(\psi)$ is *not* an independent postulate in the scalar–tensor completion; it is *derived* from the Jordan-frame coupling function $F(\psi)$.

It is useful to define the chronometric potential

$$\mathcal{C}(\psi) \equiv \ln A(\psi) = -\frac{1}{2} \ln F(\psi), \quad (22)$$

so that small variations are $\delta \mathcal{C}(\psi) = -\frac{1}{2} \delta \ln F$.

Clock-time relation (frame conversion). Physical clock time is Jordan proper time $d\tau_J$. In Einstein-frame variables,

$$d\tau_J^2 \equiv -g_{\mu\nu}^J dx^{\mu} dx^{\nu} = -A(\psi)^2 g_{\mu\nu}^E dx^{\mu} dx^{\nu} \equiv A(\psi)^2 d\tau_E^2 \quad \Rightarrow \quad d\tau_J = A(\psi) d\tau_E. \quad (23)$$

Equation (??) is the precise sense in which a scalar–tensor completion induces a “chronometric factor”: it is the conversion between Einstein-frame proper time and the proper time measured by matter clocks.

Homogeneous limit vs. observer-local imprints. In a homogeneous FLRW background, the measured cosmological redshift is determined by the Jordan scale factor: $1+z = a_J(t_0)/a_J(t_e)$. Section 7 re-derives the SN/BAO distance ladder in the Jordan frame based on the Jordan-frame expansion $H_J(z)$ implied by (??). Local time-acceleration effects correspond to *inhomogeneous* perturbations $\psi = \bar{\psi}(t) + \delta\psi(x, t)$, which can generate endpoint and integrated contributions to the observed redshift and time-domain observables. The phenomenological LTA mapping of earlier versions of this work is recovered as a controlled observer-local limit (Sec. 7.2).

4.6 Model hierarchy used in this paper: Jordan-frame scalar–tensor background $\rightarrow$ observer-local imprint

To keep the theory logically unified yet empirically testable, we distinguish three layers:

1. **Jordan-frame scalar–tensor completion (theory level).** We posit an action of the form (??) in which the order scalar $\psi(x)$ enters the gravitational sector through a coupling $F(\psi)$ and is sourced by a nonequilibrium insertion $\mathcal{S}_{\text{neq}}$. In this completion, the *chronometric factor* is not a separate axiom: it is *derived* under the standard Jordan $\rightarrow$ Einstein conformal transformation, $A(\psi) = F(\psi)^{-1/2}$ (Sec. 4.5).
2. **Homogeneous background (cosmology level).** The homogeneous mode $\bar{\psi}(t)$ backreacts on the Jordan-frame expansion history $H_J(z)$, which fully determines the standard SN/BAO distance ladder in the physical frame (Secs. ??–??). This is the channel through which the theory can mimic dark-energy-like expansion without introducing Λ as a separate component.
3. **Observer-local imprint (controlled phenomenology).** Inhomogeneities $\delta\psi(x)$ and their associated chronometric potential perturbations $\delta(x)$ can generate endpoint and integrated contributions to the observed redshift beyond the homogeneous background. In the observer-local regime tested by our current SN+BAO pipeline, we package this net perturbative contribution into an effective mapping $1 + z_{\text{obs}} = (1 + z_{\text{cos}}) \exp[I(\chi)]$ (Sec. 7.2), which is a controlled approximation to the inhomogeneous scalar–tensor propagation problem.

This hierarchy clarifies what is and is not assumed in any given fit: the SN+BAO mapping layer tests whether the data prefer an observer-local distortion, while the Jordan-frame background layer is the self-consistent route to explaining dark-energy-like effects via the same field.

5 Order field dynamics, causality, and the observer-retarded approximation

The chronometric completion in Sec. 4 identifies the redshift mapping with a scalar chronometric potential $\mathcal{C}(\psi) = \ln A(\psi)$. To connect the mapping to causal sourcing by organized nonequilibrium activity, we introduce an effective dynamics for $\psi(x)$.

5.1 Effective sourced scalar dynamics

At coarse-grained level, we treat $\psi(x)$ as a covariant scalar sourced by a nonnegative proxy for sustained nonequilibrium activity, denoted $\sigma(x) \geq 0$ (e.g. an entropy production density or dissipated free-energy throughput divided by an effective temperature). A minimal causal model is a sourced Klein–Gordon form,

$$(\square - m_\psi^2) \psi(x) = \kappa \sigma(x), \quad (24)$$

with retarded boundary conditions. The mass scale m_ψ sets an effective range/time scale over which the influence of sources persists. This is not asserted to be fundamental microphysics; it is the minimal effective field-theory statement required to enforce locality and causality in the sourcing of ψ .

5.2 Observer-local dominant limit and the retarded-time mapping

In the general universal-law setting, $\sigma(x)$ is distributed throughout the Universe and $I[\gamma]$ is path-dependent (Eq. (17)). The empirical anomaly isolated in this paper is, however, localized to the low- z Hubble-flow regime. We therefore test a controlled approximation in which the *redshift-dependent* part of the imprint is dominated by the observer’s local environment and its causal reach.

Operationally, this observer-local dominant limit is implemented by evaluating the activation history as a function of an observer-retarded time parameter $t_{\text{ret}}(\chi)$ (Sec. 7.3). The specific two-leg construction $\chi(z_e) = 2\chi$ and $t_{\text{ret}} = t_{\text{lb}}(z_e)$ is used as a convenient causal bookkeeping proxy for the retarded solution of Eq. (21) in an expanding background. This yields a smooth, bounded $I(\chi)$ that can be fit to SN+BAO and falsified via the BAO Jacobian.

6 Thermodynamic foundation: from energy throughput to a bounded constraint inventory

The empirical LTA mapping in this paper is written as an accumulated contribution to $\ln(1+z)$ along the past light cone. To connect that phenomenology to established physics, we need a quantitative, well-behaved scalar that captures “order / constraint creation” in a way that (i) is mathematically precise under coarse-graining, (ii) is directly tied to the second law, and (iii) admits hard bounds from energy throughput.

The key choice is to represent sustained “constraint” as *relative entropy* (KL divergence) with respect to an appropriate maximum-entropy (reference) distribution. This is standard in nonequilibrium statistical mechanics and stochastic thermodynamics, where relative entropy quantifies the *excess* thermodynamic potential available in a nonequilibrium state [26, 27].

Throughout this section, we assume a coarse-grained description in which microstates are labeled by x and the system has an instantaneous state distribution $p(x, t)$. We also assume a chosen reference distribution $\pi(\xi)$ representing the maximally unstructured (maximum-entropy) baseline given the constraints one is *not* counting as “order” (e.g. a canonical equilibrium, or a MaxEnt state under specified macroscopic constraints).

6.1 Lemma 1: A constraint inventory φ that is nonnegative, monotone under coarse-graining, and additive under composition

Definition (constraint inventory). Define the dimensionless *constraint inventory*

$$\phi(t) \equiv D_{\text{KL}}(p(\cdot, t) \| \pi(\cdot)) = \int dx \, p(x, t) \ln \frac{p(x, t)}{\pi(\xi)}, \quad (25)$$

and its entropy-unit version

$$\varphi(t) \equiv k_B \phi(t) \quad [\varphi] = \text{J K}^{-1}. \quad (26)$$

The KL divergence was introduced in [24].

We will refer to φ as the “constraint inventory” because it measures how many *nats* (or, dividing by $\ln 2$, *bits*) of information are required to specify the additional structure in p relative to the chosen reference π .

Lemma 1a (nonnegativity).

$$\varphi(t) \geq 0 \quad \text{for all } t, \quad \varphi(t) = 0 \iff p(\cdot, t) = \pi(\cdot) \text{ a.e.} \quad (27)$$

Reason: $D_{\text{KL}}(\cdot \| \cdot)$ is nonnegative by Gibbs’ inequality.

Lemma 1b (monotonicity under coarse-graining). Let $\mathcal{M}$ be any stochastic coarse-graining (Markov) map from microstates x to macro-cells y ; denote $p_y = \mathcal{M}p$ and $\pi_y = \mathcal{M}\pi$. Then

$$D_{\text{KL}}(p \| \pi) \geq D_{\text{KL}}(p_y \| \pi_y), \quad \text{equivalently} \quad \varphi \geq \varphi_y. \quad (28)$$

This is the *data-processing inequality* for KL divergence: coarse-graining cannot increase distinguishability from the reference and therefore cannot increase the counted constraint inventory [25].

Lemma 1c (additivity / extensivity under independent composition). If the system decomposes into weakly interacting subsystems with $p(x_1, x_2) = p_1(x_1)p_2(x_2)$ and $\pi(x_1, x_2) = \pi_1(x_1)\pi_2(x_2)$, then

$$D_{\text{KL}}(p\|\pi) = D_{\text{KL}}(p_1\|\pi_1) + D_{\text{KL}}(p_2\|\pi_2), \quad \varphi = \varphi_1 + \varphi_2. \quad (29)$$

Thus φ is *extensive* under composition when correlations between components are negligible at the chosen coarse-graining scale.

Microstate-space interpretation (explicit link to “constraining microstates”). In the special case where π is uniform on an allowed set of size Ω_0 and p is uniform on a constrained subset of size $\Omega_1 \leq \Omega_0$, one finds

$$D_{\text{KL}}(p\|\pi) = \ln \frac{\Omega_0}{\Omega_1}, \quad \varphi = k_B \ln \frac{\Omega_0}{\Omega_1}. \quad (30)$$

So φ literally counts the log-compression of accessible microstates induced by the constraints being maintained.

Comment (why we do *not* anchor on entropy alone). Total entropy production is a throughput and is always nonnegative, but it does not by itself distinguish “order creation” from generic dissipation. By contrast, φ is a *state variable* that measures how far the maintained state is from a maximum-entropy reference and thus tracks constraint maintenance directly; the second law will then bound how fast φ can grow.

6.2 Lemma 2: $T\varphi$ is the nonequilibrium free-energy budget stored in constraints

To connect φ to energetics, specialize to a canonical reference distribution

$$\pi(\xi) = \frac{e^{-\beta E(\xi)}}{Z}, \quad \beta \equiv (k_B T)^{-1}, \quad (31)$$

at temperature T for the same energy function $E(x)$.

Define the (instantaneous) nonequilibrium Helmholtz free energy functional

$$F[p] \equiv \langle E \rangle_p - TS[p], \quad S[p] \equiv -k_B \int dx p(x) \ln p(x), \quad (32)$$

and the equilibrium free energy

$$F_{\text{eq}} \equiv -k_B T \ln Z. \quad (33)$$

Lemma 2 (relative entropy identity). Under the canonical choice (28),

$$F[p] - F_{\text{eq}} = k_B T D_{\text{KL}}(p\|\pi) = T\varphi. \quad (34)$$

Proof sketch: Insert $\ln \pi = -\beta E - \ln Z$ into the definition of D_{KL} , then use $S[p] = -k_B \int p \ln p$ and $F_{\text{eq}} = -k_B T \ln Z$.

Interpretation. Equation (31) states that φ is not a metaphor: it is exactly the *excess free energy above equilibrium* stored in the maintained constraints. This is a standard result in nonequilibrium thermodynamics and stochastic thermodynamics [26, 27].

Bits and Landauer scaling (optional but useful). Define $\varphi_{\text{bits}} \equiv \varphi / (k_B \ln 2) = \phi / \ln 2$. Then $T\varphi$ corresponds to an energy scale of $k_B T \ln 2$ per bit of maintained constraint, matching the Landauer cost scale for logically irreversible information erasure [28, 29]. We emphasize: this does *not* claim biology saturates Landauer bounds; it is a hard lower bound and provides a clean dimensional bridge between constraint maintenance and energetic cost.

6.3 Lemma 3: The second law bounds $\dot{\varphi}$ by energy throughput

We now connect constraint growth to the second law in a form that yields a quantitative bound from power flow.

Scope of the bound. The inequalities below are stated for an effective *isothermal* driven setting at a single temperature T , with a fixed reference distribution π (equivalently fixed F_{eq}) for the degrees of freedom whose constraint inventory is being tracked. In more general settings with multiple reservoirs, chemical work, or time-dependent reference constraints, the same structure holds with P_{in} interpreted as the appropriate generalized free-energy/work influx, but we restrict to the isothermal form here because it is sufficient for the conservative Earth-budget envelope used later.

Consider a system coupled to an (effective) isothermal reservoir at temperature T . Let $\dot{Q}$ be heat flow *into* the system and $\dot{W}$ be work done *on* the system. The first law reads

$$\frac{d}{dt}\langle E \rangle = \dot{Q} + \dot{W}. \quad (35)$$

Define total entropy production rate (system + environment) as

$$\dot{\Sigma} \equiv \frac{dS}{dt} - \frac{\dot{Q}}{T} \geq 0, \quad (36)$$

where the inequality is the (local) second law for this driven isothermal setting [26].

Using $F = \langle E \rangle - TS$ and Eqs. (32)–(33), one obtains the standard work–free-energy–dissipation identity:

$$\dot{W} = \frac{dF}{dt} + T\dot{\Sigma}. \quad (37)$$

Lemma 3a (power bounds constraint growth). If the reference equilibrium F_{eq} is fixed (same $E(x)$ and T), then Lemma 2 implies $d(F - F_{\text{eq}})/dt = T\dot{\varphi}$. Combining with (34) and $\dot{\Sigma} \geq 0$ gives

$$T\dot{\varphi} \leq \dot{W} \equiv P_{\text{in}}, \quad \text{equivalently} \quad \dot{\varphi} \leq \frac{P_{\text{in}}}{T}. \quad (38)$$

This is the rigorous form of the chain

$$\text{energy consumption} \Rightarrow \text{bounded constraint accumulation}.$$

Lemma 3b (relation to entropy production). From (34) and $d(F - F_{\text{eq}})/dt = T\dot{\varphi}$, we also have

$$\dot{\Sigma} = \frac{P_{\text{in}}}{T} - \dot{\varphi}. \quad (39)$$

Therefore, $\dot{\Sigma} \leq P_{\text{in}}/T$ holds *whenever* $\dot{\varphi} \geq 0$ (net constraint accumulation), and in steady maintenance regimes where $\langle \dot{\varphi} \rangle \simeq 0$ one has $\langle \dot{\Sigma} \rangle \simeq \langle P_{\text{in}} \rangle / T$.

Earth-bound specialization (what we will use later). Once a conservative upper bound on $P_{\text{in}}(t)$ is specified (e.g. solar free-energy throughput available to the biosphere/technosphere) and a conservative effective temperature T is chosen, Eq. (35) provides a hard upper bound on the growth rate of the constraint inventory that any Earth-local “order field” can be fed by without violating the second law.

Transition to gravity. Lemmas 1–3 complete the “energy $\rightarrow$ constraints” portion of the chain with standard, well-established nonequilibrium/statistical thermodynamics. The remaining step (Lemma 4) is *not* thermodynamics: it is the additional postulate specifying how a covariant scalar constraint inventory φ couples into the gravitational sector (metric potentials / time metric), and how that coupling produces an accumulated frequency shift along null geodesics that reduces to the LTA mapping used in the SN+BAO analysis.

Our goal is to properly integrate this math into known field equations for GR such that we can then use these equations to evaluate whether they can be used to calculate timing arrays on pulsars or other cosmological datapoints by inputting the relative order of the system in question.

6.4 Lemma 4 (Postulate): a chronometric coupling that biases clock rates without modifying spacetime geometry

Lemmas 6.1–6.3 provide a standard, thermodynamically rigorous chain from energy throughput to a bounded constraint inventory φ (and its dimensionless version $\psi = \varphi/\varphi$). We now introduce the *single new postulate* required to connect this inventory to the observed redshift mapping used in the LTA analysis, while preserving the standard GR spacetime geometry and therefore avoiding new lensing/deflection signatures.

Postulate 4 (chronometric field). Spacetime geometry is governed by standard GR (the same $g_{\mu\nu}$ used for Λ CDM distances and null geodesics), but the *realized physical clock time* τ_{clk} measured by matter clocks is related to GR proper time τ by a dimensionless *time-flow factor* $A(\psi)$:

$$d\tau_{\text{clk}} = A(\psi) d\tau, \quad A(0) = 1, \quad \frac{\partial A}{\partial \psi} > 0. \quad (40)$$

Thus sustained organization ($\psi > 0$) increases the local accumulation rate of clock time relative to GR proper time (“local time acceleration”).

For small ψ we parameterize

$$A(\psi) = \exp(\epsilon \psi) \simeq 1 + \epsilon \psi + \mathcal{O}(\psi^2), \quad (41)$$

with a dimensionless coupling ϵ .

Frequency measurement and redshift mapping. Given the chronometric postulate (37), the measured frequency is rescaled as $\nu_{\text{meas}} = \nu_{\text{GR}}/A(\psi)$, and the observed redshift acquires an additional factor $A(\psi_{\text{obs}})/A(\psi_{\text{emit}})$. The full derivation (including the integrated $\Delta \ln(1+z)$ form, its local drift representation, and the general path-dependent expression $I[\gamma]$) is given in Sec. 4. Here we record only the resulting mapping used in the SN+BAO likelihood:

$$1 + z_{\text{obs}} = (1 + z_{\text{cos}}) \exp[I], \quad I \equiv \ln \frac{A(\psi_{\text{obs}})}{A(\psi_{\text{emit}})}. \quad (42)$$

Causality in the observer-local dominant limit. In the low- z anomaly tested here, the redshift-dependent part of I is modeled as being dominated by the observer’s local environment and its causal reach; this motivates evaluating the activation through an observer-retarded time argument $t_{\text{ret}}(\chi)$ (Sec. 5 and Sec. 7.3).

Relation to the phenomenological drift field $s(\chi)$. Using $I(\chi)$ as defined in (??), we define the drift field exactly as in the main analysis:

$$s(\chi) \equiv c \frac{dI}{d\chi}. \quad (43)$$

With $A(\psi) = \exp(\epsilon\psi)$ this becomes

$$s(\chi) = -c\epsilon \frac{d\psi_{\text{emit}}(\chi)}{d\chi}, \quad (44)$$

where the minus sign reflects that more distant emission events typically correspond to earlier (retarded) epochs when ψ was smaller. Under the simple causal bookkeeping used in this work, $t_{\text{ret}} \approx 2\chi/c$ at low redshift, so $d\psi_{\text{emit}}/d\chi \approx (2/c) \dot{\psi}(t_{\text{ret}})$, giving the useful approximate link

$$s(\chi) \approx -2\epsilon \dot{\psi}(t_{\text{ret}}(\chi)) \quad (\text{low-}z \text{ causal approximation}). \quad (45)$$

This connects the empirical $s(t_{\text{ret}})$ profile directly to the time-derivative of the underlying order field.

No new lensing by construction. Because we do not modify the GR metric governing null geodesics in this postulate, photon trajectories (and therefore gravitational lensing/deflection predictions) remain those of the baseline cosmology. The additional effect enters only through the realized clock time used in the frequency comparison that defines observed redshift.

Thermodynamic bound on the effect amplitude. Combining Lemma 3 with $\psi = \varphi/\varphi$ yields

$$\dot{\psi} = \frac{\dot{\varphi}}{\varphi} \leq \frac{\dot{\varphi}}{T \varphi} \frac{P_{\text{in}}}{(46)}$$

Using (42), this implies an energetics-based envelope on the LTA drift field,

$$|s(\chi)| \lesssim 2\epsilon \frac{P_{\text{in}}}{T \varphi} \quad (\text{low-}z \text{ causal approximation}), (47)$$

and correspondingly a bound on the saturation imprint $I_{\text{sat}} = \max I(\chi)$ in terms of the integrated Earth-system energy/entropy throughput over the active epoch.

7 Model definition: Jordan-frame scalar–tensor background plus local chronometric imprint

This section specifies the forward model used throughout in a form suitable for a *self-consistent* scalar–tensor treatment of both (i) dark-energy-like acceleration in the homogeneous background and (ii) observer-/environment-linked chronometric imprints in the local Universe.

We distinguish:

1. A **homogeneous Jordan-frame scalar–tensor background** that determines the expansion history $H_J(z)$ and therefore the standard distance ladder.
2. An **optional local chronometric imprint** (observer-local perturbation) that can shift the mapping between an “underlying” cosmological redshift z_{cos} and the observed redshift z_{obs} , recovering the LTA mapping as a controlled limit.

7.1 Jordan-frame background: expansion history from scalar–tensor dynamics

Background geometry and redshift. We take the physical (matter) metric to be a spatially flat FLRW line element in the Jordan frame:

$$ds_J^2 = -dt_J^2 + a_J^2(t_J) d\mathbf{x}^2, \quad H_J \equiv \frac{1}{a_J} \frac{da_J}{dt_J}. \quad (48)$$

The *cosmological* redshift is defined by

$$1 + z_{\text{cos}} \equiv \frac{a_J(t_{J0})}{a_J(t_{Je})}. \quad (49)$$

In the homogeneous limit, the observed redshift equals the cosmological redshift: $z_{\text{obs}} = z_{\text{cos}}$ up to the standard local corrections (peculiar velocities, local gravitational potentials).

Field equations (Jordan frame). Varying the Jordan-frame action (??) yields

$$M_{\text{Pl}}^2 F(\psi) G_{\mu\nu}^J = T_{\mu\nu}^{(m)} + T_{\mu\nu}^{(\psi)} + T_{\mu\nu}^{(\text{neq})} + M_{\text{Pl}}^2 (\nabla_\mu^J \nabla_\nu^J F(\psi) - g_{\mu\nu}^J \square_J F(\psi)), \quad (50)$$

with

$$T_{\mu\nu}^{(\psi)} = f^2 \nabla_\mu^J \psi \nabla_\nu^J \psi - g_{\mu\nu}^J \left[\frac{f^2}{2} (\nabla_J \psi)^2 + f^4 U(\psi) \right], \quad (51)$$

and a scalar equation

$$f^2 \square_J \psi - f^4 U(\psi)' + \frac{M_{\text{Pl}}^2}{2} F(\psi)' R_J = J_{\text{neq}}, \quad J_{\text{neq}} \equiv -\frac{1}{\sqrt{-g_J}} \frac{\delta \mathcal{S}_{\text{neq}}}{\delta \psi}. \quad (52)$$

In this Jordan-first setup, matter is minimally coupled, so $T_{\mu\nu}^{(m)}$ is covariantly conserved with respect to ∇^J (in the absence of explicit matter– Ξ exchange in $\mathcal{S}_{\text{neq}}$).

Friedmann form. For a homogeneous $\psi(t_J)$, the 00 equation may be written as

$$3M_{\text{Pl}}^2 F(\psi) H_J^2 = \rho_m + \rho_r + \rho_\psi + \rho_{\text{neq}} - 3M_{\text{Pl}}^2 H_J \frac{dF(\psi)}{dt_J}, \quad (53)$$

where

$$\rho_\psi = \frac{f^2}{2} \left(\frac{d\psi}{dt_J} \right)^2 + f^4 U(\psi), \quad (54)$$

and ρ_{neq} denotes the homogeneous energy density carried by the nonequilibrium sector (if present in the background). The corresponding acceleration equation can be written as

$$-2M_{\text{Pl}}^2 F(\psi) \frac{dH_J}{dt_J} = (\rho_m + p_m) + (\rho_r + p_r) + (\rho_\psi + p_\psi) + (\rho_{\text{neq}} + p_{\text{neq}}) + M_{\text{Pl}}^2 \left(\frac{d^2 F(\psi)}{dt_J^2} - H_J \frac{dF(\psi)}{dt_J} \right), \quad (55)$$

with $p_m \simeq 0$, $p_r = \rho_r/3$, and

$$p_\psi = \frac{f^2}{2} \left(\frac{d\psi}{dt_J} \right)^2 - f^4 U(\psi). \quad (56)$$

Practical parameterization. For inference, we continue to parameterize the late-time background by a present-day Hubble scale H_0 and a present-day matter density fraction Ω_m (defined in the Jordan frame), together with scalar-sector parameters specifying $F(\psi)$ and $U(\psi)$ (and any nonequilibrium/source parameters in $\mathcal{S}_{\text{neq}}$). In general there is no closed-form $E(z)$; instead $H_J(z)$ is obtained by solving the coupled background system.

7.2 Observable ladder in the homogeneous Jordan background

Given $H_J(z)$, the distance ladder follows the standard FLRW relations *in the Jordan frame* (where matter defines clocks and rulers).

Comoving distance and lookback time. The line-of-sight comoving distance is

$$\chi(z) = c \int_0^z \frac{dz'}{H_J(z')}, \quad (57)$$

and the proper lookback time is

$$t_{\text{lb}}(z) = \int_0^z \frac{dz'}{(1+z')H_J(z')}. \quad (58)$$

(These reduce to the familiar Λ CDM expressions when the model reduces to GR+ Λ .)

Luminosity and angular-diameter distances. For spatial flatness,

$$D_M(z) = \chi(z), \quad D_A(z) = \frac{\chi(z)}{1+z}, \quad d_L(z) = (1+z)\chi(z). \quad (59)$$

The SN distance modulus prediction is therefore

$$\mu_{\text{pred}}(z) = 5 \log_{10} \left(\frac{d_L(z)}{\text{Mpc}} \right) + 25. \quad (60)$$

BAO observables. The standard BAO distance combinations in the Jordan background are

$$D_H(z) \equiv \frac{c}{H_J(z)}, \quad D_V(z) \equiv [z D_H(z) D_M^2(z)]^{1/3}, \quad (61)$$

and the comparison to data is made through ratios to the sound horizon r_d . In this paper we retain a phenomenological scaling parameter $\alpha_{\text{rd}} \equiv r_d/r_{d,\text{fid}}$ in the BAO likelihood; in the fully self-consistent program, r_d is computed from the Jordan-frame early-time dynamics.

7.3 Local chronometric imprint and connection to the LTA mapping

In a strictly homogeneous Jordan-frame cosmology, $z_{\text{obs}} = z_{\text{cos}}$ and the ladder above is complete. Observer-local time-acceleration effects correspond to *inhomogeneous* chronometric structure (e.g. $\psi = \bar{\psi}(t) + \delta\psi(x, t)$) that modifies clock comparisons and therefore the operationally inferred redshift.

A convenient way to connect this local-perturbation picture back to the phenomenological LTA analysis is to package the net chronometric imprint into an additive contribution to $\ln(1+z)$:

$$1 + z_{\text{obs}} = (1 + z_{\text{cos}}) \exp[I(\chi)(z_{\text{cos}})]. \quad (62)$$

Here z_{cos} is the cosmological redshift defined by the homogeneous Jordan background (??), while z_{obs} is the measured redshift including the observer-local imprint.

As in the phenomenological formulation, we define

$$I(\chi) \equiv \frac{1}{c} \int_0^\chi s(\chi)' d\chi', \quad (63)$$

where $s(\chi)$ has dimensions of a Hubble rate (e.g. $\text{km s}^{-1} \text{Mpc}^{-1}$) and $s(\chi)/c$ plays the role of an effective fractional frequency drift per unit comoving distance in the observer-local approximation.

Differentiating Eq. (49) and using $d\chi/dz_{\text{cos}} = c/H_J(z_{\text{cos}})$ gives

$$\frac{dI(\chi)}{dz_{\text{cos}}} = \frac{1}{c} s(\chi)(z_{\text{cos}}) \frac{d\chi}{dz_{\text{cos}}} = \frac{s(\chi)(z_{\text{cos}})}{H_J(z_{\text{cos}})}. \quad (64)$$

7.4 Earth-retarded lookback time construction

To encode the motivating “observer-sourced” (Earth-anchored) character in a causally consistent way, we evaluate the activation along the photon path using an *Earth-retarded lookback time* argument $t_{\text{ret}}(\chi)$ computed in the same Jordan-frame background.

For a given line-of-sight distance χ (corresponding to z_{cos}), we define an “Earth emission” redshift z_e implicitly by

$$\chi(z_e) = 2\chi, \quad (65)$$

and then set

$$t_{\text{ret}}(\chi) \equiv t_{\text{lb}}(z_e). \quad (66)$$

At low redshift, where $\chi \approx (c/H_0)z$ and $t_{\text{lb}} \approx H_0^{-1}z$, Eq. (52) implies $z_e \approx 2z$ and $t_{\text{ret}} \approx 2t_{\text{lb}}$, making the causal bookkeeping interpretation explicit.

7.5 Bounded activation and parameterization of $s(\chi)$

We parameterize the drift-rate profile as a bounded activation scaled by an amplitude defined at an explicit *anchor time* t_{anchor} :

$$s(\chi) = s_{\text{anchor}} \frac{g(t_{\text{ret}}(\chi))}{g(t_{\text{anchor}})}. \quad (67)$$

Here $g(t) \in [0, 1]$ is an activation function of Earth-retarded lookback time, normalized so that $g(0) = 1$ and $g(t) = 0$ for $t \geq t_{\text{life}}$, where t_{life} is a finite cutoff epoch.

Anchor-time choice (leakage control). To avoid any dependence of the LTA normalization on the calibrator subset, we compute t_{anchor} using only the Hubble-flow subset (excluding calibrators). Concretely, we define t_{anchor} as the 5th percentile of the t_{ret} distribution over Hubble-flow objects. This rule is stable under resampling and keeps the normalization tied to the Hubble-flow regime that drives the cosmological fit.

Power-law activation family (fiducial choice). In this work we adopt a smooth power-law decay. For $0 \leq t \leq t_{\text{life}}$ define

$$r(t) = \left(1 + \frac{t}{B}\right)^{-p}, \quad (68)$$

with timescale $B > 0$ (in Gyr) and slope $p \geq 0$, and set

$$g(t) = \frac{r(t) - r(t_{\text{life}})}{1 - r(t_{\text{life}})}. \quad (69)$$

This guarantees $g(0) = 1$ and $g(t_{\text{life}}) = 0$, yielding a continuous, bounded activation with a tunable early-time scale (B) and tail steepness (p).

Fiducial degrees of freedom (what is actually fit). Because the observer-local imprint enters through a line-of-sight integral, there are strong degeneracies between overall amplitude and detailed curve-shape parameters. Our primary analysis therefore fixes the activation *shape* to representative smooth values and fits only the amplitude s_{anchor} as the additional imprint degree of freedom beyond the background parameters (including α_{rd} in the BAO likelihood).

Optional modulation exponents (fixed in this work). The analysis code supports additional multiplicative modulation exponents (e.g. “complexity” and “life” weights) that rescale the activation family. In the fiducial results emphasized in this paper, these exponents are fixed to unity so that they do not contribute additional degrees of freedom.

7.6 Inverse mapping and redshift Jacobian

Data are indexed by z_{obs} (or by a measured redshift proxy such as z_{HD} for SNe), whereas the homogeneous background distances are naturally functions of z_{cos} . Therefore, for each observed redshift z_{obs} we obtain $z_{\text{cos}}(z_{\text{obs}})$ by numerically inverting Eq. (49) under the monotonicity condition

$$\frac{dz_{\text{obs}}}{dz_{\text{cos}}} > 0. \quad (70)$$

Differentiating Eq. (49) gives

$$\frac{dz_{\text{obs}}}{dz_{\text{cos}}} = \exp[I(\chi)] \left[1 + (1 + z_{\text{cos}}) \frac{dI(\chi)}{dz_{\text{cos}}} \right] = \exp[I(\chi)] \left[1 + (1 + z_{\text{cos}}) \frac{s(\chi)(z_{\text{cos}})}{H_{\text{J}}(z_{\text{cos}})} \right], \quad (71)$$

where the final equality uses Eq. (51). This Jacobian enters directly in radial BAO predictions and serves as an important diagnostic of mapping regularity.

7.7 Distances and effective expansion observables as functions of z_{obs}

The comoving distance as a function of observed redshift is evaluated as

$$\chi(z_{\text{obs}}) \equiv \chi(z_{\text{cos}}(z_{\text{obs}})), \quad (72)$$

with $\chi(z)$ computed in the Jordan background via Eq. (46). For radial derivatives,

$$\frac{d\chi}{dz_{\text{obs}}} = \frac{d\chi}{dz_{\text{cos}}} \frac{dz_{\text{cos}}}{dz_{\text{obs}}} = \frac{c}{H_{\text{J}}(z_{\text{cos}})} \left(\frac{dz_{\text{obs}}}{dz_{\text{cos}}} \right)^{-1}. \quad (73)$$

Accordingly, the *effective* Hubble rate inferred from the $z_{\text{obs}}-\chi$ relation is

$$H_{\text{eff}}(z_{\text{obs}}) \equiv \frac{c}{d\chi/dz_{\text{obs}}} = H_{\text{J}}(z_{\text{cos}}) \left(\frac{dz_{\text{obs}}}{dz_{\text{cos}}} \right). \quad (74)$$

Here “effective” means “as inferred from the z_{obs} -distance mapping in the presence of an observer-local chronometric imprint”; it does *not* redefine the homogeneous background law $H_{\text{J}}(z)$.

In the joint SN+BAO analysis below, transverse distances depend primarily on $\chi(z_{\text{obs}})$, while radial observables depend explicitly on the Jacobian factor $dz_{\text{obs}}/dz_{\text{cos}}$ through $H_{\text{eff}}(z_{\text{obs}})$ (Sec. 10.3).

8 Consistency constraints and falsification channels

A Jordan-frame scalar–tensor completion has sharp falsification channels in: (i) local gravity tests (PPN, fifth forces, and time variation of the effective Newton constant), (ii) clock and redshift comparisons (as operational probes of inhomogeneous endpoint/integrated terms), and (iii) early-universe calibration (sound horizon, recombination, and CMB inference). This section records the principal channels that must be confronted as the analysis moves from the observer-local mapping limit to a fully self-consistent cosmology.

8.1 Local gravity constraints: PPN, fifth forces, and $\dot{G}_{\text{eff}}/G_{\text{eff}}$

In the Jordan-first completion, matter is minimally coupled in the physical frame, while the effective gravitational coupling scales as

$$G_{\text{eff}} \propto \frac{1}{F(\psi)}. \quad (75)$$

Therefore any late-time evolution of $\bar{\psi}(t)$ generically implies a time variation of G_{eff} and potentially fifth-force/PPN signatures unless the model includes a screening mechanism or a parameter regime compatible with local tests.

In addition to cosmological fits, any viable parameter region must satisfy:

1. **Bounds on $\dot{G}/G$** from solar-system, binary pulsar, and other timing constraints, interpretable as $\dot{F}/F$ bounds in the model.
2. **PPN constraints / equivalence principle tests** that limit scalar-mediated deviations from GR in weak-field environments (see, e.g., [34] and modern satellite redshift tests [35, 36]).

8.2 Time-domain consistency: spectroscopy vs. time dilation in transients

A particularly crisp self-consistency check is that the redshift inferred from spectroscopy and the time-dilation inferred from light-curve stretching (or other transient timescales) must agree once all observables are computed in the same physical frame.

In a homogeneous Jordan-frame FLRW cosmology,

$$\Delta t_{\text{obs}} = (1 + z) \Delta t_{\text{emit}}, \quad (76)$$

with the same z as inferred from spectral line ratios. In the presence of an observer-local inhomogeneous imprint (the $I(\chi)$ mapping limit), the theory must specify whether the imprint acts as a purely spectroscopic endpoint/integrated frequency bias, a genuine modification of propagation, or a combination. This is therefore a primary falsification channel for any model that attempts to explain substantial cosmological effects via chronometric structure.

8.3 Early-universe calibration: sound horizon, recombination, and Planck priors

Once the scalar–tensor completion is treated as the cosmological background (rather than a local mapping), the sound horizon r_d and recombination-era observables are no longer external inputs: they are *model outputs* that depend on the physical expansion history $H_J(z)$ and on any energy exchange implied by the Einstein-frame representation of the theory. Accordingly:

1. The BAO ruler calibration (α_{rd} or r_d) should ultimately be computed from the model, not treated purely phenomenologically.
2. Any use of Planck-derived priors extracted under Λ CDM must be treated as *approximate* until recombination and CMB inference are recomputed within the scalar–tensor model.

For completeness we provide in Appendix E the frame bookkeeping and equations needed for the early-time extension; the current SN+BAO analysis retains a phenomenological α_{rd} while the full early-universe closure is implemented.

9 Data

9.1 Pantheon+SH0ES: calibrators + Hubble-flow subset with full covariance

We use the Pantheon+SH0ES public release with the corrected SN Ia magnitude observable

$$y \equiv m_b^{\text{corr}}, \quad (77)$$

and the accompanying *full* covariance matrix (including cross-correlations between calibrators and Hubble-flow objects). Our selected sample is the SH0ES global subset consisting of

$$N_{\text{SN}} = 354 \text{ total SNe, with } N_{\text{cal}} = 77 \text{ calibrators, } N_{\text{HF}} = 277 \text{ Hubble-flow objects.} \quad (78)$$

The selected Hubble-diagram redshifts satisfy

$$z_{\text{HD}} \in [0.00122, 0.14898], \quad (79)$$

and we retain the standard distinction between heliocentric redshift z_{HEL} (for the $(1 + z_{\text{HEL}})$ luminosity-distance prefactor) and the Hubble-diagram redshift z_{HD} (for the cosmology-dependent distance).

For calibrators, we use the externally determined distance moduli provided by the release (e.g. Cepheid-based; column CEPH_DIST in our implementation), spanning approximately

$$\mu_{\text{cal}} \in [29.177, 34.526], \quad (80)$$

and we keep calibrators present in every fit (and in every refit / cross-validation split when relevant) to preserve the absolute-distance anchor.

Covariance diagnostics. The released covariance is used in full (file-provided mode). As a compact indicator of correlation strength, we report an “effective dimension”

$$N_{\text{eff}} \equiv \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}, \quad (81)$$

where $\{\lambda_i\}$ are eigenvalues of the SN covariance; for the selected sample we obtain $N_{\text{eff}} \approx 149.5$ (out of 354), illustrating that correlations materially reduce the number of effectively independent modes.

9.2 BAO consensus compilation

We use the DESI 2024 Gaussian BAO compilation (GCcomb)[3], which provides D_V/r_d , D_M/r_d , and D_H/r_d constraints in multiple tracer/redshift bins, with a provided covariance.

We jointly fit a consensus BAO compilation providing transverse and radial distance information at discrete redshifts. The BAO data vector is treated as Gaussian with a provided covariance matrix. In our parameterization we introduce a single scaling parameter α_{rd} to absorb uncertainty in the sound horizon:

$$\alpha_{\text{rd}} \equiv \frac{r_d}{r_{d,\text{fid}}}. \quad (82)$$

To match the convention used in our code, the BAO observables are taken in the rescaled form

$$D_M^{(\text{data})}(z) \equiv \left(\frac{D_M(z)}{r_d} \right) r_{d,\text{fid}}, \quad H^{(\text{data})}(z) \equiv (H(z) r_d) \frac{1}{r_{d,\text{fid}}}, \quad (83)$$

so that the theory predictions become simple multiplicative scalings by α_{rd} (see Sec. 10.3). We assume the BAO and SN datasets are independent so that the joint likelihood covariance is block diagonal between SN and BAO.

10 Likelihood and inference

10.1 Supernova distance model and residual vector

For each SN, we compute the luminosity distance using the comoving distance evaluated at the cosmological redshift corresponding to the *observed* Hubble-diagram redshift:

$$d_L = (1 + z_{\text{HEL}}) \chi(z_{\text{cos}}(z_{\text{HD}})), \quad \mu_{\text{pred}} = 5 \log_{10} \left(\frac{d_L}{\text{Mpc}} \right) + 25. \quad (84)$$

The corrected magnitude observable is modeled as

$$m_{b,i}^{\text{corr}} = \mu_{\text{pred}}(z_i; \boldsymbol{\theta}) + M + \epsilon_i, \quad (85)$$

where M is an intercept (absolute magnitude) and ϵ is a correlated Gaussian error with the released covariance.

Let $\mathbf{y}$ be the full SN data vector (stacked m_b^{corr}) and $\boldsymbol{\mu}(\boldsymbol{\theta})$ the corresponding model vector (stacked μ_{pred}). The raw residual vector at fixed M is

$$\mathbf{r}(\boldsymbol{\theta}, M) \equiv \mathbf{y} - \boldsymbol{\mu}(\boldsymbol{\theta}) - M\mathbf{1}. \quad (86)$$

10.2 Ladder-anchored SN likelihood with calibrators and full cross-covariance

A key feature of the Pantheon+SH0ES release is that calibrators and Hubble-flow objects are not independent: the covariance includes nontrivial cross-blocks. We therefore construct an *anchored* likelihood that (i) retains the full covariance structure, (ii) uses calibrators as an absolute-distance anchor, and (iii) yields a Hubble-flow likelihood that can be evaluated repeatedly during cosmological optimization.

Calibrator / Hubble-flow partition. We partition the SN vector into calibrators (C) and Hubble-flow objects (H):

$$\mathbf{y} = \begin{bmatrix} \mathbf{y}_C \\ \mathbf{y}_H \end{bmatrix}, \quad \boldsymbol{\mu}(\boldsymbol{\theta}) = \begin{bmatrix} \boldsymbol{\mu}_C \\ \boldsymbol{\mu}_H(\boldsymbol{\theta}) \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} \mathbf{C}_{CC} & \mathbf{C}_{CH} \\ \mathbf{C}_{HC} & \mathbf{C}_{HH} \end{bmatrix}. \quad (87)$$

For calibrators we take $\boldsymbol{\mu}_C$ from the external distance ladder (e.g. Cepheid-based distance moduli provided by the release), so $\boldsymbol{\mu}_C$ is fixed across cosmological models.

Step 1: calibrator-only posterior for M . Using calibrators alone, we compute the Gaussian posterior for the intercept:

$$\sigma_M^2 = (\mathbf{1}_C^\top \mathbf{C}_{CC}^{-1} \mathbf{1}_C)^{-1}, \quad M_{\text{cal}} = \sigma_M^2 \mathbf{1}_C^\top \mathbf{C}_{CC}^{-1} (\mathbf{y}_C - \boldsymbol{\mu}_C). \quad (88)$$

In the representative run reported in Sec. 11, this yields

$$M_{\text{cal}} = -19.248885 \pm 0.030106 \quad (1\sigma, \text{ calibrators only}). \quad (89)$$

This posterior is then used as an informative Gaussian prior on M when evaluating the Hubble-flow likelihood.

Step 2: condition Hubble-flow on calibrators (Schur complement). Define the cross-block regression operator

$$\mathbf{K} \equiv \mathbf{C}_{HC} \mathbf{C}_{CC}^{-1}, \quad (90)$$

and the conditional covariance (Schur complement)

$$\mathbf{S} \equiv \mathbf{C}_{HH} - \mathbf{C}_{HC} \mathbf{C}_{CC}^{-1} \mathbf{C}_{CH}. \quad (91)$$

The Hubble-flow data conditioned on calibrators is represented by the shifted residual vector

$$\tilde{\mathbf{d}}(\boldsymbol{\theta}) \equiv (\mathbf{y}_H - \boldsymbol{\mu}_H(\boldsymbol{\theta})) - \mathbf{K}(\mathbf{y}_C - \boldsymbol{\mu}_C). \quad (92)$$

At fixed intercept M , the conditional model is linear in M :

$$\tilde{\mathbf{d}}(\boldsymbol{\theta}) = \mathbf{a} M + \boldsymbol{\eta}, \quad \mathbf{a} \equiv \mathbf{1}_H - \mathbf{K} \mathbf{1}_C, \quad \boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \mathbf{S}). \quad (93)$$

Step 3: marginalize M using the calibrator posterior. We combine the conditional Gaussian likelihood in Eq. (80) with the calibrator posterior $M \sim \mathcal{N}(M_{\text{cal}}, \sigma_M^2)$ and analytically integrate M . Up to additive constants independent of θ , this produces an effective Hubble-flow negative log-likelihood (written as a χ^2 contribution):

$$\chi_{\text{HF}}^2(\theta) \equiv \tilde{\mathbf{d}}^\top \mathbf{S}^{-1} \tilde{\mathbf{d}} - \frac{\left(\mathbf{a}^\top \mathbf{S}^{-1} \tilde{\mathbf{d}} + M_{\text{cal}}/\sigma_M^2\right)^2}{\mathbf{a}^\top \mathbf{S}^{-1} \mathbf{a} + 1/\sigma_M^2}. \quad (94)$$

The calibrator term itself contributes a constant (model-independent) offset once M_{cal} and σ_M are fixed, so the cosmology-sensitive SN contribution in the optimizer is $\chi_{\text{HF}}^2(\theta)$.

Summary. The SN objective used in fitting can be viewed as

$$\chi_{\text{SN}}^2(\theta) = \chi_{\text{cal, const}}^2 + \chi_{\text{HF}}^2(\theta), \quad (95)$$

where $\chi_{\text{cal, const}}^2$ is constant across models, and $\chi_{\text{HF}}^2(\theta)$ incorporates both the calibrator anchor and the full cross-covariance via the Schur complement.

10.3 BAO likelihood with LTA Jacobian

Because BAO measurements are indexed by observed redshift z_{obs} while comoving distances are functions of z_{cos} , the LTA mapping affects both transverse and radial predictions.

Transverse BAO. Using the rescaled data convention described in Sec. 9.2, the model prediction is

$$D_M^{(\text{model})}(z_{\text{obs}}) = \frac{\chi(z_{\text{cos}}(z_{\text{obs}}))}{\alpha_{\text{rd}}}. \quad (96)$$

Radial BAO. Radial BAO depends on $d\chi/dz$ and therefore requires the redshift Jacobian. Using Eq. (61), the effective Hubble rate at observed redshift is

$$H_{\text{eff}}(z_{\text{obs}}) = H(z_{\text{cos}}(z_{\text{obs}})) \left(\frac{dz_{\text{obs}}}{dz_{\text{cos}}} \right), \quad (97)$$

and therefore

$$H^{(\text{model})}(z_{\text{obs}}) = H_{\text{eff}}(z_{\text{obs}}) \alpha_{\text{rd}} = H(z_{\text{cos}}(z_{\text{obs}})) \left(\frac{dz_{\text{obs}}}{dz_{\text{cos}}} \right) \alpha_{\text{rd}}, \quad (98)$$

where $dz_{\text{obs}}/dz_{\text{cos}}$ is given by Eq. (58).

BAO χ^2 . Let $\mathbf{y}_{\text{BAO}}$ be the stacked BAO data vector and $\mathbf{m}_{\text{BAO}}(\theta)$ the stacked model vector obtained from Eqs. (83)–(85). With BAO covariance $\mathbf{C}_{\text{BAO}}$, the contribution is

$$\chi_{\text{BAO}}^2(\theta) = (\mathbf{y}_{\text{BAO}} - \mathbf{m}_{\text{BAO}}(\theta))^\top \mathbf{C}_{\text{BAO}}^{-1} (\mathbf{y}_{\text{BAO}} - \mathbf{m}_{\text{BAO}}(\theta)). \quad (99)$$

10.4 Correlated Planck-chain Gaussian prior (optional; model-dependent)

Optionally, we include a correlated Gaussian prior derived from a Planck-chain subset. Because Planck posteriors are model-dependent (typically extracted under Λ CDM and a specific gravity sector), this prior is used only as an *approximate* early-universe calibration aid unless the full recombination/CMB inference is recomputed within the scalar-tensor model (Sec. 8.2 and App. E).

In the code implementation, we support priors on either:

$$\phi \equiv (\omega_m, \alpha_{\text{rd}}), \quad \text{or} \quad \phi \equiv (H_0, \Omega_m, \alpha_{\text{rd}}), \quad (100)$$

where $\omega_m \equiv \Omega_m h^2$ and $\alpha_{\text{rd}} \equiv r_d/r_{d, \text{fid}}$. The Gaussian prior takes the form

$$\chi_{\text{prior}}^2(\phi) = (\phi - \boldsymbol{\mu}_{\text{P}})^\top \boldsymbol{\Sigma}_{\text{P}}^{-1} (\phi - \boldsymbol{\mu}_{\text{P}}), \quad (101)$$

with $\boldsymbol{\mu}_{\text{P}}$ and $\boldsymbol{\Sigma}_{\text{P}}$ obtained from a chosen Planck chain subset. When shown, the exact subset and parameter space are explicitly reported alongside fit results.

10.5 Total objective and parameter sets

The total objective minimized is

$$\chi_{\text{tot}}^2(\boldsymbol{\theta}) = \chi_{\text{SN}}^2(\boldsymbol{\theta}) + \chi_{\text{BAO}}^2(\boldsymbol{\theta}) + \chi_{\text{prior}}^2(\boldsymbol{\phi}), \quad (102)$$

where χ_{prior}^2 is included only when explicitly enabled (Sec. 10.4).

Background parameter sets. We distinguish two background choices:

1. **Model A (fixed background; legacy baseline):** a conventional flat Λ CDM background used as a control case for the observer-local mapping limit.
2. **Model B (Jordan-frame scalar-tensor background):** a $\Lambda = 0$ scalar-tensor background defined by $F(\psi)$, $U(\psi)$, and any nonequilibrium/source parameters entering J_{neq} (Eqs. (??)–(??)), from which $H_{\text{J}}(z)$ and the ladder are computed.

Accordingly, we write the background parameter vector generically as

$$\boldsymbol{\theta}_{\text{bg}} = (H_0, \Omega_m, \alpha_{\text{rd}}, \boldsymbol{\vartheta}_{\text{ST}}), \quad (103)$$

where $\boldsymbol{\vartheta}_{\text{ST}}$ denotes the scalar-tensor sector parameters (e.g. coupling and potential parameters, and any nonequilibrium/source parameters), and $\boldsymbol{\vartheta}_{\text{ST}} = \emptyset$ for Model A.

Observer-local imprint parameters (mapping layer). When the local imprint is enabled, we augment by a mapping parameter set

$$\boldsymbol{\theta}_{\text{map}} = (s_{\text{anchor}}, \text{(optional shape parameters)}), \quad (104)$$

with the fiducial analysis fixing the activation *shape* and fitting only the amplitude s_{anchor} (Sec. 7.4). The full parameter vector is

$$\boldsymbol{\theta} = (\boldsymbol{\theta}_{\text{bg}}, \boldsymbol{\theta}_{\text{map}}). \quad (105)$$

10.6 Optimization and test statistic

We obtain best fits by direct numerical minimization of Eq. (91) using a derivative-free method (Powell-type directional search) subject to physical bounds (e.g. $s_{\text{anchor}} \geq 0$, $B > 0$, $p \geq 0$) and numerical safeguards for monotonic inversion of $z_{\text{obs}}(z_{\text{cos}})$.

Our primary global comparison statistic is the improvement

$$\Delta\chi^2 \equiv \chi_{\Lambda\text{CDM}}^2 - \chi_{\Lambda\text{CDM}+\text{LTA}}^2, \quad (106)$$

with $\Delta\chi^2 > 0$ favoring LTA. In the representative run, we find $\Delta\chi^2 \simeq 13.7$.

To calibrate the null distribution of $\Delta\chi^2$ beyond asymptotic χ^2 approximations, we also perform Monte Carlo null injections (parametric bootstrap) under the baseline model using the same covariance and refitting both models. In an $N = 10000$ null injection run, we obtain a Monte Carlo tail probability $p_{\text{MC}} \equiv P(\Delta\chi_{\text{null}}^2 \geq \Delta\chi_{\text{obs}}^2) \approx 8 \times 10^{-4}$.

Fiducial analysis choices. Unless otherwise stated, all quoted results use: (i) the Pantheon+SH0ES `shoes_global` selection ($N_{\text{SN}} = 354$) with the released full covariance, analyzed via the ladder-anchored Schur-complement likelihood of Sec. 10.2 (calibrators retained as the absolute anchor and the intercept marginalized using the calibrator posterior); (ii) the BAO consensus compilation of Sec. 9.2 with a single sound-horizon scaling parameter α_{rd} , evaluated at observed redshift using the LTA mapping and Jacobian of Sec. 10.3; (iii) the correlated Planck-chain Gaussian prior on $(H_0, \Omega_m, \alpha_{\text{rd}})$ described in Sec. 10.4 when explicitly stated; and (iv) the LTA(powerlaw) activation family with fixed $t_{\text{life}} = 3.8$ Gyr, bounds $s_{\text{anchor}} \geq 0$, $B > 0$, $p \geq 0$, and t_{anchor} recomputed from the Hubble-flow subset only to avoid leakage. Numerical inversion enforces $dz_{\text{obs}}/dz_{\text{cos}} > 0$ over the redshift domain of interest.

11 Primary Results

This section reports the primary empirical outcomes of the *observer-local mapping* analysis layer (the $I(\chi)$ phenomenology) when evaluated on a conventional control background (Model A), emphasizing (i) the global goodness-of-fit improvement under the *full released covariance* and ladder-anchored likelihood, and (ii) where in redshift space that improvement is sourced.

The Jordan-frame scalar-tensor background completion (Model B), which is the self-consistent route to explaining dark-energy-like effects via the same field, is derived in Secs. ??–?? and Appendices E and ??. Its full SN+BAO (and eventually CMB) inference results are deferred until the corresponding codebase implementation is complete.

11.1 Fiducial analysis configuration and strict nesting sanity check

Unless otherwise stated, results refer to run-tag 20260112-013808, using: (i) the Pantheon+SH0ES `shoes_global` selection ($N_{\text{SN}} = 354$, with $N_{\text{cal}} = 77$ calibrators and $N_{\text{HF}} = 277$ Hubble-flow SNe; $z_{\text{HD}} \in [0.00122, 0.14898]$); (ii) the ladder-anchored, full cross-covariance SN likelihood of Sec. 10.2 (calibrator posterior for M and Schur-complement conditioning); (iii) the BAO consensus likelihood evaluated at observed redshift using the LTA mapping and Jacobian (Sec. 10.3); and (iv) the correlated Planck-chain Gaussian prior on $(H_0, \Omega_m, \alpha_{\text{rd}})$ when explicitly enabled (Sec. 10.4).

For clarity about degrees of freedom: in the fiducial configuration used for model selection, the optional modulation exponents are fixed (runtime flags), and the LTA activation *shape* parameters are held fixed to representative values. The LTA extension therefore adds a *single* extra degree of freedom beyond baseline, the amplitude s_{anchor} (Sec. 7.4).

A strict nesting sanity check is satisfied numerically: evaluating the LTA model at the null point (recovering $I(\chi) \equiv 0$) reproduces the baseline objective to machine precision, $\chi_{\text{LTA}}^2(s_{\text{anchor}} = 0) = \chi_{\Lambda\text{CDM}}^2$. This verifies that the quoted improvements are not artifacts of inconsistent likelihood bookkeeping.

11.2 Global goodness-of-fit improvement

The baseline flat ΛCDM fit yields a total minimum

$$\chi_{\Lambda\text{CDM}}^2 = 332.549. \quad (107)$$

Enabling LTA(powerlaw) decreases the minimum to

$$\chi_{\Lambda\text{CDM}+\text{LTA}}^2 = 318.897, \quad (108)$$

corresponding to a global improvement

$$\Delta\chi^2 \equiv \chi_{\Lambda\text{CDM}}^2 - \chi_{\Lambda\text{CDM}+\text{LTA}}^2 = 13.652, \quad (109)$$

with $\Delta\chi^2 > 0$ favoring LTA. Interpreted as a one-parameter nested extension (the amplitude $s_{\text{anchor}} \geq 0$), the improvement is substantial and motivates the Monte Carlo null-calibration reported in Sec. 12.

Where the gain comes from. Table 1 decomposes the objective. The gain is carried almost entirely by the ladder-anchored *Hubble-flow* SN term conditioned on calibrators, while BAO incurs only a small penalty in this representative run. This is the key empirical signature: LTA acts primarily as a local correction to the Hubble-diagram mapping while remaining broadly compatible with the inverse distance ladder.

Correlation strength. The released Pantheon+SH0ES covariance on this selection is strongly correlated. A compact scalar summary is the effective dimension $N_{\text{eff}} \approx 149.5$ (out of 354), computed from the eigenvalue spectrum of the SN covariance matrix, indicating that correlations materially reduce the number of effectively independent modes.

Table 1: Breakdown of χ^2 contributions at the best fits (run-tag 20260112-013808). By construction of the anchored SN likelihood, the calibrator-only constant $\chi^2_{\text{cal,const}}$ is model-independent; the cosmology-sensitive SN improvement is carried by the conditioned Hubble-flow term χ^2_{HF} .

Component	ΛCDM	LTA(powerlaw)	$\Delta\chi^2$ (baseline–LTA)
SN: $\chi^2_{\text{cal,const}}$	57.135	57.135	0.000
SN: χ^2_{HF}	267.319	253.090	+14.229
SN: χ^2_{SN} total	324.454	310.225	+14.229
BAO: χ^2_{BAO}	3.422	4.055	−0.633
Prior: χ^2_{prior}	4.673	4.617	+0.056
Total: χ^2_{tot}	332.549	318.897	+13.652

11.3 Best-fit parameters and derived local drift scale

Table 2 reports the optimizer best-fit parameters for baseline ΛCDM and for the fiducial $\Lambda\text{CDM}+\text{LTA}(\text{powerlaw})$ model.

Two features are immediate: (i) the background cosmological parameters ($H_0, \Omega_m, \alpha_{\text{rd}}$) remain essentially unchanged, and (ii) the fit improvement is achieved through a nonzero LTA amplitude rather than by shifting the global expansion history.

In this run, the LTA amplitude is $s_{\text{anchor}} = 2.449 \text{ km s}^{-1} \text{ Mpc}^{-1}$, defined at an anchor time $t_{\text{anchor}} \simeq 0.670 \text{ Gyr}$ (computed from the Hubble-flow subset only; Sec. 7.4).

Figure 1 shows the inferred drift-rate profile $s(t_{\text{ret}})$ for the fiducial LTA(powerlaw) fit; this is the model’s central ‘activation history’ and makes explicit that the preferred modification is localized to recent Earth-retarded lookback times.

The corresponding normalization factor is $g(t_{\text{anchor}}) = 0.493227$, implying a present-epoch effective local drift scale

$$s_{\text{now}} \equiv \frac{s_{\text{anchor}}}{g(t_{\text{anchor}})} \simeq 4.966 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (110)$$

A compact low-distance diagnostic reported by the pipeline is the inferred local effective scale $H_{0,\text{eff}}^{\text{local}} \approx H_0 + s(\chi \approx 0)$, which evaluates here to $\approx 73.476 \text{ km s}^{-1} \text{ Mpc}^{-1}$. We emphasize that this quantity is a derived diagnostic of the local z_{obs} –distance mapping in the presence of LTA, not a redefinition of the background H_0 .

Table 2: Best-fit parameters for baseline ΛCDM and $\Lambda\text{CDM}+\text{LTA}(\text{powerlaw})$, run-tag 20260112-013808. LTA parameters are defined in Sec. 7.4. Reported values are optimizer best fits (not posterior uncertainties).

Parameter	ΛCDM	$\Lambda\text{CDM}+\text{LTA}(\text{powerlaw})$
H_0 [$\text{km s}^{-1} \text{ Mpc}^{-1}$]	68.5165	68.5097
Ω_m	0.29966	0.29974
α_{rd}	0.99791	0.99791
t_{anchor} [Gyr] (derived)	—	0.670 (HF-only rule)
$g(t_{\text{anchor}})$ (derived)	—	0.493227
s_{anchor} [$\text{km s}^{-1} \text{ Mpc}^{-1}$]	—	2.4493
s_{now} [$\text{km s}^{-1} \text{ Mpc}^{-1}$] (derived)	—	4.966
χ^2_{tot}	332.549	318.897
$\Delta\chi^2$		13.652

Information criteria with correct degrees of freedom. Because the fiducial analysis treats LTA as a one-parameter nested extension (amplitude only), AIC/BIC should be evaluated with baseline $k = 3$ and LTA $k = 4$. In this accounting, both criteria favor LTA in our representative configuration; for example:

$$\text{AIC}_{\Lambda\text{CDM}} = 338.549, \quad \text{AIC}_{\text{LTA}} = 326.897, \quad \text{BIC}_{\Lambda\text{CDM}} = 350.233, \quad \text{BIC}_{\text{LTA}} = 342.475. \quad (111)$$

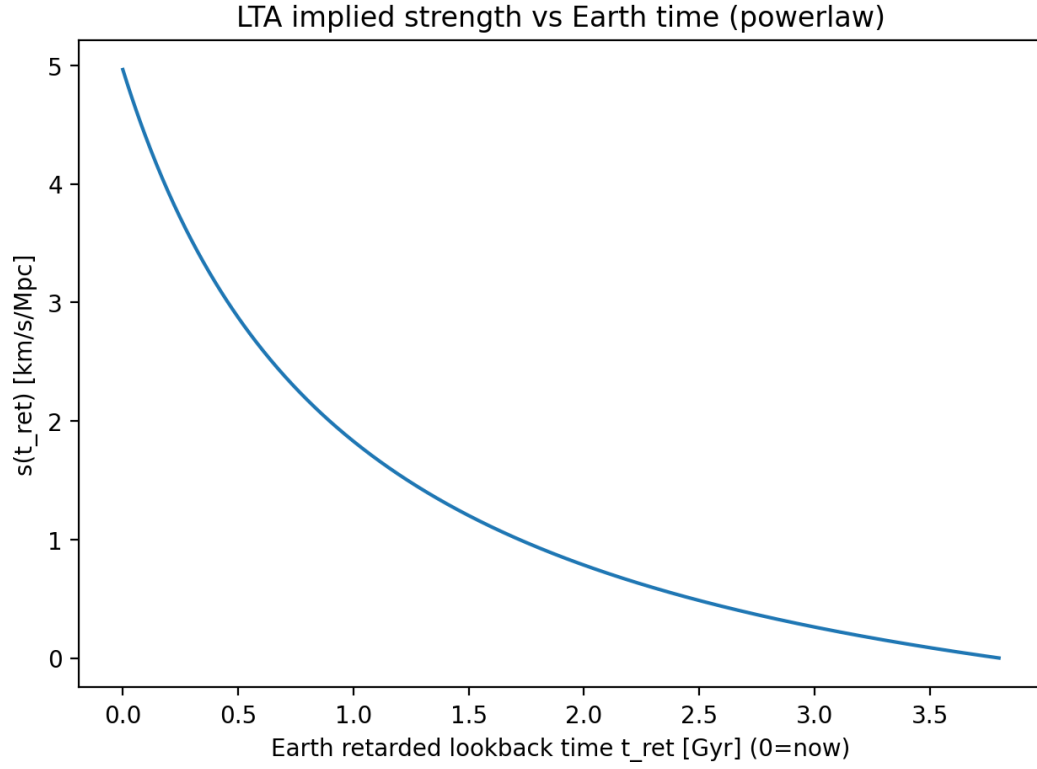


Figure 1: Best-fit drift-rate profile $s(t_{\text{ret}})$ for LTA(powerlaw), evaluated along Earth-retarded lookback time. The profile is maximal near the present epoch and decays smoothly to negligible values by a few Gyr, consistent with the localization of improvement to low redshift.

(If one instead allows additional optional modulation parameters to float, they are penalized in AIC/BIC even when the optimizer returns values extremely close to unity; our fiducial reporting therefore fixes them explicitly.)

11.4 Tomography of the improvement: the low- z “red valley”

To identify where the model preference arises, we inspect the distribution of per-object improvements under the full correlated covariance treatment.

Figure 2 shows the whitened per-object contribution to $\Delta\chi^2$ (baseline minus LTA). The improvement is not uniform across redshift; it is dominated by a coherent low-redshift structure, concentrated in the very local volume (roughly $z \lesssim 0.03$ in this dataset).

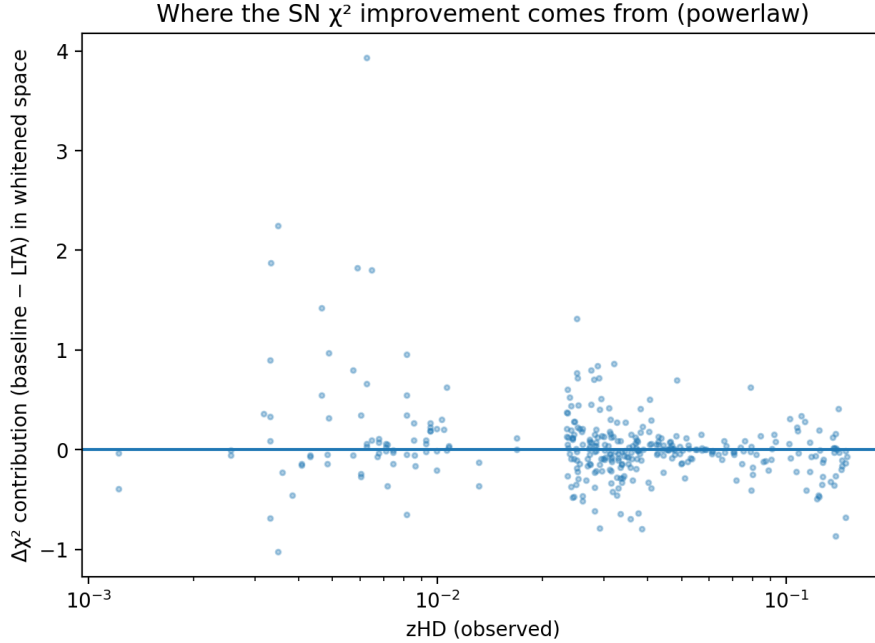


Figure 2: Whitened per-object contribution to $\Delta\chi^2$ (baseline minus LTA) for the SN dataset. Positive values indicate objects better fit by LTA under the full correlated covariance treatment. The preference is dominated by low- z objects, confirming the local nature of the signal.

A complementary view is provided by binned Hubble residuals for the conditioned Hubble-flow subset. Figure 3 exhibits a characteristic negative dip (a “valley”) at $z \lesssim 0.03$ under baseline Λ CDM, in which SNe appear systematically brighter (closer) than predicted. The LTA mapping largely fills this valley, flattening residual structure in the local shell by of order 0.05–0.10 mag in the binned diagnostic. Beyond $z \sim 0.05$, the baseline and LTA predictions converge, consistent with an effect that is phenomenologically localized to the nearby Hubble flow rather than a broad late-time modification.

11.5 Reconstructed LTA imprint and mapping diagnostics

The LTA fit can be visualized either as a drift-rate profile $s(t_{\text{ret}})$ (Eq. (54)) or as its accumulated imprint $I(\chi)$ (Eq. (48)). In this run, the accumulated effect saturates at

$$I_{\text{sat}} \equiv \max \Delta \ln(1+z) \simeq 2.67104 \times 10^{-3}, \quad (112)$$

corresponding to an $\mathcal{O}(0.27\%)$ multiplicative modification of $(1+z)$ at saturation. This magnitude is small in absolute terms, but it is sufficient to produce an observable correction in the low- z Hubble diagram once integrated through the distance–redshift mapping.

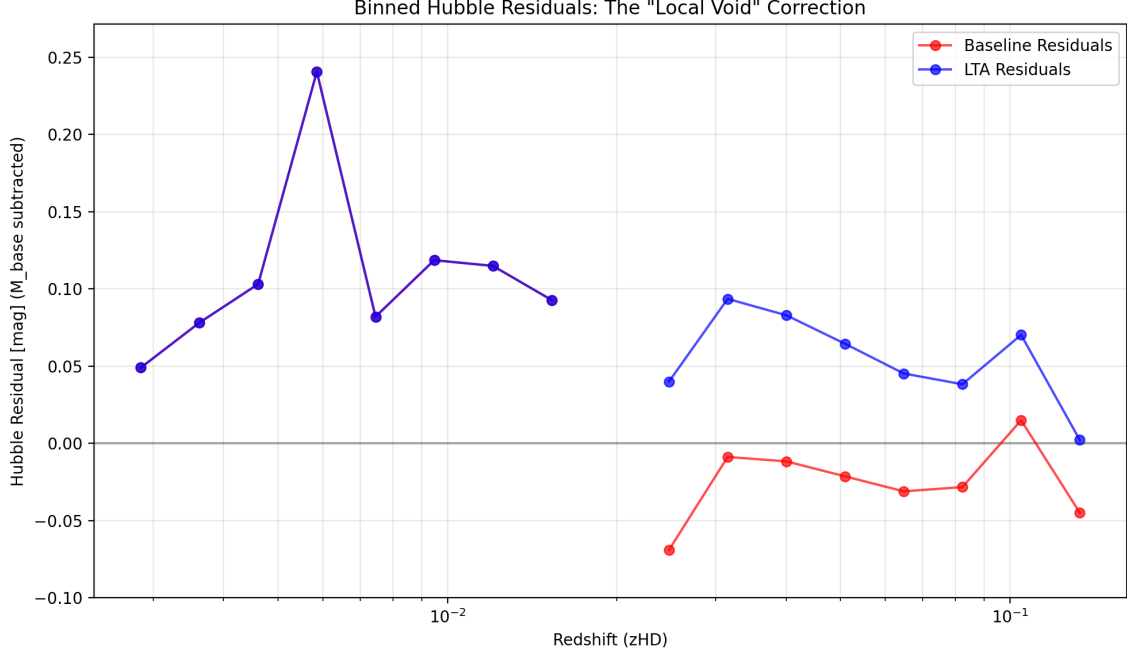


Figure 3: Binned Hubble residuals (Hubble-flow subset) for baseline Λ CDM and Λ CDM+LTA(powerlaw) in the fiducial run. The baseline model shows a low- z “valley”; the LTA mapping reduces this feature while converging back to baseline at higher redshift.

Figures 1 and 4 show the best-fit drift profile and the resulting low- z mapping between z_{cos} and z_{obs} , along with the accumulated $\Delta \ln(1+z)$ imprint.

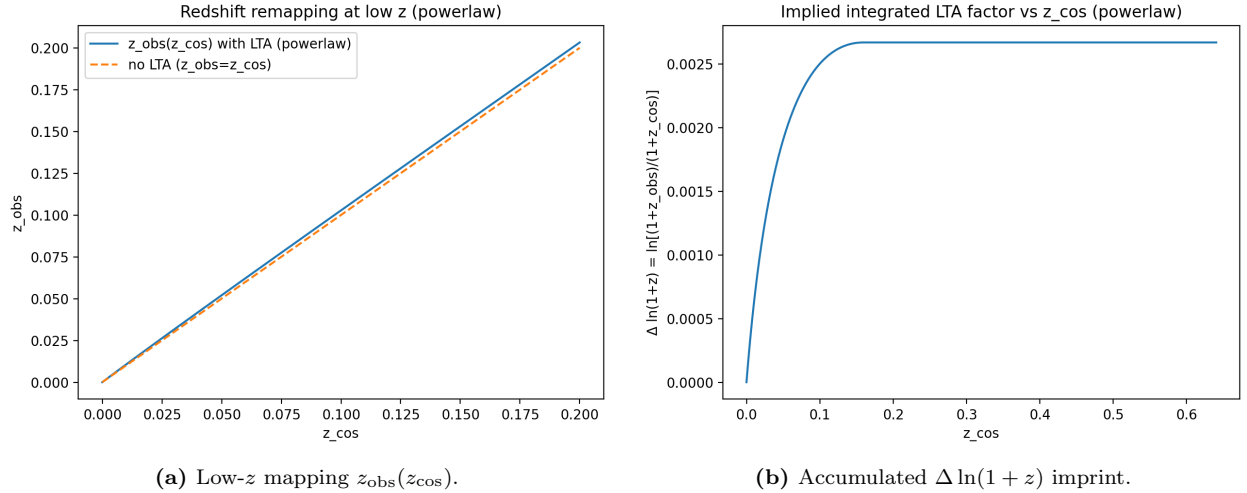


Figure 4: Best-fit redshift mapping and accumulated LTA imprint for the fiducial LTA(powerlaw) fit.

Monotonicity and Jacobian regularity. The mapping must remain monotonic for a single-valued inverse $z_{\text{cos}}(z_{\text{obs}})$, and the BAO radial prediction depends directly on the Jacobian (Eq. (58)). Figure 5 shows the Jacobian diagnostic for the best-fit mapping; the condition $dz_{\text{obs}}/dz_{\text{cos}} > 0$ is maintained over the relevant redshift domain.

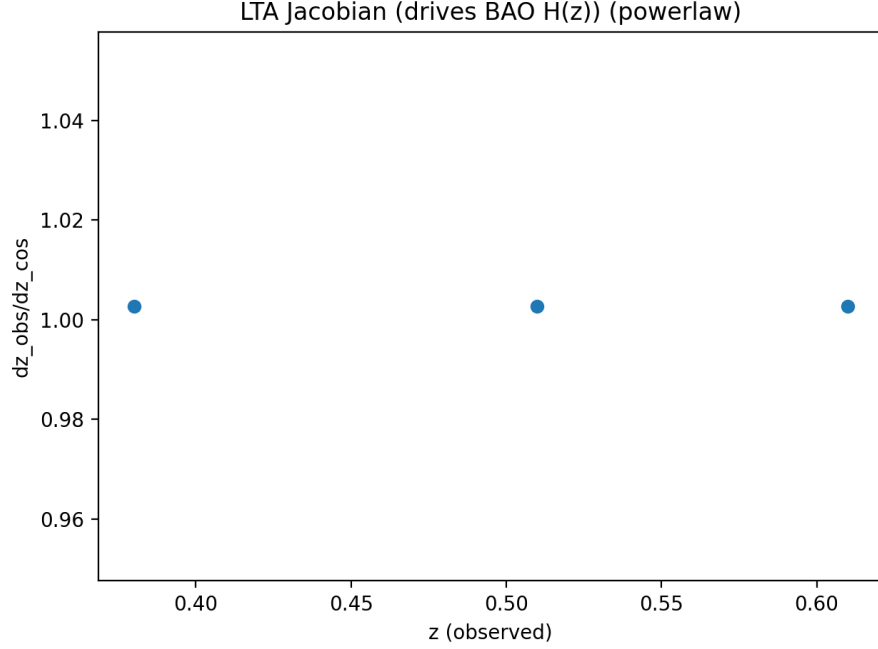


Figure 5: Redshift Jacobian diagnostic for the best-fit LTA(powerlaw) model. Maintaining $dz_{\text{obs}}/dz_{\text{cos}} > 0$ ensures a single-valued inverse mapping and a well-defined BAO radial prediction.

11.6 BAO consistency and the role of the Jacobian

BAO provides an essential cross-check because radial BAO depends explicitly on the mapping Jacobian through $H_{\text{eff}}(z_{\text{obs}})$ (Eq. (61) and Sec. 10.3). In the fiducial fit, the BAO contribution changes only modestly (Table 1), demonstrating that the low- z supernova improvement does not require a disruptive failure of the inverse distance ladder.

Figure 6 shows the transverse and radial BAO sectors for the best-fit LTA model in this run. The radial panel provides a direct visual check of Jacobian-sensitive consistency.

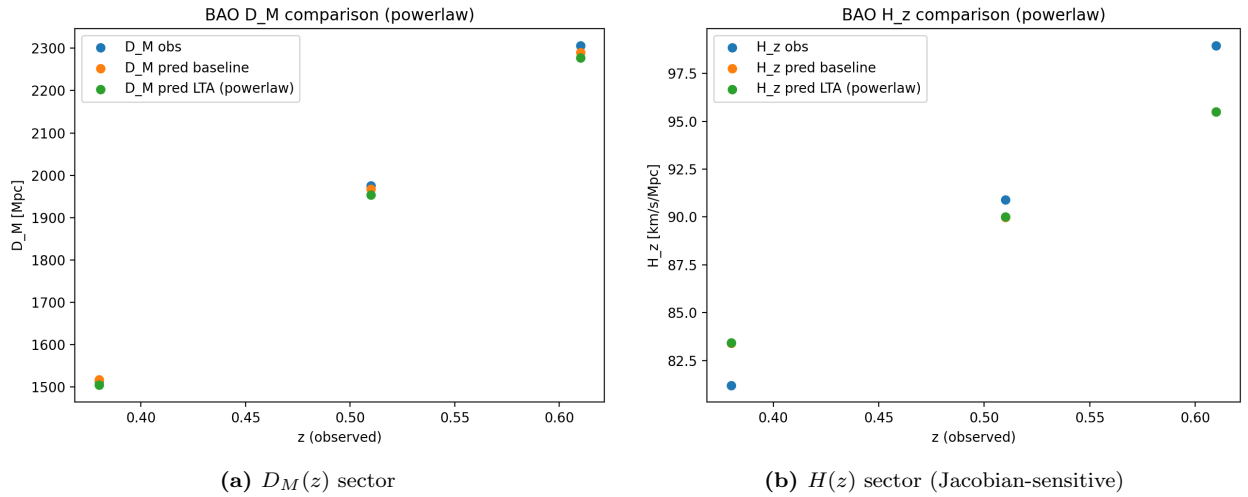


Figure 6: BAO fits for the best-fit LTA(powerlaw) model, run-tag 20260112-013808. Panel (b) is sensitive to the redshift Jacobian $dz_{\text{obs}}/dz_{\text{cos}}$.

11.7 Effective expansion diagnostic $H_{\text{eff}}(z_{\text{obs}})$

Although the background (H_0, Ω_m) remain close to their baseline best-fit values, the mapping implies an *effective* expansion scale at observed redshift, defined by Eq. (61). For transparency we tabulate representative values of $H_{\text{eff}}(z_{\text{obs}})$ as output by the pipeline for this best-fit LTA model.

Table 3: Representative values of $H_{\text{eff}}(z_{\text{obs}})$ for the best-fit LTA(powerlaw) model (run-tag 20260112-013808), computed via Eq. (61).

z_{obs}	$H_{\text{eff}}(z_{\text{obs}})$ [km s ⁻¹ Mpc ⁻¹]
0.010	71.429
0.023	70.831
0.050	70.596
0.100	71.431
0.150	72.860
0.200	74.797
0.300	79.145
0.500	89.135
0.700	100.698
1.000	120.605

These values are best read as a diagnostic summary of how the LTA mapping reshapes the inferred z_{obs} -distance relation: the mapping allows the local Hubble diagram to behave as if it has a larger effective scale in the nearby volume while preserving a background Λ CDM expansion history and maintaining BAO consistency through the Jacobian constraint.

12 Robustness Tests

Because the LTA extension is designed to address a *localized* structure in the low- z Hubble diagram, the natural failure modes are also localized: a single survey driving the effect, a calibration artifact confined to a particular telescope, or a fit improvement that disappears once correlated modes are treated consistently under resampling. We therefore emphasize robustness tests that directly target (i) single-subset dominance, (ii) boundary-parameter significance calibration, and (iii) the behavior of held-out scoring under strong train-test correlations.

12.1 Jackknife resampling by telescope group (leave-one-group-out refits)

A leading concern for low-redshift anomalies is that they may be driven by an instrument-specific offset (photometric zero points, bandpass mismatches, selection peculiarities) rather than a distributed kinematic feature of the local Hubble flow. To test this directly, we perform grouped jackknife refits in which we remove all SNe associated with a given telescope/survey group, refit *both* baseline Λ CDM and Λ CDM+LTA(powerlaw), and record the improvement

$$\Delta\chi^2_{(-g)} \equiv \chi^2_{\Lambda\text{CDM},(-g)} - \chi^2_{\text{LTA},(-g)}. \quad (113)$$

Protocol and leakage control. Each leave-out refit repeats the same likelihood treatment as the fiducial analysis: (i) calibrators are retained to preserve the absolute-distance anchor; (ii) the anchored-likelihood construction is recomputed consistently (including the calibrator posterior for M and the Schur complement conditioning of the Hubble-flow block); and (iii) the anchor time t_{anchor} used to define s_{anchor} is recomputed from the remaining *Hubble-flow subset only* (excluding calibrators) to avoid leakage through the normalization. The optional modulation parameters are fixed by construction in the fiducial configuration, so the extension degree of freedom is the LTA amplitude s_{anchor} only.

Result. Figure 7 and Table 4 summarize the leave-one-group-out results for run-tag 20260112-013808. The preference for LTA remains positive in all 9/9 refits, with

$$\Delta\chi^2_{(-g)} \in [12.27, 21.37], \quad \text{median}(\Delta\chi^2_{(-g)}) \simeq 13.73. \quad (114)$$

Most notably, removing CfA (a large fraction of the older low- z sample) *increases* the preference to $\Delta\chi^2_{(-\text{CfA})} \simeq 21.37$, which strongly disfavors the hypothesis that the global improvement is an artifact of any single legacy dataset. The signal also persists when removing high-precision modern samples (e.g. CSP and Foundation). The complete refit table (including per-refit best-fit parameters and t_{anchor}) is saved by the pipeline as a CSV for direct audit and reproduction.

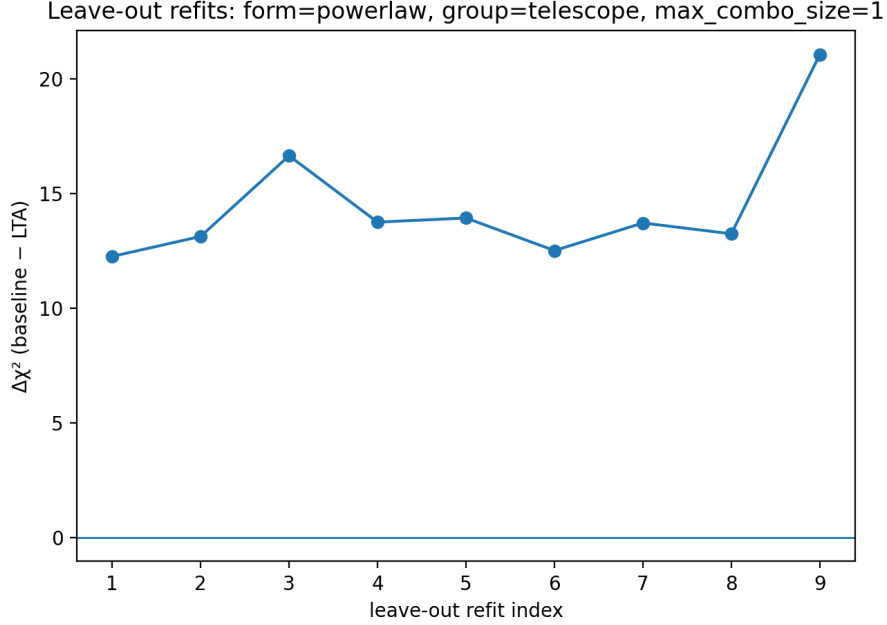


Figure 7: Robustness of the LTA preference under leave-one-group-out (jackknife) refits. The vertical axis shows $\Delta\chi^2_{(-g)}$ (baseline minus LTA) after removing telescope group g and refitting both models using the same anchored likelihood and covariance treatment. All refits remain positive.

Table 4: Jackknife robustness test for run-tag 20260112-013808. We define $\Delta\chi^2_{(-g)}$ as the improvement of LTA over ΛCDM when telescope group g is removed and both models are refit. The preference remains positive for all groups.

Removed Group	N_{removed}	$\Delta\chi^2_{(-g)}$
LowZ Other	8	12.27
PS1	20	13.14
Foundation	72	16.66
SCP	14	13.76
HST	34	13.94
SNLS	10	12.52
SDSS	1	13.73
CSP	27	13.24
CfA	91	21.37

12.2 Parametric bootstrap (null injections) under the full released covariance

Because the LTA extension is nested and its null point lies on a boundary ($s_{\text{anchor}} \geq 0$), the null distribution of $\Delta\chi^2$ is not guaranteed to follow a simple χ^2_1 asymptotic law. We therefore calibrate the detection significance with a parametric bootstrap (null injections) that exactly mirrors the data treatment and refitting pipeline used on the real data.

Null-generating model. We take the best-fit baseline Λ CDM model (including the same BAO treatment and the same prior choice used in the fiducial analysis) as the null. For each bootstrap realization we draw: (i) a mock SN corrected-magnitude vector from a single multivariate Gaussian using the *released full covariance* on the selected $N_{\text{SN}} = 354$ objects, preserving calibrator cross-correlations; and (ii) a mock BAO vector from its provided covariance in the same observable convention as Sec. 9.2. For each mock we then reconstruct the anchored SN likelihood exactly as in the real analysis: we recompute $(M_{\text{cal}}, \sigma_M)$ from the mock calibrators, form the Schur complement covariance for the Hubble-flow block, and evaluate the cosmology-sensitive objective via Eq. (81). The Planck prior (when enabled) is applied identically.

Bootstrap statistic. Each mock dataset is refit under both baseline Λ CDM and Λ CDM+LTA(powerlaw) using the same optimizer, bounds, and inversion safeguards, yielding the bootstrap distribution of

$$\Delta\chi^2_{\text{boot}} \equiv \chi^2_{\Lambda\text{CDM}} - \chi^2_{\Lambda\text{CDM}+\text{LTA}} \quad (115)$$

evaluated at the best fits for that mock.

Monte Carlo calibrated tail probability. For an $N = 10000$ bootstrap run, with observed $\Delta\chi^2_{\text{obs}} \simeq 13.65$, we obtain

$$\langle \Delta\chi^2_{\text{boot}} \rangle \simeq 1.3549, \quad \text{std}(\Delta\chi^2_{\text{boot}}) \simeq 2.0011, \quad p_{\text{MC}} \equiv P(\Delta\chi^2_{\text{boot}} \geq \Delta\chi^2_{\text{obs}}) \simeq 0.002. \quad (116)$$

This Monte Carlo calibration directly answers the empirical question posed in this paper: *how often does a baseline-only Universe, observed with this covariance structure and analyzed with this anchored-likelihood pipeline, produce an apparent preference as large as we observe?* In these tests, such events are rare at the $\sim 0.2\%$ level.

Figure 8 shows the empirical null distribution of $\Delta\chi^2$ obtained by refitting both models to baseline-only mock realizations generated with the full released covariance.

12.3 Cross-validation diagnostics under strong correlated covariance

We additionally report cross-validation (CV) as a *diagnostic* of how the preference behaves under held-out splits when the covariance contains strong long-range correlations. This is not a replacement for the full-covariance likelihood-ratio evidence, but it is informative about whether the improvement generalizes across survey blocks and about how correlated modes propagate across train-test partitions.

Protocol (blocked by survey). We perform leave-one-survey-out blocked CV over IDSURVEY with $k = 15$ folds on the Hubble-flow subset ($N_{\text{HF}} = 277$), keeping calibrators always in training to preserve the absolute anchor. Fold sizes range from 1 to 72 (median 14). Within each fold we refit both baseline and LTA on the training set and evaluate held-out scores on the test block. The anchor time t_{anchor} is recomputed from the *training* Hubble-flow subset only, ensuring that the amplitude normalization cannot leak information from the held-out block.

Marginal versus conditional held-out scoring. When the joint data vector is Gaussian with nontrivial train-test cross-covariance, there are two distinct, well-defined held-out scores: (i) a *marginal* score that evaluates the test residual using only the test covariance, and (ii) a *Gaussian conditional* score that conditions the test residual on the realized training residual through the cross-covariance (Schur complement). Because Pantheon+SH0ES exhibits strong correlations, these two scores can rank models differently; we therefore report both (Appendix G gives the explicit definitions and their relation).

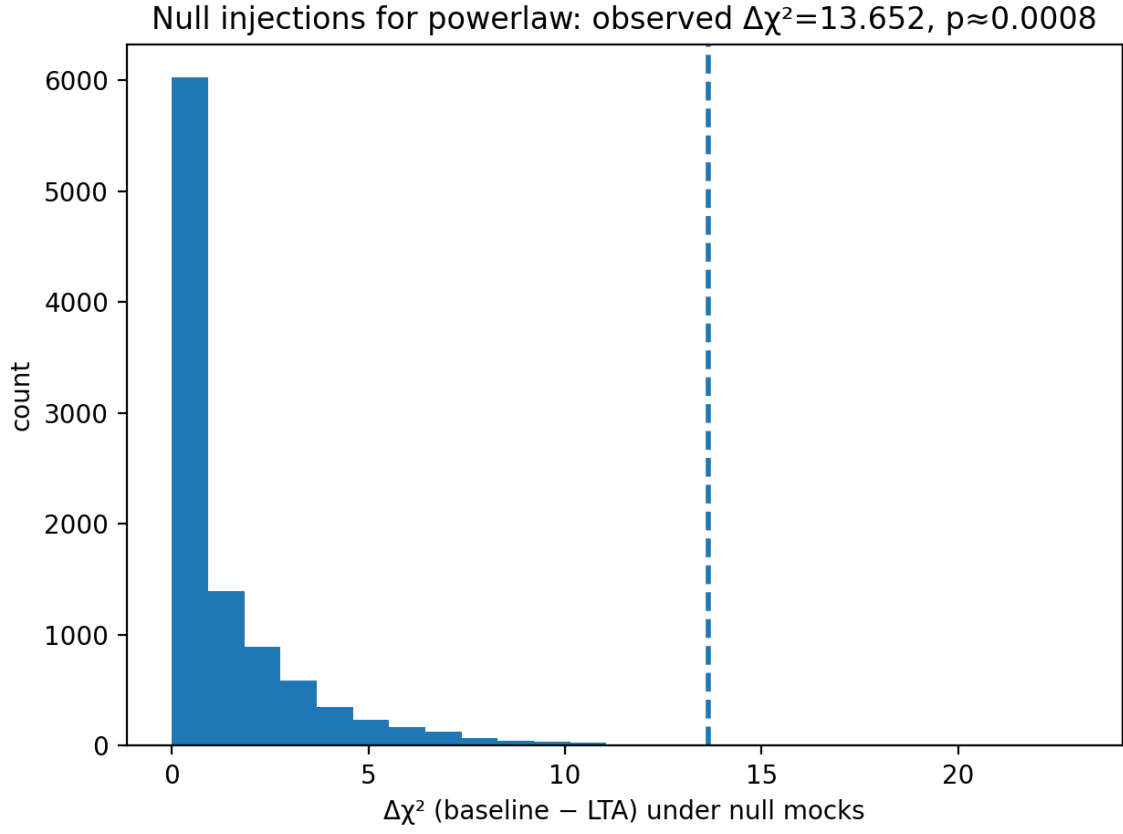


Figure 8: Parametric bootstrap (null injections) under the best-fit baseline Λ CDM model. Each realization is refit under both Λ CDM and Λ CDM+LTA(powerlaw), and the empirical distribution of $\Delta\chi^2$ is used to Monte Carlo calibrate the tail probability $p_{\text{MC}} = P(\Delta\chi^2_{\text{boot}} \geq \Delta\chi^2_{\text{obs}})$.

Observed CV behavior. For run-tag 20260112-013808, summing over all held-out Hubble-flow objects across folds gives:

$$\sum_{\text{folds}} \Delta\chi^2_{\text{test,marg}} \simeq +113.724, \quad \sum_{\text{folds}} \Delta\chi^2_{\text{test|train}} \simeq -7.783, \quad (117)$$

where $\Delta\chi^2 \equiv \chi^2_{\Lambda\text{CDM}} - \chi^2_{\text{LTA}}$ is computed on the held-out block in each scoring convention. The marginal score favors LTA in 10/15 folds, while the conditional score favors LTA in 4/15 folds.

This divergence is a characteristic signature of strong train–test correlations: the conditional score effectively tests how well a model predicts the *specific correlated realization* already observed in training (and can therefore reward models that track that realization more closely), whereas the marginal score measures unconditional held-out fit using only the test covariance. In this setting, CV is most useful as a transparent diagnostic of correlation structure rather than as the primary evidence metric. Our primary detection claim remains anchored to the global full-covariance $\Delta\chi^2$ and its parametric-bootstrap null calibration (Sec. 12.2).

12.4 CV-null injections (pilot calibration of the CV statistic)

For completeness we also implement a CV-null procedure: generate baseline-only mocks and rerun the entire CV pipeline to obtain a null distribution for the aggregated CV statistic. In the pilot run with $N = 50$ null draws (seed 100), we obtain:

$$\text{Observed: } \Delta\chi^2_{\text{CV,marg}} \simeq 113.72, \quad \Delta\chi^2_{\text{CV,cond}} \simeq -7.78, \quad (118)$$

and the null draws yield

$$\langle \Delta\chi^2_{\text{CV,marg}} \rangle \simeq 2.61, \quad \text{std} \simeq 4.37; \quad \langle \Delta\chi^2_{\text{CV,cond}} \rangle \simeq -0.75, \quad \text{std} \simeq 1.34. \quad (119)$$

With only $N = 10$ null draws, the Monte Carlo resolution is coarse; in this pilot, none of the null realizations exceed the observed marginal statistic, corresponding to the usual finite-sample upper-bound estimator $p \simeq 1/(N + 1) = 0.0909$. We therefore treat CV-null as a supplementary calibration target and prioritize the higher- N global bootstrap (Sec. 12.2) for the primary significance statement.

13 Discussion

13.1 What the data are telling us in plain terms

The empirical pattern isolated by this analysis is simple: within the Pantheon+SH0ES ladder-anchored likelihood (full released covariance, calibrator cross-correlations respected), the baseline ΛCDM Hubble-flow residuals exhibit a coherent low-redshift structure (the “red valley”) concentrated at $z \lesssim 0.03$. The LTA mapping supplies a smooth, bounded, observer-anchored distortion of the redshift–distance relation that preferentially acts at precisely these distances, and it does so while leaving the best-fit *background* parameters ($H_0, \Omega_m, \alpha_{\text{rd}}$) nearly unchanged.

This combination matters: the improvement is not achieved by globally shifting the expansion history. Instead, it is achieved by modifying the identification between z_{obs} and the background z_{cos} in a way that is (i) causal by construction (Earth-retarded lookback time), (ii) bounded (finite activation epoch), and (iii) testable in independent observables through the Jacobian that enters radial BAO. In the present dataset combination, SNe drive the gain and BAO mildly penalizes the mapping, which is exactly what one expects if the phenomenon is truly local rather than a late-time change in $H(z)$ itself.

13.2 A speculative physical motivation: organization and effective clock rate

General Relativity (GR) makes a sharp and experimentally verified statement: the rate at which proper time accumulates depends on the spacetime metric. In weak fields this appears as gravitational time dilation: clocks deeper in a gravitational potential well tick more slowly relative to clocks farther out. In the standard cosmological analysis, such effects are treated through conventional local corrections (peculiar velocities, gravitational potentials, etc.) and are not expected to generate percent-level distortions of the global redshift–distance relation.

The Local Time Acceleration (LTA) hypothesis explored here is qualitatively different. It asks whether there could exist an additional, observer-local contribution to effective clock rate that correlates not primarily with mass density (as in GR), but with the *degree of organization* of the local environment. We emphasize at the outset that this is speculative: the present paper does not derive such a coupling from established microphysics. Rather, it tests whether the *phenomenology* of an observer-anchored, causal, bounded redshift mapping is preferred by late-time data when the published covariances are treated consistently.

A useful heuristic lens is provided by Stephen Wolfram’s “computational theory of time,” in which the passage of time is associated with the updating of relational structure in an underlying discrete system. In that viewpoint, as the number of degrees of freedom and the density of their interactions increase, the number of effective constraints or “relationships” can grow rapidly, and one may expect the rate of consistent updating (i.e. the “clock rate” of the system) to be reduced. Whether or not one accepts this computational framing literally, it suggests a concrete intuition: time dilation can be viewed as arising when the local state is harder to update, in the sense of having more entangled constraints among constituents.

The inverse question then becomes natural: if one increases *order*—in the sense of reducing the effective complexity of interactions by building persistent correlations and structured organization—could an opposite-sign, observer-local effect arise in which the local pace of physical processes is *slightly faster* relative to a cosmological reference? In thermodynamic language, living systems and engineered systems are paradigmatic examples of far-from-equilibrium “negentropic bubbles”: they maintain low internal entropy (high organization) by consuming free energy and exporting entropy to their surroundings, so that total entropy production remains positive. The relevant speculative ingredient is therefore not a violation of the second law, but a hypothesized coupling between *local, sustained organization* (or some proxy such as long-lived correlations, mutual information, or algorithmic complexity measures) and the rate at which local proper time accumulates relative to a cosmological standard.

If such a coupling existed at a very small level, it could manifest observationally because redshift is fundamentally a comparison between an emitted photon frequency and the receiver’s local frequency standard. Any systematic drift in the receiver’s effective clock rate relative to the background time coordinate can be re-expressed as a multiplicative contribution to $(1+z)$, i.e. an additive contribution to $\ln(1+z)$, precisely the structure implemented in Eq. (49).

In the fiducial analysis we fix the cutoff epoch to $t_{\text{life}} = 3.8 \text{ Gyr}$ as a motivated order-of-magnitude proxy for the timescale over which complex terrestrial biospheric organization has existed. We emphasize that t_{life} is *not* inferred by the optimizer in the configuration reported here; rather, the data show that a cutoff on this scale is compatible with, and can support, a substantial fit improvement given the other model assumptions. Allowing t_{life} (and the activation shape parameters) to vary and mapping the resulting degeneracies with s_{anchor} is an important extension left to future work.

If the observer-local interpretation of LTA were correct, it would not be a minor adjustment to standard cosmological modeling. It would require extending our current framework in a way that introduces an additional state variable (or effective degree of freedom) encoding some measure of organization/order, rather than sourcing clock behavior solely through stress–energy. Constructing a consistent covariant completion—and determining whether such a completion can be made compatible with locality, Lorentz invariance, and precision metrology constraints—is an open theoretical problem. In this paper we do not attempt that construction; we restrict ourselves to a minimal, falsifiable phenomenological mapping and test it against SN+BAO data.

Thermodynamic closure and the role of clocks. In Jacobson’s thermodynamic derivation of general relativity, the Einstein field equations arise as a local equilibrium condition enforcing the Clausius relation $\delta Q = T \delta S$ across all local causal horizons [16]. A complementary and thematically related viewpoint is the “thermal time” hypothesis, in which a notion of time flow is associated with the statistical state of a generally covariant system via modular structure rather than being taken as primitive [22]. While our chronometric postulate is different in content, the shared motif is that the operational time experienced by physical systems may be linked to coarse-grained state structure and thermodynamic bookkeeping. In this framework, spacetime curvature is fixed by the requirement that horizon entropy, taken to be proportional to area, evolves consistently under local energy flux.

The Local Time Acceleration (LTA) hypothesis explored here is most naturally interpreted as a *nonequilibrium extension* of this picture. Sustained organization in driven subsystems does not modify the equilib-

rium area–entropy relation directly, but instead introduces continuous entropy production $\sigma = \nabla_\mu s^\mu \geq 0$ [17]. Rather than sourcing large curvature gradients, such nonequilibrium contributions can enter dominantly through the *clock sector*: they bias the local rate at which proper time accumulates relative to the background cosmological time coordinate.

Operationally, redshift is a comparison between an emitted photon frequency and the observer’s local frequency standard. A persistent, observer-local bias in clock rate therefore appears as a multiplicative correction to $(1+z)$, independent of any large modification to null geodesics or lensing. This motivates treating LTA primarily as an effective clock-rate perturbation sourced by entropy production, with metric backreaction remaining subleading in the weak-field regime probed by low-redshift supernovae.

13.3 Earth budget closure: magnitude consistency with planetary free-energy throughput

A central question for any observer-local, organization-sourced clock-bias hypothesis is whether the inferred magnitude is even remotely compatible with known terrestrial energy and entropy budgets. In our fiducial fit, the LTA mapping saturates at

$$I_{\text{sat}} \simeq 2.67 \times 10^{-3}, \quad (120)$$

corresponding to a fractional multiplicative factor in $(1+z)$ of $\exp(I_{\text{sat}}) - 1 \approx 2.7 \times 10^{-3}$ (a $\sim 0.27\%$ effect).

In the constraint-field formulation developed in Appendix ?? (energy $\rightarrow$ constraints $\rightarrow$ clock-bias chain), the relevant thermodynamic control parameter is not “entropy” in the sense of disorder, but the *nonequilibrium free-energy throughput that can be stored in, and continually used to maintain, constraints* (i.e. microstate-space restrictions). The second law then supplies a robust bound: maintaining (or increasing) any coarse-grained constraint inventory requires dissipation, and the rate of constraint maintenance is bounded by available power and the effective dissipation temperature.

Conservative biospheric throughput estimate. A conservative proxy for Earth’s sustained constraint-building throughput is the chemical free-energy stored by global net primary production (NPP) [20, 19]. Using a global NPP of $\dot{M}_C \sim 10^2 \text{ PgC yr}^{-1}$ and a standard free-energy requirement $\Delta G^\circ \simeq 4.78 \times 10^5 \text{ J mol}^{-1}$ per mole of CO_2 reduced to the carbohydrate level [21], the implied biospheric free-energy throughput is of order

$$P_{\text{bio}} \sim 10^{14} \text{ W}, \quad (121)$$

and the corresponding entropy-production scale at ambient temperature is

$$\dot{S}_{\text{bio}} \sim \frac{P_{\text{bio}}}{T_{\text{amb}}} \sim 10^{11} \text{ W K}^{-1} \quad (T_{\text{amb}} \sim 300 \text{ K}). \quad (122)$$

This is a *throughput* bound: it does not assert that NPP equals total biospheric dissipation, only that it sets a conservative scale for sustained, order-linked chemical free-energy processing.

Planetary entropy-production reference scale. A natural comparison scale for any Earth-sourced mechanism is the planetary entropy production associated with the solar-to-infrared energy degradation of the Earth system, which is of order [18]

$$\dot{S}_{\oplus} \sim 10^{14} \text{ W K}^{-1}. \quad (123)$$

Published radiative-entropy-budget estimates also report a smaller but still large internal irreversible entropy-production scale, $\dot{S}_{\oplus, \text{int}} \sim 10^{13} \text{ W K}^{-1}$, associated with non-radiative irreversible processes within the climate system.

Closure in a dimensionless form. Because I_{sat} is dimensionless, a minimal (and explicitly bounded) closure consistent with second-law bookkeeping is to write the saturation depth as proportional to a dimensionless source strength defined by an entropy-throughput fraction:

$$I_{\text{sat}} \equiv \epsilon_{\text{tc}} \frac{\dot{S}_{\text{src}}}{\dot{S}_{\text{ref}}}, \quad (124)$$

where $\dot{S}_{\text{src}}$ is the entropy-production throughput associated with the organized, constraint-maintaining source region (conservatively $\dot{S}_{\text{src}} \sim \dot{S}_{\text{bio}}$), $\dot{S}_{\text{ref}}$ is a planetary reference throughput (e.g. $\dot{S}_{\oplus}$ or $\dot{S}_{\oplus,\text{int}}$), and ϵ_{tc} is a dimensionless transduction parameter encoding the microphysical efficiency with which organized nonequilibrium throughput biases the local clock sector.

With $\dot{S}_{\text{bio}}/\dot{S}_{\oplus} \sim 10^{-3}$ (radiative reference) or $\dot{S}_{\text{bio}}/\dot{S}_{\oplus,\text{int}} \sim 10^{-2}$ (internal reference), the observed $I_{\text{sat}} \sim 2.7 \times 10^{-3}$ corresponds to ϵ_{tc} in the broad range $\sim 10^{-1}$ to a few, i.e. not an astronomically large or obviously impossible efficiency parameter. The important point is not that this fixes ϵ_{tc} , but that the *required clock-bias amplitude is compatible, at the level of order-of-magnitude energy/entropy throughput, with what Earth can sustain without violating the second law.*

Full arithmetic and explicit numerical values used for the above ratios are given in Appendix B.

13.4 Spacetime interpretation: an observer-anchored frequency-drift imprint on the past light cone

Operationally, LTA is implemented as a multiplicative correction to the observed redshift,

$$1 + z_{\text{obs}} = (1 + z_{\text{cos}}) \exp[I(\chi)], \quad (125)$$

with $I(\chi)$ defined as a line-of-sight integral of an effective drift-rate field $s(\chi)$ (Eq. (48)). This form is deliberately chosen because redshift factors compose multiplicatively, while their logarithms add; an accumulated, small fractional frequency drift along a trajectory naturally appears as an additive contribution to $\ln(1 + z)$.

A physically transparent way to read this mapping is to recall that a redshift measurement compares a received photon frequency to the receiver’s local time standard. If the receiver’s effective clock rate is biased relative to a cosmological reference (for whatever reason), then the measured frequency ratio acquires an extra factor that is indistinguishable from an additional redshift contribution. In the present framework, that extra factor is encoded by $\exp[I(\chi)]$, where the integral representation provides a controlled, causal way to localize the effect along the past light cone and to enforce saturation.

The “Earth-retarded lookback time” argument t_{ret} (Sec. 7.3) then supplies a specific causal bookkeeping device: it ties the strength of $s(\chi)$ at each point on the photon path to a retarded-time assignment relative to the observer, with the intent that an observer-local influence cannot affect regions outside its causal domain. In this language one may visualize the activation as defining a retarded domain of influence expanding outward at (at most) the speed of light. Incoming photons traverse this domain and accumulate a small net imprint in $\ln(1 + z)$. Because the activation is bounded and saturating, the accumulated imprint is also bounded: in the fiducial best fit we reconstruct a saturation scale $I_{\text{sat}} \simeq 2.67 \times 10^{-3}$ (Sec. 11.5), i.e. an $\mathcal{O}(10^{-3})$ multiplicative correction in $(1 + z)$.

This picture is sometimes described colloquially as a “time wave.” We stress that nothing in the analysis requires a literal propagating wave in a medium. Rather, it is a compact visualization of a causal, observer-anchored domain within which an additional, small frequency-drift contribution is accumulated. In the specific interpretation motivating the present model, the “steps” are not physical displacements of the observer away from the sources, but incremental changes in the local rate at which the observer’s clock accumulates proper time relative to the cosmological reference. Photons emitted earlier then arrive at an observer whose effective time standard is slightly different from what would be expected under baseline assumptions, and this mismatch appears as an extra contribution to the measured redshift. The reason this can preferentially affect the low-redshift Hubble diagram is precisely the bounded activation: the imprint grows with distance up to a finite epoch, after which additional path length no longer increases $I(\chi)$.

13.5 Relation to dark-energy phenomenology and broader implications

The analysis in this paper is intentionally scoped. The best-fit LTA profiles favored by the Pantheon+SH0ES + BAO combination in our fiducial configuration are *local and saturating*: they are designed to act primarily in the low- z regime where the Hubble ladder is anchored and where coherent residual structure is observed, while becoming negligible at larger lookback times. This is why the model can improve the low- z Hubble-flow residuals without demanding a wholesale rewrite of high-redshift cosmology.

With that said, it is difficult to discuss any redshift-mapping modification without acknowledging the broader question it raises: could some portion of the phenomenology commonly attributed to late-time

acceleration be connected with assumptions about how observed redshift maps to the background redshift used in distance predictions? In standard Λ CDM, dark energy (modeled as a cosmological constant Λ) is inferred from the redshift dependence of distances and expansion observables across a wide redshift lever arm. Our present implementation does not attempt to re-fit that entire lever arm with a generalized mapping, and it does not claim to replace Λ as a successful effective description at high redshift. Indeed, in the data combination used here, the most informative signature is that the improvement is localized to the nearby Hubble flow and is already constrained by the Jacobian-sensitive radial BAO sector.

Nevertheless, if an organization-linked clock-rate bias were real in some form, one must at least ask whether analogous effects could arise in other environments or epochs. For example, during specific phases of structure formation—when baryons condensed into stars and galaxies and when large, long-lived correlations emerged—it is conceivable (as a hypothesis to be tested) that the Universe exhibited different levels of sustained “order” than it does in the late-time diffuse intergalactic medium. In such a scenario, photons from high- z sources could accumulate additional frequency-shift contributions as they traverse regions with different organization histories, producing something akin to a “redshift lens” (a propagation-induced frequency mapping) rather than a true modification of the background expansion rate. Whether any realistic version of this idea can reproduce the suite of high- z constraints (CMB peak structure, high- z BAO, growth of structure, lensing) is a stringent question, and nothing in our present analysis answers it.

For clarity, we state our intent explicitly in the strongest terms. We chose not to resolve both the Hubble tension and the existence of Dark Energy in one-shot, but if our result is to be trusted then it opens the door to new explorations for alternatives to Λ CDM. In practice, the next step would be to generalize the mapping (or its physical interpretation) beyond an Earth-anchored activation, and then confront that generalization with the full high-redshift dataset hierarchy. That program is well beyond the scope of the present work.

13.6 A speculative aside: the Fermi paradox under “time-acceleration wells”

The Fermi paradox, like the Hubble tension, remains an open question in cosmological physics: if life is common, why is clear evidence scarce? If negentropic time acceleration exists and scales with sustained organization, then a sufficiently advanced civilization could inhabit a deeper local “time-acceleration well” than ours. In that scenario, outbound signals would need to propagate out of that well, and from our perspective their society could appear “in the future”—not because information travels faster than light, but because their local proper-time accumulation per unit of our coordinate time would be larger. This is speculative, but it illustrates a sharp point: an LTA-like effect, if real and scalable, would have implications beyond cosmology, including how we should model timebases and frequency drifts in SETI-style searches.

13.7 Paths toward falsification with timing and metrology

Because the LTA hypothesis is, at root, a claim about frequency comparisons and effective clock rates, precision timekeeping is a natural arena for independent tests. Several possibilities are worth highlighting, along with the corresponding difficulties:

- **GNSS and terrestrial time-transfer networks.** GNSS clocks are continuously steered to a coordinate time standard, and slow drifts are routinely absorbed into control models and treated as systematics. This makes them challenging probes: a genuine slow local drift could be “calibrated away.” Still, with access to sufficiently raw telemetry and careful modeling, one can search for residuals that correlate with environmental or organizational proxies not captured by standard relativistic corrections.
- **Pulsar timing arrays (PTAs).** PTAs are exquisitely sensitive to timing residuals, but they also inherit a fundamental ambiguity if the hypothesized effect couples to the receiver clock: a drift in the terrestrial time standard is largely degenerate with broad classes of timing model errors. In addition, pulsars and their magnetospheres are themselves highly structured systems, so PTAs could in principle probe a superposition of source-side, propagation-side, and receiver-side effects. This does not rule PTAs out; it highlights the need for careful null tests and independent time standards.
- **Differential optical clock experiments across controlled “order” environments.** The most direct laboratory strategy is differential: compare two or more state-of-the-art optical clocks (linked

by fiber or transportable) placed in environments engineered to differ strongly in their degree of sustained organization while controlling conventional influences (gravitational potential, temperature, EM environment, pressure). Candidate environments include dense biospheres (e.g. tropical rainforest sites versus deserts), high-throughput bioreactors with controlled metabolic flux, and large-scale engineered information-processing systems (e.g. data centers). The goal would be to bound (or detect) any reproducible differential drift beyond known systematics.

- **Additional experimental directions.** Other controlled comparisons that could be informative as timekeeping technology improves include: (i) clocks operated in high-activity chemical plants or electrochemical storage facilities versus quiet controls; (ii) clocks colocated with high-coherence electromagnetic systems (high- Q cavities, large interferometers) versus controls; (iii) deep underground laboratories versus surface sites to change environmental noise and shielding; and (iv) long-duration campaigns designed to test correlations with seasonal biosphere cycles, regional power throughput, or other measurable proxies for sustained organization.

These tests are ambitious, but they have a decisive virtue: they are not cosmological, and they do not rely on SN calibration pipelines. If an LTA-like effect is real and couples to local time standards at the magnitude implied by the cosmological fit, precision metrology should either constrain it sharply or reveal correlated residuals in a controlled setting. If, conversely, no such signal exists in the laboratory, then the natural interpretation of the cosmological LTA preference would shift toward unmodeled low- z astrophysical systematics (local flows, selection, calibration structure) rather than new clock-rate physics.

In summary, the conservative conclusion is that the present work isolates a specific empirical pattern in the low- z Hubble diagram that is well described by a bounded, causal, observer-anchored mapping. Whether that mapping is merely an effective proxy for local kinematics or a hint of a deeper connection between organization and time standards is not resolved here. What *is* resolved is the form of the question: the mapping makes concrete predictions for redshift Jacobians and for clock/frequency drift observables, and those predictions can be tested—and potentially falsified—by independent data.

14 Conclusion

We have introduced and tested a phenomenological extension of flat Λ CDM in which the observed redshift is related to the background cosmological redshift by a bounded, causal, observer-anchored mapping. In this “Local Time Acceleration” (LTA) framework, photons accumulate a small additional contribution to $\ln(1+z)$ along the line of sight via an effective drift rate $s(\chi)$ whose amplitude is defined at an anchor time computed from the Hubble-flow subset only, and whose activation decays to zero at a finite epoch. The model is nested: $s_{\text{anchor}} \rightarrow 0$ recovers standard Λ CDM exactly.

Confronting this mapping with the Pantheon+SH0ES public release using the *full* published covariance (including calibrator–Hubble-flow cross-correlations) and a ladder-anchored Schur-complement likelihood, and jointly fitting a BAO consensus compilation [3] (with the LTA Jacobian properly propagated into the radial observable), we find a substantial improvement in global goodness-of-fit relative to the baseline model in the fiducial configuration. In the representative run reported here (run-tag 20260112–013808), the best-fit LTA(powerlaw) model improves the total objective by $\Delta\chi^2 \simeq 13.7$ while leaving the best-fit background parameters ($H_0, \Omega_m, \alpha_{\text{rd}}$) essentially unchanged. The gain is carried almost entirely by the anchored Hubble-flow SN likelihood, with only a mild BAO penalty, consistent with a genuinely local modification to the redshift–distance identification rather than a global change in $H(z)$.

Residual and whitening diagnostics show that the improvement is not uniformly distributed in redshift: it is localized to the very local volume ($z \lesssim 0.03$), where the baseline Hubble diagram exhibits a coherent “valley” often discussed in the context of local kinematics. Within the LTA framework, this structure is naturally modeled by a smooth, saturating mapping that becomes negligible at higher redshift, yielding convergence to the standard relation for $z \gtrsim 0.05$.

Crucially, the preference is not fragile to single-subset removal. Grouped jackknife refits that remove each major telescope/survey group in turn preserve a positive improvement in every case in the tests reported here, disfavoring the hypothesis that the signal is driven by one faulty instrument or one calibration subset. Because the LTA amplitude is bounded and the null point lies on a parameter boundary, we further calibrate

the detection statistic using parametric bootstrap (null injections) that preserve the full released covariance and refit both models; with a high-statistics run ($N = 10000$ null draws) a tail probability of order $p \sim 0.0008$ was obtained for a $\Delta\chi^2$ of this magnitude, and ongoing larger- N runs are used to refine the Monte Carlo resolution.

The present work is intentionally not a “one-shot” replacement for the standard cosmological model. It establishes a narrower empirical claim: within a conservative full-covariance data treatment, a bounded, causal redshift-mapping extension provides a better description of the low- z Hubble diagram than baseline Λ CDM in the configurations tested, and it does so in a way that is robust to obvious single-survey explanations. Whether this mapping is best interpreted as a proxy for unmodeled local kinematics (e.g. Local Void dynamics) or as evidence for a genuinely observer-local clock-rate contribution remains open. The decisive next step is therefore falsification via independent observables: precision clock networks, standard sirens, time-delay cosmography, and sky-dependent correlations with local structure, all of which can sharply constrain (or exclude) an LTA-like mechanism.

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A Thermodynamic and chronometric foundations of LTA

This appendix provides a single continuous derivation chain connecting:

energy throughput $\implies$ bounded constraint inventory $\implies$ order field $\psi(x) \implies$ scalar–tensor coupling $F(\psi) \implies$ (derived) and clarifies how these ingredients feed into two distinct observational channels:

1. **Homogeneous cosmology:** the background mode $\bar{\psi}(t)$ modifies the Jordan-frame expansion history $H_J(z)$, enabling dark-energy-like effects without introducing Λ as a separate component (Sec. ??).
2. **Observer-local imprint:** inhomogeneous perturbations $\delta\psi(x)$ can generate endpoint/integrated frequency-comparison effects that, in a controlled approximation, may be packaged into an effective $\Delta \ln(1+z)$ mapping (Sec. 7.2).

Everything up to the covariant insertion point S_{neq} uses standard nonequilibrium/statistical thermodynamics and conservative bookkeeping. The nonstandard physics is localized to: (i) the choice of scalar–tensor functions $F(\psi)$, $U(\psi)$, and (ii) the form of the nonequilibrium source J_{neq} that links sustained constraint-maintenance to the scalar sector.

Conventions (fixed reference; GR recovery in vacuum). Throughout Appendix A we choose the maximum-entropy reference distribution π *once* and hold it fixed in time. Equivalently, the macroscopic conditions defining the “unconstrained baseline” (bath temperature T , energy function $E(\xi)$, conserved quantities, etc.) are taken time-independent on the timescales of interest. With this convention the reference equilibrium free energy F_{eq} is constant, and φ can be interpreted as a true “constraint inventory” whose growth is bounded by input work (Sec. A.3).

In the absence of matter/nonequilibrium sources (ideal vacuum), the local state coincides with the reference ($p = \pi$), so $\varphi = \psi = 0$ and realized clock time reduces to standard Jordan-frame proper time. All LTA phenomenology is sourced by nonzero ψ in material environments.

A.1 A.1 Constraint inventory as microstate-space reduction

We require a mathematically precise scalar quantifying “constraint / order” that: (i) is nonnegative; (ii) behaves conservatively under coarse-graining; and (iii) has a direct energetic meaning under the second law.

Let ξ denote a microstate at a chosen coarse-graining and let $p(\xi, t)$ be the system’s (coarse-grained) microstate distribution. Let $\pi(\xi, t)$ denote a reference maximum-entropy distribution representing the *unconstrained* baseline given whatever macroscopic constraints we are *not* counting as “order” (e.g. fixed energy, particle content, etc.). In the simplest canonical case at temperature T , π is Gibbs; more generally it can be a MaxEnt reference.

Definition (constraint inventory). Define

$$\phi(t) \equiv D_{\text{KL}}(p(\cdot, t) \parallel \pi) = \int d\xi p(\xi, t) \ln \frac{p(\xi, t)}{\pi(\xi)}. \quad (126)$$

and the entropy-unit inventory

$$\varphi(t) \equiv k_B \phi(t) \quad [\varphi] = \text{J K}^{-1}. \quad (127)$$

Core properties (why this is a conservative notion of “order”). KL divergence satisfies:

$$\varphi(t) \geq 0, \quad \varphi(t) = 0 \iff p = \pi \text{ a.e.}, \quad (128)$$

$$\varphi(t) \geq \varphi_{\text{coarse}}(t) \quad \text{under any coarse-graining map (data-processing)}, \quad (129)$$

$$\varphi_{AB} = \varphi_A + \varphi_B \quad \text{for independent composition } p_{AB} = p_A p_B, \pi_{AB} = \pi_A \pi_B. \quad (130)$$

Thus φ is nonnegative, cannot be inflated by coarse-graining, and is additive under independent composition.

Explicit microstate-space link. If π is uniform on an allowed set of size Ω_0 and p is uniform on a constrained subset of size $\Omega_1 \leq \Omega_0$, then

$$\phi = \ln \frac{\Omega_0}{\Omega_1}, \quad \varphi = k_B \ln \frac{\Omega_0}{\Omega_1}. \quad (131)$$

In this special case, φ literally counts the log-reduction in accessible microstates. This is the precise sense in which “constraints reduce microstate space.”

A.2 A.2 Nonequilibrium free energy: $T\varphi$ is a stored constraint budget

In an isothermal setting at temperature T , define the entropy functional

$$S[p] \equiv -k_B \int d\xi p(\xi, t) \ln p(\xi, t), \quad (132)$$

and the nonequilibrium Helmholtz free energy functional

$$F[p] \equiv \langle E \rangle_p - TS[p]. \quad (133)$$

We assume the bath temperature T and the energy function $E(\xi)$ defining the canonical reference are time-independent, so that F_{eq} is a constant and $d(F - F_{\text{eq}})/dt = dF/dt$.

For a canonical reference $\pi(\xi) \propto e^{-\beta E(\xi)}$ with $\beta = (k_B T)^{-1}$, the equilibrium free energy is $F_{\text{eq}} = -k_B T \ln Z$, and one has the standard identity

$$F[p] - F_{\text{eq}} = k_B T D_{\text{KL}}(p \parallel \pi) = T\varphi. \quad (134)$$

In the fixed-reference convention adopted here, F_{eq} is constant in time. Therefore $T\varphi$ is exactly the *excess free energy above equilibrium* stored in the maintained constraints/correlations relative to π .

Bits language (optional). Define $\varphi_{\text{bits}} \equiv \varphi/(k_B \ln 2) = \phi/\ln 2$. Then $T\varphi$ corresponds to an energy scale of $k_B T \ln 2$ per bit, matching the Landauer scale for logically irreversible operations. We do *not* assume biological systems saturate this bound; it is used only as a clean dimensional benchmark.

A.3 A.3 Second-law throughput bound: power limits constraint growth

Assume an effective isothermal reservoir at temperature T . Let $\dot{Q}$ be heat flow into the system and $\dot{W}$ work done on the system; the first law gives

$$\frac{d}{dt}\langle E \rangle = \dot{Q} + \dot{W}. \quad (135)$$

Define total entropy production rate (system + environment) by

$$\dot{\Sigma} \equiv \frac{dS}{dt} - \frac{\dot{Q}}{T} \geq 0, \quad (136)$$

the local second law for this setting.

Using $F = \langle E \rangle - TS$ yields the standard balance

$$\dot{W} = \frac{dF}{dt} + T\dot{\Sigma}. \quad (137)$$

Because the reference π (hence F_{eq}) and the bath temperature T are fixed in the present convention, combining (122) with (125) gives

$$T\dot{\varphi} = \frac{d}{dt}(F - F_{\text{eq}}) \leq \dot{W} \equiv P_{\text{in}}, \quad \Rightarrow \quad \dot{\varphi} \leq \frac{P_{\text{in}}}{T}. \quad (138)$$

This is the key throughput bound used later: *constraint inventory cannot grow faster than available power divided by temperature*.

In steady maintenance regimes where $\langle \dot{\varphi} \rangle \simeq 0$, one has $\langle \dot{\Sigma} \rangle \simeq \langle P_{\text{in}} \rangle / T$.

A.4 A.4 From constraint throughput to a causal order field $\psi(x)$

The cosmological LTA mapping is a function of spacetime position along the observer past light cone. To connect (126) to a lightcone imprint, we introduce a coarse-grained, covariant scalar *order field* $\psi(x)$ sourced by local nonequilibrium activity.

Order field definition (local inventory density). To obtain a spacetime field, apply the same fixed-reference KL construction to a coarse-graining cell of physical volume V_c centered at spacetime event x , yielding a cell inventory $\varphi_c(x) \equiv k_B D_{\text{KL}}(p_c(\cdot; x) \| \pi_c)$ with units JK^{-1} . Define the corresponding local inventory *density*

$$\varphi_{\text{loc}}(x) \equiv \frac{\varphi_c(x)}{V_c}, \quad [\varphi_{\text{loc}}] = \text{JK}^{-1} \text{m}^{-3}. \quad (139)$$

The total inventory associated with a spatial region $\mathcal{R}$ on a constant-time slice is then $\varphi_{\mathcal{R}}(t) = \int_{\mathcal{R}} d^3x \sqrt{h} \varphi_{\text{loc}}(x, t)$, where h is the induced 3-metric determinant.

Define a dimensionless order field by

$$\psi(x) \equiv \frac{\varphi_{\text{loc}}(x)}{\varphi_*}, \quad (140)$$

where φ_* is a fixed reference *density* scale.

Entropy-field interpretation and current. It is often convenient to keep track of an entropy-unit scalar by undoing the normalization, $S_{\text{ord}}(x) \equiv S_* \psi(x)$ with $[S_{\text{ord}}] = \text{JK}^{-1}$ (or $S_* \equiv \varphi_* V_c$ if one prefers the density normalization used above). With metric signature $(-, +, +, +)$, a future-directed “informational/entropy” current built from S_{ord} is naturally defined as

$$J_{\text{ord}}^\mu \equiv -\nabla^\mu S_{\text{ord}}, \quad \sigma_{\text{ord}} \equiv \nabla_\mu J_{\text{ord}}^\mu = -\square S_{\text{ord}}. \quad (141)$$

For a homogeneous field on FLRW this choice gives $J_{\text{ord}}^0 = +\dot{S}_{\text{ord}}$ and $\sigma_{\text{ord}} = \ddot{S}_{\text{ord}} + 3H\dot{S}_{\text{ord}}$.

Causal propagation (minimal effective field). To encode finite-range causal influence and saturation, we treat ψ as an effective scalar propagating on the standard GR background with a retarded Green’s function. The minimal local form is a sourced Klein–Gordon-type equation

$$(\square - m_\psi^2)\psi(x) = \kappa \sigma(x), \quad (142)$$

where $\sigma(x) \geq 0$ is a scalar proxy for nonequilibrium throughput (e.g. local entropy production density $\sigma \equiv \nabla_\mu s^\mu$), m_ψ sets an effective range / saturation scale, and κ is a coupling constant. Retarded boundary conditions ensure causality: $\psi(x)$ depends only on past sources.

Observer-local source approximation and retarded-time bookkeeping. If $\sigma(x)$ is strongly localized near the observer world-tube (Earth), then on the past light cone the dominant dependence reduces to a function of a retarded time parameter, $\psi \approx \psi(t_{\text{ret}}(\chi))$. In the low- z FLRW limit, the standard “two-leg” causal intersection geometry motivates the approximation $t_{\text{ret}} \approx 2\chi/c$, and in the main text this is implemented via the Earth-retarded lookback-time mapping defined by $\chi(z_e) = 2\chi$ and $t_{\text{ret}} = t_{\text{lb}}(z_e)$ (Sec. 7.3). This step is a bookkeeping approximation of the retarded solution of (129); it is not a modification of the background expansion.

B Earth budget closure: numerical entropy-throughput ratios

This appendix provides the explicit arithmetic supporting the order-of-magnitude statements in Sec. 13.3. The purpose is a *feasibility bound*: demonstrate that the required LTA saturation magnitude $I_{\text{sat}} \sim 10^{-3}$ does not immediately contradict known terrestrial free-energy and entropy-production scales.

B.1 Biospheric chemical free-energy throughput from NPP

We take a representative global net primary production (NPP) value [20, 19]

$$\dot{M}_C \simeq 104.9 \text{ PgC yr}^{-1}. \quad (143)$$

Convert to moles of carbon fixed per unit time using 12 g mol^{-1} :

$$\dot{n}_C = \frac{104.9 \times 10^{15} \text{ g yr}^{-1}}{12 \text{ g mol}^{-1}} \simeq 8.74 \times 10^{15} \text{ mol yr}^{-1}. \quad (144)$$

Using a standard free-energy requirement for reduction of one mole of CO_2 to the carbohydrate level,

$$\Delta G_{\text{red}}^\circ \simeq 4.78 \times 10^5 \text{ J mol}^{-1} [21], \quad (145)$$

the global chemical free-energy throughput is

$$P_{\text{bio}} = \frac{\dot{n}_C \Delta G_{\text{red}}^\circ}{(1 \text{ yr})} \simeq \frac{(8.74 \times 10^{15})(4.78 \times 10^5) \text{ J yr}^{-1}}{3.156 \times 10^7 \text{ s yr}^{-1}} \simeq 1.32 \times 10^{14} \text{ W}. \quad (146)$$

At $T_{\text{amb}} \simeq 288 \text{ K}$, the associated entropy-throughput scale is

$$\dot{S}_{\text{bio}} \sim \frac{P_{\text{bio}}}{T_{\text{amb}}} \simeq \frac{1.32 \times 10^{14} \text{ W}}{288 \text{ K}} \simeq 4.6 \times 10^{11} \text{ W K}^{-1}. \quad (147)$$

We stress: this is a conservative scale-setting calculation; gross primary production and additional biogeochemical and technospheric throughput can increase the true total organized-dissipation budget.

B.2 Planetary entropy-production reference scales

Representative entropy-budget calculations (reviewed, e.g., in [18]) report global-mean radiative entropy production of order $\sim 0.9 \text{ W m}^{-2} \text{ K}^{-1}$ and internal irreversible entropy production of order $\sim 0.076 \text{ W m}^{-2} \text{ K}^{-1}$. Using Earth’s surface area $A_\oplus \simeq 4\pi R_\oplus^2$ with $R_\oplus = 6.371 \times 10^6 \text{ m}$ gives $A_\oplus \simeq 5.10 \times 10^{14} \text{ m}^2$. Thus:

$$\dot{S}_\oplus \sim (0.93) A_\oplus \simeq 4.7 \times 10^{14} \text{ W K}^{-1}, \quad (148)$$

$$\dot{S}_{\oplus, \text{int}} \sim (0.076) A_\oplus \simeq 3.9 \times 10^{13} \text{ W K}^{-1}. \quad (149)$$

B.3 Dimensionless ratios and implied transduction parameter

From Eqs. (134)–(above), the biospheric ratios are

$$\frac{\dot{S}_{\text{bio}}}{\dot{S}_{\oplus}} \simeq \frac{4.6 \times 10^{11}}{4.7 \times 10^{14}} \simeq 9.7 \times 10^{-4}, \quad (150)$$

$$\frac{\dot{S}_{\text{bio}}}{\dot{S}_{\oplus, \text{int}}} \simeq \frac{4.6 \times 10^{11}}{3.9 \times 10^{13}} \simeq 1.2 \times 10^{-2}. \quad (151)$$

For the fiducial best-fit saturation $I_{\text{sat}} \simeq 2.67 \times 10^{-3}$, the closure relation in Eq. (112) implies a transduction parameter ϵ_{tc} of order

$$\epsilon_{\text{tc}} \simeq \frac{I_{\text{sat}}}{\dot{S}_{\text{bio}}/\dot{S}_{\oplus}} \simeq 2.7 \quad (\text{radiative reference}), \quad (152)$$

$$\epsilon_{\text{tc}} \simeq \frac{I_{\text{sat}}}{\dot{S}_{\text{bio}}/\dot{S}_{\oplus, \text{int}}} \simeq 0.23 \quad (\text{internal reference}). \quad (153)$$

Thus, at the level of second-law throughput accounting, the saturation magnitude inferred from the SN+BAO mapping is not obviously inconsistent with what Earth can sustain.

Interpretation and limitations. This calculation is a *closure check*, not a derivation. It shows that an $I_{\text{sat}} \sim 10^{-3}$ clock-bias amplitude does not require an implausibly large terrestrial free-energy or entropy-production budget. A complete theory must still specify (and constrain) the microphysical meaning of ϵ_{tc} and satisfy laboratory tests (precision clocks, equivalence principle, Lorentz invariance).

B.4 Biophysical mechanisms: how life maintains constraints on microstate space

This paper’s thermodynamic completion does not treat “entropy” as the fundamental measure of organization. Instead, we define an explicit *constraint inventory* φ (Appendix A) that quantifies how strongly a system’s coarse-grained microstate distribution p is restricted relative to an unconstrained maximum-entropy reference π , $\varphi = k_B D_{\text{KL}}(p||\pi)$. Entropy production enters only as the *accounting variable* demanded by the second law: it bounds how quickly such constraints can be created or maintained by any physical process.

In this section we identify concrete, well-established biophysical mechanisms by which living systems (cells, tissues, ecosystems) maintain $\varphi > 0$ for long durations by consuming free energy and exporting entropy to their environment. These mechanisms provide a physically grounded interpretation of the source term $\sigma(x)$ that drives the order field $\psi(x)$ in the chronometric completion (Appendix A), without assuming that biology violates the second law or that “entropy” itself is the order variable. [26, 27]

B.4.1 Life as a driven nonequilibrium steady state

At biological scales, a cell is accurately modeled as an *open, driven nonequilibrium system*: it maintains concentrations, gradients, macromolecular structures, and correlations that would relax toward equilibrium if energy throughput ceased. Mathematically, many such systems can be represented at mesoscopic resolution by a stochastic Markov dynamics on states i with probabilities $p_i(t)$:

$$\dot{p}_i = \sum_j (W_{ij}p_j - W_{ji}p_i), \quad (154)$$

where W_{ij} are transition rates. At equilibrium with a reference distribution π , rates satisfy detailed balance $W_{ij}\pi_j = W_{ji}\pi_i$. Living systems break detailed balance through continual chemical driving (e.g. ATP/GTP hydrolysis, redox chemistry, ion pumping), creating sustained probability currents and maintaining $p \neq \pi$ (a nonequilibrium steady state) [26, 37].

The connection to our constraint inventory is direct: the nonequilibrium free energy stored in such maintained constraints is $F - F_{\text{eq}} = T\varphi$ (Appendix A). Stochastic thermodynamics provides a standard energy/entropy balance for driven isothermal systems: work input can be decomposed into the rate of change of nonequilibrium free energy plus a strictly nonnegative entropy production term. This is the formal reason the second law constrains how large and how rapidly maintained constraints can be [26, 27].

B.4.2 Mechanism 1: compartmentalization and boundary conditions (membranes)

A defining physical act of life is the creation and maintenance of *boundaries* that impose conditional structure on accessible microstates. Lipid membranes and cell walls constrain spatial degrees of freedom and enable long-lived gradients (ionic, chemical, electrical) that would quickly disappear in an unconstrained fluid equilibrium. In the language of Appendix A, these boundaries alter the feasible macrostate constraints defining π and maintain a p that is concentrated on a restricted subset of microstates (thereby increasing φ). The energetic cost arises because boundaries are not passive: maintaining them in the presence of thermal fluctuations and chemical degradation requires continual repair, transport, and pumping, all of which dissipate free energy and generate entropy production [40, 38].

B.4.3 Mechanism 2: chemical potential gradients and chemostatted reaction networks

At the biochemical level, the most universal “constraint engine” is a driven chemical reaction network (CRN) with chemostatted species that act as fuel reservoirs (ATP/ADP/Pi, NADH/NAD⁺, proton gradients, etc.). Such networks are a canonical setting of nonequilibrium thermodynamics: one can define a nonequilibrium Gibbs free energy and show that maintaining a nonequilibrium state requires work by the chemostats, with entropy production given by flux–affinity products:

$$\dot{\Sigma} = \sum_r \frac{J_r \mathcal{A}_r}{T} \geq 0, \quad (155)$$

where J_r is the net flux of reaction channel r and $\mathcal{A}_r$ its chemical affinity. This formalism makes precise the statement that biological order is paid for by dissipation: persistent organization corresponds to persistent nonzero affinities and fluxes, hence positive $\dot{\Sigma}$ [26, 37, 41].

Crucially, this does *not* imply that “entropy” is the order variable. In our chain, order is φ (a microstate-restriction measure), while $\dot{\Sigma}$ is the second-law bookkeeping term that bounds how φ can change:

$$\dot{\varphi} \leq \frac{P_{\text{in}}}{T}, \quad (156)$$

as derived in Appendix A. A cell’s free-energy throughput is therefore a direct upper bound on how quickly it can build or refresh microstate restrictions, independent of any philosophical interpretation of entropy [26, 27].

B.4.4 Mechanism 3: polymer sequence order and kinetic proofreading (energy for specificity)

A large fraction of biological “constraints” are not merely concentration gradients, but *sequence-level correlations* in polymers: DNA/RNA sequences, folded proteins, and regulated macromolecular complexes. These are strong restrictions on microstate space because they represent highly specific arrangements among astronomically many alternatives.

A key point for rigor is that biology achieves error rates that exceed what would be expected from equilibrium discrimination alone, by using energy-driven *kinetic proofreading*. In Hopfield’s mechanism, an irreversible, energy-consuming step (e.g. via phosphate hydrolysis) allows the system to reject wrong intermediates more often than equilibrium would permit, increasing specificity at a thermodynamic cost. This is an explicit, mathematically analyzable example of

energy consumption $\Rightarrow$ conditional pathway filtering $\Rightarrow$ reduced accessible outcome microstates,

i.e. an increase in maintained constraint inventory φ [42, 43].

B.4.5 Mechanism 4: feedback control and information processing

Living systems constantly sense and regulate internal variables (chemotaxis, gene regulation, homeostasis). In physics terms, this is feedback control on stochastic degrees of freedom. A robust result across the thermodynamics-of-information literature is that *reliable* information processing and irreversible operations have thermodynamic costs (Landauer scale for erasure), and more generally that feedback-controlled systems

must still satisfy second-law inequalities when information flows are accounted for. In our framework, the relevance is structural: feedback loops implement *conditional constraints* on trajectories through state space, and maintaining those conditionals at low error requires dissipation, contributing to $\dot{\Sigma}$ and thereby bounding maintainable φ . [27, 28, 29]

B.4.6 From local constraints to a cosmology-facing source term

The previous subsections justify, in standard biophysical terms, why life plausibly sources a positive nonequilibrium throughput field $\sigma(x)$ (or an equivalent proxy such as local dissipated power density divided by temperature) that can be used as the causal source of an order field $\psi(x)$ in Appendix A. Importantly, $\sigma(x)$ is not a synonym for order: it is the second-law production term that accompanies the building and maintenance of constraints.

In this language, “life alters microstate space in front of every moment” can be stated mathematically and without mysticism: by consuming free energy, living systems actively reshape the transition structure (the effective generator W) of local chemical and mechanical degrees of freedom [26], thereby maintaining a state distribution p that is persistently restricted relative to the unconstrained reference π . The maintained restriction is quantified by $\varphi = k_B D_{\text{KL}}(p||\pi)$ [24, 25], its rate of refresh is bounded by power through $\dot{\varphi} \leq P_{\text{in}}/T$ [26], and in the chronometric completion this throughput sources $\psi(x)$, which (by postulate) biases realized clock time and thereby maps z_{obs} to z_{cos} without modifying photon trajectories or large-scale geometry [23, 33].

C Unified order–gravity framework: one field, two scales (auxiliary completion)

This appendix gives an *auxiliary* covariant completion of the Local Time Acceleration (LTA) idea in which a single “order” scalar field plays *two roles*: (i) a *cosmic* homogeneous mode that backreacts on geometry and can replace Λ (dark-energy-like acceleration), and (ii) a *local* inhomogeneous mode sourced by the observer environment that biases realized clock time and therefore induces the observer-local redshift-mapping distortion used in the main text.

The completion is written as an effective field theory (EFT). All *information/thermodynamic* statements (relative entropy, KL/entanglement measures, nonequilibrium free-energy identity, and second-law bounds on constraint maintenance) are unchanged from the main thermodynamic chain (App. A); the only new ingredient is an *EFT insertion point* that allows this thermodynamic source to enter a covariant action while keeping the gravitational field equations derivable by variation.

Scope. This appendix is not yet used in the SN+BAO inference pipeline of the main paper. Its purpose is to (1) derive the unified field equations cleanly and (2) show how the LTA mapping arises as a controlled limit of the same theory when local perturbations dominate the observable imprint.

C.1 U.1 Order as microstate restriction: a QFT-clean scalar

Pick a small region $A(x)$ around spacetime event x (e.g. a small causal diamond/ball at a chosen coarse-graining scale). Let $\rho_A(x)$ be the reduced density matrix of the actual state and $\sigma_A(x)$ a chosen reference reduced state (vacuum, local thermal, or MaxEnt under specified macroscopic constraints). Define the *relative entropy* (entanglement-relative-entropy) order measure

$$S_{\text{rel}}(x) \equiv S(\rho_A(x) || \rho_{A,\text{ref}}(x)) = \text{Tr}[\rho_A (\ln \rho_A - \ln \rho_{A,\text{ref}})] \geq 0. \quad (157)$$

Define the dimensionless *order field*

$$\psi(x) \equiv S_{\text{rel}}(x) \geq 0, \quad [\psi] = 1. \quad (158)$$

This is the cleanest “microstate restriction” scalar in QFT language. The exact identity $S(\rho||\rho_{\text{ref}}) = \Delta\langle K_{\text{ref}} \rangle - \Delta S$ (with modular Hamiltonian $K_{\text{ref}} \equiv -\ln \rho_{\text{ref}}$) and the linearized entanglement first law $\delta S = \delta\langle K \rangle$ motivate interpreting S_{rel} as a controlled nonequilibrium displacement from the reference.

Classical (commuting) limit and connection to Appendix A. If the reduced states commute and are diagonal in a common basis, $\rho_A = \sum_i p_i \ddot{u}$ and $\rho_{A,\text{ref}} = \sum_i \pi_i \ddot{u}$, then the quantum relative entropy reduces to the classical KL divergence:

$$S_{\text{rel}}(\rho_A \parallel \rho_{A,\text{ref}}) = \sum_i p_i \ln \frac{p_i}{\pi_i} \equiv D_{\text{KL}}(p \parallel \pi). \quad (159)$$

With the definition (145), this implies $\psi \rightarrow \phi$ of Eq. (114), and the entropy-unit inventory in the same coarse-graining cell is $\varphi_c = k_B \psi$, consistent with Eqs. (115) and (122). If one further defines a local density $\varphi_{\text{loc}} = \varphi_c / V_c$, then the Appendix A field normalization $\psi = \varphi_{\text{loc}} / \varphi_*$ (Eq. (128)) is recovered up to the fixed choice of the reference scale φ_* .

Classicalization assumption (EFT step). In cosmological applications we assume that, after coarse-graining on scales large compared to $A(x)$, $\psi(x)$ can be treated as a smooth classical scalar field whose dynamics are captured by an effective action.

C.2 U.2 Thermodynamic insertion: order costs free energy and produces entropy

This completion keeps the thermodynamic chain of App. A intact: relative-entropy order is an extensive “constraint inventory” and its maintenance/growth is bounded by work/entropy production.

In an isothermal driven setting (effective temperature T) with canonical reference, the standard identity gives

$$F - F_{\text{eq}} = k_B T \psi, \quad (160)$$

and the work/free-energy/dissipation balance $\dot{W} = \dot{F} + T\dot{\Sigma}$ with $\dot{\Sigma} \geq 0$ yields the bound

$$\dot{\psi} \leq \frac{P_{\text{in}}}{k_B T}. \quad (161)$$

(See Sec. 6.2–6.3 and App. A.)

Covariant nonequilibrium sector (the insertion point). To couple this thermodynamic bookkeeping into covariant field equations *without* introducing external nonconservative forcing by hand, we introduce an EFT “nonequilibrium” sector S_{neq} . Its role is to encode coarse-grained, open-system organization/dissipation physics in a diffeomorphism-invariant way. In a hydrodynamic/open-EFT limit, this sector supplies:

- a nonnegative scalar proxy $\sigma(x) \geq 0$ (e.g. entropy-production density, or dissipated power density divided by an effective temperature);
- a scalar functional derivative $J_{\text{neq}} \equiv -\frac{1}{\sqrt{-g}} \frac{\delta S_{\text{neq}}}{\delta \psi}$ that acts as the *covariant source* for ψ ;
- and a stress-energy tensor $T_{\mu\nu}^{(\text{neq})}$ for consistency with the Bianchi identity.

In a controlled coarse-grained closure one may model J_{neq} as a constitutive functional of σ , e.g. $J_{\text{neq}} \sim \kappa_\sigma \sigma$, with κ_σ setting dimensions and efficiency.

C.3 U.3 Action: Jordan-frame scalar–tensor + nonequilibrium insertion (chronometric factor derived)

We now write a unified covariant completion in which: (i) the homogeneous mode of the order scalar ψ backreacts on the physical expansion history and can mimic dark-energy-like acceleration without Λ , and (ii) local inhomogeneities of the same field can generate an observer-local imprint.

Jordan frame as the physical matter frame. We take the Jordan metric $g_{\mu\nu}^J$ to be the metric seen by matter and clocks. A minimal scalar-tensor + nonequilibrium insertion action is

$$\mathcal{S} = \int d^4x \sqrt{-g_J} \left[\frac{M_{\text{Pl}}^2}{2} F(\psi) R_J - \frac{f^2}{2} (\nabla_J \psi)^2 - f^4 U(\psi) \right] + \mathcal{S}_m[g^J, \Phi] + \mathcal{S}_{\text{neq}}[g^J, \psi, \Xi]. \quad (162)$$

This matches the main-text definition (Eq. (??)) but is repeated here for self-containedness.

Einstein frame and the derived chronometric factor. Define $g_{\mu\nu}^E = F(\psi) g_{\mu\nu}^J$. Then the matter action becomes conformally coupled, $\mathcal{S}_m[g^J, \Phi] = \mathcal{S}_m[A(\psi)^2 g^E, \Phi]$ with

$$A(\psi) \equiv F(\psi)^{-1/2}, \quad \mathcal{C}(\psi) \equiv \ln A(\psi) = -\frac{1}{2} \ln F(\psi). \quad (163)$$

Thus the “chronometric factor” is not independent; it is determined by $F(\psi)$. A convenient special case is $F(\psi) = e^{-2\epsilon\psi}$, which yields $A(\psi) = e^{\epsilon\psi}$, recovering the exponential coupling form used in earlier drafts as a specific parameterization.

C.4 U.4 Field equations (Jordan frame) and the split into background + local imprint

Varying (151) yields the Jordan-frame field equations

$$M_{\text{Pl}}^2 F(\psi) G_{\mu\nu}^J = T_{\mu\nu}^{(m)} + T_{\mu\nu}^{(\psi)} + T_{\mu\nu}^{(\text{neq})} + M_{\text{Pl}}^2 (\nabla_\mu^J \nabla_\nu^J F(\psi) - g_{\mu\nu}^J \square_J F(\psi)), \quad (164)$$

with

$$T_{\mu\nu}^{(\psi)} = f^2 \nabla_\mu^J \psi \nabla_\nu^J \psi - g_{\mu\nu}^J \left[\frac{f^2}{2} (\nabla_J \psi)^2 + f^4 U(\psi) \right], \quad (165)$$

and a scalar equation

$$f^2 \square_J \psi - f^4 U(\psi)' + \frac{M_{\text{Pl}}^2}{2} F(\psi)' R_J = J_{\text{neq}}, \quad J_{\text{neq}} \equiv -\frac{1}{\sqrt{-g_J}} \frac{\delta \mathcal{S}_{\text{neq}}}{\delta \psi}. \quad (166)$$

Background/perturbation split. Write $\psi = \bar{\psi}(t_J) + \delta\psi(x)$. The homogeneous mode determines the physical expansion $H_J(z)$ (Sec. ??). The local mode produces endpoint/integrated frequency-comparison effects, conveniently packaged into the observer-local mapping limit as $\ln[(1+z_{\text{obs}})/(1+z_{\text{cos}})] \equiv I$ (Sec. 7.2).

C.5 U.5 Cosmology: homogeneous mode can replace Λ

Assume a flat FLRW Einstein-frame geometry

$$ds^2 = -dt^2 + a^2(t) d\vec{x}^2, \quad H \equiv \frac{\dot{a}}{a}, \quad (167)$$

and a homogeneous field $\psi = \bar{\psi}(t)$.

The scalar energy density and pressure are

$$\rho_\psi = \frac{f^2}{2} \dot{\bar{\psi}}^2 + f^4 U(\bar{\psi}), \quad p_\psi = \frac{f^2}{2} \dot{\bar{\psi}}^2 - f^4 U(\bar{\psi}). \quad (168)$$

The Friedmann equations (with $\Lambda = 0$ in the action (151)) read

$$3M_{\text{Pl}}^2 H^2 = \rho_m + \rho_r + \rho_\psi + \rho_{\text{neq}}, \quad (169)$$

$$-2M_{\text{Pl}}^2 \dot{H} = (\rho_m + p_m) + (\rho_r + p_r) + (\rho_\psi + p_\psi) + (\rho_{\text{neq}} + p_{\text{neq}}). \quad (170)$$

Scalar-driven acceleration occurs when the scalar sector is potential dominated:

$$\rho_\psi + 3p_\psi = 2f^2 \dot{\bar{\psi}}^2 - 2f^4 U(\bar{\psi}) < 0 \quad \Longleftrightarrow \quad U(\bar{\psi}) > \frac{\dot{\bar{\psi}}^2}{f^2}. \quad (171)$$

Thus the *cosmic element* of the unified theory is the homogeneous order field acting like a dark-energy component through its stress-energy and potential.

The homogeneous scalar equation is obtained from (157):

$$f^2 \left(\ddot{\bar{\psi}} + 3H\dot{\bar{\psi}} \right) + f^4 U'(\bar{\psi}) = -\epsilon \bar{T}_{(m)} + \bar{J}_{\text{neq}}(t), \quad (172)$$

where $\bar{J}_{\text{neq}}(t)$ is a cosmic nonequilibrium/entropy-production history proxy. In a coarse-grained closure one may model $\bar{J}_{\text{neq}}(t) \sim \kappa_\sigma \bar{\sigma}(t)$, with $\bar{\sigma}(t)$ representing cosmological “order production” (e.g. structure formation, star formation, black-hole growth).

C.6 U.6 Local perturbations: observer-sourced order produces a local imprint

Decompose the field into a cosmic mean and a local perturbation:

$$\psi(x, t) = \bar{\psi}(t) + \delta\psi(x, t), \quad J_{\text{neq}}(x, t) = \bar{J}_{\text{neq}}(t) + \delta J_{\text{neq}}(x, t). \quad (173)$$

Linearizing (157) about $\bar{\psi}$ gives (schematically)

$$\square \delta\psi - m_{\text{eff}}^2(t) \delta\psi = -\frac{\epsilon}{f^2} \delta T_{(m)} + \frac{1}{f^2} \delta J_{\text{neq}}, \quad m_{\text{eff}}^2(t) \equiv f^2 U''(\bar{\psi}(t)). \quad (174)$$

In the observer-local dominant regime relevant to the LTA interpretation (weak curvature, quasi-static, subhorizon), this reduces to a Yukawa-type equation

$$(\nabla^2 - m_{\text{eff}}^2) \delta\psi \approx -\frac{1}{f^2} \delta J_{\text{neq,local}}, \quad (175)$$

with retarded solution

$$\delta\psi(\vec{x}, t) = \frac{1}{4\pi f^2} \int d^3x' \frac{e^{-m_{\text{eff}}|\vec{x}-\vec{x}'|}}{|\vec{x}-\vec{x}'|} \delta J_{\text{neq,local}}(\vec{x}', t_{\text{ret}}). \quad (176)$$

Equation (170) is the field-theory version of the observer-retarded causal bookkeeping used in the main LTA pipeline: the influence of a localized nonequilibrium source enters only within its causal domain and with retarded time.

C.7 U.7 Clocks, photon frequencies, and the recovered LTA mapping

Redshift is operationally a frequency ratio measured by matter clocks. Because matter proper time is $\tilde{\tau}$ (Eq. (152)), the measured frequency is cycles per unit $\tilde{\tau}$, implying that clock-referenced frequency ratios acquire an endpoint factor $A(\psi_{\text{obs}})/A(\psi_{\text{emit}})$.

Let z_{geom} denote the redshift predicted by the Einstein-frame geometry $g_{\mu\nu}$ (including any cosmological acceleration produced by ψ through (163)–(164)). Then the observed redshift inferred by matter clocks is

$$1 + z_{\text{obs}} = (1 + z_{\text{geom}}) \frac{A(\psi_{\text{obs}})}{A(\psi_{\text{emit}})}. \quad (177)$$

With $A = e^{\epsilon\psi}$ this becomes

$$1 + z_{\text{obs}} = (1 + z_{\text{geom}}) \exp\left[\epsilon(\psi_{\text{obs}} - \psi_{\text{emit}})\right]. \quad (178)$$

Define the chronometric potential $\mathcal{C} \equiv \ln A = \epsilon\psi$. Then the mapping is

$$1 + z_{\text{obs}} = (1 + z_{\text{geom}}) \exp\left[\mathcal{C}_{\text{obs}} - \mathcal{C}_{\text{emit}}\right], \quad \mathcal{C} = \epsilon\psi. \quad (179)$$

Recovered LTA notation. Define the accumulated imprint

$$I(\chi) \equiv \mathcal{C}_{\text{obs}} - \mathcal{C}_{\text{emit}}(\chi) = \epsilon(\psi_{\text{obs}} - \psi_{\text{emit}}(\chi)). \quad (180)$$

Then

$$1 + z_{\text{obs}} = (1 + z_{\text{geom}}) e^{I(\chi)}. \quad (181)$$

In the observer-local dominant limit used in the main paper, $z_{\text{geom}} \rightarrow z_{\text{cos}}$ to excellent approximation (because the background was fixed to ΛCDM), and the redshift-dependent part of $I(\chi)$ is dominated by the observer-local $\delta\psi$ perturbation sourced by $\delta J_{\text{neq,local}}$.

Define the drift-rate field as in the main text:

$$s(\chi) \equiv c \frac{dI}{d\chi}. \quad (182)$$

If $\psi_{\text{emit}}(\chi)$ is evaluated by an observer-retarded argument $\psi_{\text{emit}}(\chi) \approx \psi(t_{\text{ret}}(\chi))$ (with $t_{\text{ret}} \approx 2\chi/c$ at low z), then

$$s(\chi) \approx -2\epsilon \dot{\psi}(t_{\text{ret}}(\chi)) \quad (\text{low-}z \text{ observer-retarded approximation}). \quad (183)$$

This is the unified-theory origin of the empirical $s(t_{\text{ret}})$ profiles used in the LTA pipeline.

C.8 U.8 Second-law envelope on the imprint (energy $\rightarrow$ order $\rightarrow$ redshift)

The thermodynamic chain bounds the allowed magnitude and growth rate of ψ sourced by any nonequilibrium process. In a local rest frame, Eq. (148) implies an envelope on the drift field via (177):

$$|s(\chi)| \lesssim 2\epsilon \frac{P_{\text{in}}}{k_B T} \quad (\text{low-}z \text{ observer-retarded approximation}). \quad (184)$$

Integrating yields a corresponding bound on the saturation depth

$$I_{\text{sat}} \sim \epsilon \Delta\psi \lesssim \epsilon \int \frac{P_{\text{in}}(t)}{k_B T(t)} dt, \quad (185)$$

where the integral is over the active epoch of the source. This is the unified version of the ‘‘Earth budget closure’’ logic: the same second-law bookkeeping that limits constraint maintenance limits the achievable clock-bias/redshift imprint.

Two-scale interpretation (the unified story). Within this auxiliary completion:

- The **cosmic element** is the homogeneous mode $\bar{\psi}(t)$ sourced by a cosmological nonequilibrium history $\bar{J}_{\text{neq}}(t)$ (or $\bar{\sigma}(t)$) and backreacting on geometry through $T_{\mu\nu}^{(\psi)}$, potentially replacing Λ if $U(\bar{\psi})$ becomes potential dominated at late times.
- The **local element** is a localized perturbation $\delta\psi$ sourced by $\delta J_{\text{neq,local}}$ (e.g. biosphere/technosphere nonequilibrium throughput), producing an observer-local endpoint bias $e^{\epsilon(\psi_{\text{obs}} - \psi_{\text{emit}})}$ and thus an LTA-style redshift mapping distortion concentrated in the nearby Hubble flow.

A fully self-consistent cosmological inference program would fit both effects jointly: solve for the background expansion from (163)–(164) (cosmic mode) and for the observer-local mapping from (175)–(176) (local mode), with thermodynamic throughput bounds (148)–(179) providing physically grounded envelopes.

Two-epoch sourcing within one field. For the unified narrative we allow $J_{\text{neq}} = J_{\text{early}} + J_{\text{late}}$, where J_{early} captures organization associated with early structure/region formation (cosmological-scale sourcing), and J_{late} captures observer-local nonequilibrium throughput (e.g. biosphere/technosphere).

D Observer-retarded time as an approximation to the retarded Green's-function solution

This appendix justifies the specific “two-leg” retarded-time bookkeeping used in Sec. 7.3,

$$\chi(z_e) = 2\chi \quad \Rightarrow \quad t_{\text{ret}}(\chi) \equiv t_{\text{lb}}(z_e), \quad (186)$$

by showing that it is a controlled approximation to the *retarded* Green's-function solution of a causally sourced order field. In words: if the dominant source of the order field is localized near the observer, then the field encountered by an incoming photon at comoving radius χ is set primarily by the observer-source history evaluated at a retarded time corresponding (to leading order) to a *round-trip* light-travel construction, yielding the factor of 2χ .

D.1 V.1 Retarded solution of the sourced scalar model

The effective sourced scalar dynamics used in the main text (Sec. 5.1) is

$$(\square - m_\psi^2) \psi(x) = \kappa \sigma(x), \quad (187)$$

with retarded boundary conditions. Here $\sigma(x) \geq 0$ is a coarse-grained nonequilibrium throughput proxy and κ is a coupling.

Let $G_{\text{ret}}(x; x')$ be the retarded Green's function satisfying

$$(\square - m_\psi^2) G_{\text{ret}}(x; x') = \frac{\delta^{(4)}(x - x')}{\sqrt{-g}}, \quad G_{\text{ret}}(x; x') = 0 \text{ if } x' \notin J^-(x), \quad (188)$$

where $J^-(x)$ denotes the causal past of x . The unique retarded solution is then

$$\psi(x) = \psi_{\text{hom}}(x) + \kappa \int d^4x' \sqrt{-g(x')} G_{\text{ret}}(x; x') \sigma(x'). \quad (189)$$

In an observer-local dominant limit we drop ψ_{hom} (or absorb it into a background), and focus on the sourced part.

D.2 V.2 Specialize to FLRW and an observer-local source

Assume a spatially flat FLRW background for the geometry used to define light propagation and comoving distance:

$$ds^2 = -c^2 dt^2 + a^2(t) d\vec{x}^2 = a^2(\eta) (-c^2 d\eta^2 + d\vec{x}^2), \quad d\eta \equiv \frac{dt}{a(t)}. \quad (190)$$

Radial null curves satisfy

$$d\chi = c d\eta, \quad \Rightarrow \quad \chi(z) = c(\eta_0 - \eta(z)), \quad (191)$$

where η_0 is the conformal time at observation.

Now assume the dominant source is localized in a small world-tube around the observer at $\vec{x} = 0$. A convenient idealization is a pointlike comoving source with time-dependent strength $\sigma_\oplus(\eta)$:

$$\sigma(\eta, \vec{x}) \approx \sigma_\oplus(\eta) \frac{\delta^{(3)}(\vec{x})}{a^3(\eta)}. \quad (192)$$

The a^{-3} factor ensures correct covariant normalization under the physical volume element $\sqrt{-g} d^4x = a^4(\eta) c d\eta d^3x$.

Substituting (186) into (183) collapses the spatial integral:

$$\psi(\eta, \chi) \approx \kappa c \int d\eta' a(\eta') G_{\text{ret}}((\eta, \chi); (\eta', 0)) \sigma_\oplus(\eta'). \quad (193)$$

Thus, in the observer-local limit, the field along the line of sight is a *retarded convolution* of the observer history with the retarded Green's function.

D.3 V.3 Light-cone dominance and the “two-leg” retarded time

Equation (187) is exact given the model assumptions. The step that produces the paper’s $\chi(z_e) = 2\chi$ rule is the *light-cone dominance* approximation: the retarded response kernel is sharply peaked on (or near) the null cone, so the dominant contribution comes from the source evaluated at the retarded time corresponding to propagation at speed c .

A transparent way to see this is to recall the flat-spacetime (Minkowski) massless retarded Green’s function in $3 + 1$ dimensions:

$$G_{\text{ret}}^{(\text{M})}(t, r; t', 0) = \frac{\delta(t - t' - r/c)}{4\pi r} \Theta(t - t'). \quad (194)$$

In conformal coordinates of a conformally flat spacetime, the leading causal support still tracks the null condition $c(\eta - \eta') \approx \chi$.¹

Accordingly, the dominant contribution to (187) is obtained by evaluating the source at

$$\eta' \approx \eta - \frac{\chi}{c}. \quad (195)$$

Now evaluate the field *at the spacetime event on the incoming photon worldline* that lies at comoving radius χ from the observer. For an incoming photon observed at $(\eta_0, \chi = 0)$, the event at radius χ along the same null ray occurs at conformal time

$$\eta_\gamma(\chi) = \eta_0 - \frac{\chi}{c}. \quad (196)$$

Plugging (190) into the retarded-time condition (189) gives the source time (on the observer worldline) that dominantly sets the field encountered by the photon at radius χ :

$$\eta'_\oplus(\chi) \approx \eta_\gamma(\chi) - \frac{\chi}{c} = \eta_0 - \frac{2\chi}{c}. \quad (197)$$

This is the central point:

Along the observer’s past light cone, an observer-local source influences the photon at radius χ primarily through the observer history evaluated at a conformal time offset $\Delta\eta \approx 2\chi/c$.

In Minkowski space this “two-leg” factor is exact under the delta-function Green’s function: the observer influence must propagate *out* to the interaction point (distance χ) and the photon must then propagate *in* to the observer (distance χ), so the relevant source epoch precedes observation by $2\chi/c$.

D.4 V.4 From conformal-time retardation to the paper’s lookback-time construction

To connect (191) to the specific implementation used in Sec. 7.3, use the exact FLRW identity (185):

$$\chi(z) = c(\eta_0 - \eta(z)). \quad (198)$$

Define a redshift z_e such that the comoving distance to z_e equals 2χ :

$$\chi(z_e) = 2\chi \quad \Longleftrightarrow \quad \eta(z_e) = \eta_0 - \frac{2\chi}{c}. \quad (199)$$

Comparing (193) to (191) shows that z_e is precisely the redshift label of the retarded conformal time $\eta'_\oplus(\chi)$.

Finally, convert this retarded conformal time to the observer’s proper lookback time. By definition, the lookback time to redshift z is

$$t_{\text{lb}}(z) = \int_{t(z)}^{t_0} dt = \int_0^z \frac{dz'}{(1+z')H(z')}, \quad (200)$$

which is exactly Eq. (47) in the main text. Therefore, the observer-retarded time corresponding to $\eta(z_e) = \eta_0 - 2\chi/c$ is

$$t_{\text{ret}}(\chi) \equiv t_0 - t(z_e) = t_{\text{lb}}(z_e), \quad \text{with } \chi(z_e) = 2\chi, \quad (201)$$

which is exactly the construction adopted in Sec. 7.3.

¹For massive fields and/or non-conformal couplings in curved spacetime, the retarded Green’s function generally includes a “tail” inside the light cone. The approximation used here retains the leading null-cone timing and treats tails and amplitude factors as subleading corrections in the observer-local regime.

Low- z limit. At low redshift, $\chi \simeq (c/H_0)z$ and $t_{\text{lb}} \simeq z/H_0$, so $\chi(z_e) = 2\chi$ implies $z_e \simeq 2z$ and therefore $t_{\text{ret}} \simeq 2t_{\text{lb}}$, matching the intuitive “round-trip” picture.

D.5 V.5 What is being approximated (and how to systematically improve it)

The paper’s t_{ret} mapping (195) is an intentionally minimal, controlled proxy for the exact retarded convolution (187). It corresponds to the following approximations:

1. **Observer-local source.** We approximate $\sigma(x)$ as localized near the observer worldline (Eq. (186)). If sources are spatially extended (e.g. distributed across the Local Group / Local Sheet), then the retarded integral becomes a weighted superposition over source locations, and the effective retarded-time assignment becomes a mild smearing of the $2\chi/c$ rule.
2. **Light-cone dominance (kernel collapse).** We approximate the retarded kernel by its dominant null-cone timing, collapsing the convolution to an evaluation at a single retarded time. Tail terms (inside-the-light-cone support) arising from massive propagation and/or curvature coupling are neglected at the level of *timing* (they can be absorbed phenomenologically into the smooth activation function $g(t)$).
3. **Isotropy.** We reduce $\psi(x)$ on the past light cone to a function of χ alone. If the true source is anisotropic (e.g. local structure aligned with the Local Void), then $I[\gamma]$ becomes direction-dependent. (This is a natural extension and suggests sky-split tests.)
4. **Amplitude factors folded into $g(t)$.** Even when timing is set by $2\chi/c$, the exact Green’s function supplies additional amplitude factors (e.g. $1/\chi$, scale-factor prefactors, and Yukawa suppression $e^{-m_\psi\chi}$). In the present paper these are not modeled explicitly because (i) the inference is anchored to low- z and (ii) $s(\chi)$ is treated phenomenologically. A more microscopic model could include these factors directly in $s(\chi)$.

A systematic improvement path (optional). If one wishes to go beyond the single-time evaluation, replace the phenomenological activation $g(t_{\text{ret}}(\chi))$ with an explicit time-kernel convolution derived from an assumed retarded Green’s function:

$$\psi(\eta_\gamma(\chi), \chi) \approx \kappa c \int d\eta' \mathcal{K}(\chi; \eta_\gamma - \eta') \sigma_\oplus(\eta'), \quad (202)$$

with $\mathcal{K}$ read off from $a(\eta')G_{\text{ret}}$ (Eq. (187)). The present paper corresponds to the narrow-kernel limit $\mathcal{K} \sim \delta(\eta' - \eta_0 + 2\chi/c)$ in timing, plus a bounded phenomenological envelope.

D.6 V.6 Connection back to the LTA drift profile $s(\chi)$

In the chronometric mapping used in the main text, the accumulated imprint is $I(\chi) = \mathcal{C}_{\text{obs}} - \mathcal{C}_{\text{emit}}(\chi)$ and

$$s(\chi) \equiv c \frac{dI}{d\chi}. \quad (203)$$

In the observer-local dominant regime, the χ -dependence comes primarily from the retarded evaluation of the order field, schematically

$$\psi_{\text{emit}}(\chi) \approx \psi_\oplus(t_{\text{ret}}(\chi)), \quad t_{\text{ret}}(\chi) \approx t_{\text{lb}}(z_e), \quad \chi(z_e) = 2\chi, \quad (204)$$

so that at low z one recovers the useful approximation already recorded in the main text (Eq. (42)):

$$s(\chi) \approx -2\epsilon \dot{\psi}(t_{\text{ret}}(\chi)), \quad (\text{low-}z \text{ observer-retarded approximation}). \quad (205)$$

This is the precise sense in which the t_{ret} prescription is a *proxy for the causal retarded solution*: it captures the leading timing of how an observer-local source history can influence the field encountered along the past light cone.

E Full cosmological propagation of the order/chronometric field: background expansion, sound horizon r_d , and recombination-era observables

This appendix derives the equations required to propagate the unified order-field theory into: (i) the *background expansion* $H(z)$ (and the physical expansion rate relevant to matter), and (ii) the *early-time calibration* of the sound horizon r_d and recombination-era observables (visibility function, decoupling/drag epochs, and the geometric quantities entering the CMB).

The key point is that once the chronometric factor $A(\psi)$ is promoted from a local phenomenology to a *cosmological field*, it cannot be treated as a mere redshift remapping. It changes the relation between the gravitational (Einstein-frame) background and the *matter-frame* expansion history that controls the photon–baryon plasma, recombination, and hence the calibration of r_d and the entire CMB inference chain.

E.1 W.1 Unified action and frame bookkeeping (Jordan-first; A derived)

We start from the Jordan-frame scalar–tensor completion (Eq. (??) or Eq. (151)), where $g_{\mu\nu}^J$ is the physical metric seen by matter and by the photon–baryon plasma.

Jordan $\rightarrow$ Einstein transformation. Define the Einstein-frame metric

$$g_{\mu\nu}^E \equiv F(\psi) g_{\mu\nu}^J, \quad A(\psi) \equiv F(\psi)^{-1/2}, \quad \mathcal{C}(\psi) \equiv \ln A(\psi) = -\frac{1}{2} \ln F(\psi). \quad (206)$$

In Einstein variables, matter couples to $A(\psi)^2 g_{\mu\nu}^E$. A convenient special case is $F(\psi) = e^{-2\epsilon\psi} \Rightarrow A(\psi) = e^{\epsilon\psi}$, but the bookkeeping below is valid for general $F(\psi)$.

Scale factors and Hubble rates. For homogeneous FLRW,

$$ds_E^2 = -dt_E^2 + a_E^2(t_E) d\mathbf{x}^2, \quad ds_J^2 = -dt_J^2 + a_J^2(t_J) d\mathbf{x}^2, \quad (207)$$

with

$$a_J = A(\psi) a_E, \quad dt_J = A(\psi) dt_E. \quad (208)$$

Therefore the physical (matter-frame) expansion rate relevant to recombination is

$$H_J \equiv \frac{1}{a_J} \frac{da_J}{dt_J} = \frac{1}{A(\psi)} \left(H_E + \frac{d \ln A(\psi)}{dt_E} \right), \quad (209)$$

where $H_E \equiv \dot{a}_E/a_E$ and the derivative is with respect to t_E .

Observed redshift in the homogeneous limit. In the homogeneous FLRW limit, the physically observed redshift is

$$1 + z \equiv \frac{a_J(t_{J0})}{a_J(t_J)}. \quad (210)$$

Expressed in Einstein variables this becomes

$$1 + z = \frac{A(\psi)_0 a_E(t_{E0})}{A(\psi) a_E(t_E)}, \quad (211)$$

which is the usual Einstein-frame expression with an endpoint $A(\psi)$ factor, but here it is not a new postulate: it is algebraically equivalent to the Jordan-frame scale-factor law.

The remaining subsections use Einstein-frame equations as a computational convenience while keeping track of which combinations (H_J , a_J , r_d) correspond to physical matter-frame observables.

E.2 W.2 Background field equations in Einstein frame

Varying (201) gives the Einstein equation

$$M_{\text{Pl}}^2 G_{\mu\nu} = T_{\mu\nu}^{(\psi)} + T_{\mu\nu}^{(m)} + T_{\mu\nu}^{(\text{neq})}, \quad (212)$$

with

$$T_{\mu\nu}^{(\psi)} = f^2 \nabla_\mu \psi \nabla_\nu \psi - g_{\mu\nu} \left[\frac{f^2}{2} (\nabla_\lambda \psi)^2 + f^4 U(\psi) \right]. \quad (213)$$

The scalar equation of motion is

$$f^2 \square \psi - f^4 U'(\psi) = -\frac{d \ln A}{d\psi} T_{(m)} + J_{\text{neq}}, \quad J_{\text{neq}} \equiv -\frac{1}{\sqrt{-g}} \frac{\delta S_{\text{neq}}}{\delta \psi}. \quad (214)$$

For $A = e^{\epsilon\psi}$, $d \ln A / d\psi = \epsilon$:

$$f^2 \square \psi - f^4 U'(\psi) = -\epsilon T_{(m)} + J_{\text{neq}}. \quad (215)$$

FLRW background. Take Einstein-frame FLRW

$$ds^2 = -dt^2 + a^2(t) d\vec{x}^2, \quad H \equiv \frac{\dot{a}}{a}, \quad \psi = \bar{\psi}(t), \quad (216)$$

and define scalar energy density and pressure (Einstein frame)

$$\rho_\psi = \frac{f^2}{2} \dot{\bar{\psi}}^2 + f^4 U(\bar{\psi}), \quad p_\psi = \frac{f^2}{2} \dot{\bar{\psi}}^2 - f^4 U(\bar{\psi}). \quad (217)$$

Then the Friedmann equations are

$$3M_{\text{Pl}}^2 H^2 = \rho_\psi + \rho_m + \rho_r + \rho_{\text{neq}}, \quad (218)$$

$$-2M_{\text{Pl}}^2 \dot{H} = (\rho_m + p_m) + (\rho_r + p_r) + (\rho_\psi + p_\psi) + (\rho_{\text{neq}} + p_{\text{neq}}). \quad (219)$$

The homogeneous scalar equation from (208) is

$$f^2 \left(\ddot{\bar{\psi}} + 3H \dot{\bar{\psi}} \right) + f^4 U'(\bar{\psi}) = -\epsilon \bar{T}_{(m)} + \bar{J}_{\text{neq}}(t), \quad (220)$$

where $\bar{T}_{(m)} \equiv -\rho_m + 3p_m$ (Einstein-frame trace).

E.3 W.3 Matter-frame expansion $\tilde{H}$: the quantity that controls recombination

Define Jordan/matter-frame cosmic time and scale factor by

$$d\tilde{t} \equiv A(\bar{\psi}) dt, \quad \tilde{a} \equiv A(\bar{\psi}) a. \quad (221)$$

Then the *matter-frame* Hubble parameter is

$$\tilde{H} \equiv \frac{1}{\tilde{a}} \frac{d\tilde{a}}{d\tilde{t}} = \frac{1}{A} \left(H + \frac{\dot{A}}{A} \right) = \frac{1}{A} \left(H + \epsilon \dot{\bar{\psi}} \right), \quad (222)$$

and the observed redshift in the homogeneous theory is

$$1 + z \equiv 1 + z_{\text{obs}} = \frac{\tilde{a}_0}{\tilde{a}(t)}. \quad (223)$$

Why $\tilde{H}(z)$ is the correct input for r_d and recombination. The photon–baryon plasma, Thomson scattering, and recombination kinetics are governed by the physical expansion experienced by matter and radiation (the frame in which particle masses and cross sections are time-independent). In the present completion, that is the matter/Jordan frame, so the correct expansion rate appearing in recombination ODEs and horizon integrals is $\tilde{H}(z)$.

E.4 W.4 Energy-momentum exchange and density scalings

Diffeomorphism invariance implies conservation of the *total* stress-energy, but the matter sector is not separately conserved in Einstein frame when $A(\psi)$ varies:

$$\nabla_\mu T_{(m)}^{\mu\nu} = + \frac{d \ln A}{d\psi} T_{(m)} \nabla^\nu \psi + Q_{\text{neq}}^\nu, \quad (224)$$

where Q_{neq}^ν captures exchange between matter and the nonequilibrium sector.

Ignoring Q_{neq}^ν for the early-time plasma (a controllable assumption; reinstate if needed), the background continuity equation becomes

$$\dot{\rho}_i + 3H(\rho_i + p_i) = -\epsilon \dot{\bar{\psi}} (-\rho_i + 3p_i) = +\epsilon (1 - 3w_i) \dot{\bar{\psi}} \rho_i. \quad (225)$$

Thus:

$$\text{Radiation } (w = 1/3) : \quad \dot{\rho}_r + 4H\rho_r = 0, \quad \rho_r \propto a^{-4}, \quad (226)$$

$$\text{Nonrelativistic matter } (w = 0) : \quad \dot{\rho}_m + 3H\rho_m = +\epsilon \dot{\bar{\psi}} \rho_m \Rightarrow \rho_m \propto a^{-3} e^{+\epsilon(\bar{\psi} - \bar{\psi}_0)}. \quad (227)$$

In the matter frame, densities follow the standard scalings with $\tilde{a}$:

$$\tilde{\rho}_r \propto \tilde{a}^{-4}, \quad \tilde{\rho}_m \propto \tilde{a}^{-3}, \quad (228)$$

consistent with $1 + z = \tilde{a}_0/\tilde{a}$.

E.5 W.5 Sound speed, sound horizon, and the drag scale r_d

The baryon-photon sound speed is (matter frame)

$$c_s(z) = \frac{c}{\sqrt{3(1 + R(z))}}, \quad R(z) \equiv \frac{3\tilde{\rho}_b(z)}{4\tilde{\rho}_\gamma(z)}. \quad (229)$$

Using the standard matter-frame scalings $\tilde{\rho}_b \propto (1 + z)^3$ and $\tilde{\rho}_\gamma \propto (1 + z)^4$,

$$R(z) = R_0(1 + z)^{-1}, \quad R_0 \equiv \frac{3\tilde{\rho}_{b0}}{4\tilde{\rho}_{\gamma 0}}. \quad (230)$$

Sound horizon at an arbitrary redshift. Define the comoving sound horizon at redshift z by

$$r_s(z) \equiv \int_z^\infty \frac{c_s(z')}{\tilde{H}(z')} dz'. \quad (231)$$

Equivalently in scale factor,

$$r_s(\tilde{a}) = \int_0^{\tilde{a}} \frac{c_s(\tilde{a}')}{\tilde{a}'^2 \tilde{H}(\tilde{a}')} d\tilde{a}'. \quad (232)$$

Drag epoch z_d and r_d . The BAO ruler relevant for late-time BAO analyses is the sound horizon at the *drag epoch* z_d , when Compton drag ceases to efficiently couple baryons to photons. A clean definition is via a drag optical depth: the baryon-photon momentum-exchange rate is proportional to $\dot{\tau}/R$, where τ is Thomson optical depth. Define

$$\frac{d\tau}{d\eta} = \tilde{a} n_e \sigma_T c, \quad R(\eta) = \frac{3\tilde{\rho}_b}{4\tilde{\rho}_\gamma}, \quad (233)$$

and the drag optical depth

$$\tau_{\text{drag}}(z) \equiv \int_z^\infty \frac{c \sigma_T n_e(z')}{(1 + z') \tilde{H}(z')} \frac{1}{R(z')} dz'. \quad (234)$$

Then z_d is defined implicitly by

$$\tau_{\text{drag}}(z_d) = 1, \quad (235)$$

and the BAO sound horizon is

$$r_d \equiv r_s(z_d) = \int_{z_d}^\infty \frac{c_s(z')}{\tilde{H}(z')} dz'. \quad (236)$$

What changes relative to Λ CDM. The microphysics in c_s is unchanged in the matter frame, but the *entire* early calibration changes through $\tilde{H}(z)$, which is modified by: (i) the order-field energy density in (211)–(212), and (ii) the chronometric relation (215) linking H and $\tilde{H}$ via $\dot{\psi}$.

Therefore, to replace α_{rd} with a derived r_d in a global fit, one must solve the background system (211)–(213) and then compute (228)–(229).

E.6 W.6 Recombination: $x_e(z)$, optical depth, and the visibility function

The recombination-era observables needed for CMB prediction are controlled by the free-electron fraction $x_e(z) \equiv n_e/n_H$ and the Thomson optical depth $\tau(z)$.

Thomson optical depth. Define the Thomson optical depth (from z to today) by

$$\tau(z) \equiv \int_0^z \frac{c \sigma_T n_e(z')}{(1+z') \tilde{H}(z')} dz', \quad (237)$$

where $n_e(z) = x_e(z) n_H(z)$ and $n_H(z)$ is the hydrogen number density.

Visibility function and photon decoupling. The visibility function is

$$g(z) \equiv e^{-\tau(z)} \frac{d\tau}{dz}, \quad \frac{d\tau}{dz} = \frac{c \sigma_T n_e(z)}{(1+z) \tilde{H}(z)}. \quad (238)$$

The recombination redshift z (last scattering) is conventionally defined as the peak of $g(z)$ (or approximately where $\tau(z) \sim 1$); in a full computation, z emerges from $x_e(z)$ and $\tilde{H}(z)$.

Recombination ODE: schematic Peebles/Recfast form. A standard effective recombination equation can be written schematically as

$$\frac{dx_e}{dz} = \frac{1}{(1+z) \tilde{H}(z)} [\mathcal{R}(z) x_e^2 n_H(z) - \mathcal{I}(z) (1 - x_e)] \mathcal{C}(z), \quad (239)$$

where:

- $\mathcal{R}(z)$ is an effective case-B recombination coefficient (depends on $T_m \simeq T_\gamma$ prior to thermal decoupling),
- $\mathcal{I}(z)$ is an effective photoionization coefficient,
- $\mathcal{C}(z)$ is the Peebles “ C factor” encoding Lyman- α trapping and 2s–1s two-photon decay.

The *only* structural change forced by the unified order field at this level is the replacement

$$H(z) \longrightarrow \tilde{H}(z)$$

in the Hubble-dilution/time-conversion factor $1/[(1+z)\tilde{H}(z)]$. (If one instead works in Einstein frame, then particle masses and rates become ψ -dependent and the microphysical coefficients acquire additional $A(\psi)$ factors; the matter-frame formulation avoids that.)

Temperature scaling. Because $1+z = \tilde{a}_0/\tilde{a}$, the photon temperature obeys the standard scaling in terms of the observed redshift:

$$T_\gamma(z) = T_0(1+z). \quad (240)$$

Thus the CMB monopole scaling remains standard in the *fully self-consistent* matter-frame cosmology. What changes is the mapping between redshift and cosmic time via $\tilde{H}(z)$.

E.7 W.7 CMB geometric observables: θ , damping scale, and distances

Even before computing the full anisotropy spectrum, several key recombination-era observables depend directly on the background and recombination outputs:

Comoving distance and angular diameter distance. The comoving distance to redshift z (matter frame) is

$$\chi(z) = \int_0^z \frac{c}{\tilde{H}(z')} dz'. \quad (241)$$

For spatial curvature $K = 0$ (your baseline assumption), the transverse comoving distance is $D_M(z) = \chi(z)$, and the angular diameter distance is $D_A(z) = D_M(z)/(1+z)$.

Acoustic scale. The acoustic angular scale is

$$\theta \equiv \frac{r_s(z)}{D_M(z)}. \quad (242)$$

Planck strongly constrains θ ; therefore any unified model must reproduce it by jointly adjusting $\tilde{H}(z)$ (via ψ dynamics and matter content) and z (via recombination).

Photon diffusion (Silk) damping scale. The comoving diffusion scale depends on the mean free path and expansion rate; schematically,

$$r_D^2(z) \sim \int_z^\infty dz' \frac{c^2}{(1+z')\tilde{H}(z')} \frac{1}{\sigma_T n_e(z')} \times (\text{order-unity baryon-loading factors}), \quad (243)$$

again showing that the unified theory enters through $\tilde{H}(z)$ and $n_e(z)$.

Implication. A correct early-time calibration of r_d and recombination observables *requires* solving:

1. the background ODE system (211)–(213) (to get $H(t)$ and $\bar{\psi}(t)$),
2. the frame relation (215) to obtain $\tilde{H}(z)$,
3. the recombination ODE (232) to obtain $x_e(z)$, hence $\tau(z)$ and z ,
4. the drag condition (228) to obtain z_d and then r_d from (229),
5. and finally the geometric integrals (234)–(235).

E.8 W.8 (Optional, but needed for full C_ℓ) Linear perturbations: what must be modified

To propagate the unified theory into the *full* recombination-era observables (CMB anisotropy spectra), one must modify the Einstein–Boltzmann system.

A minimal specification is:

Perturbed Einstein equation (Einstein frame). Linearize (205):

$$\delta G_{\mu\nu} = \frac{1}{M_{\text{Pl}}^2} \left(\delta T_{\mu\nu}^{(\psi)} + \delta T_{\mu\nu}^{(m)} + \delta T_{\mu\nu}^{(\text{neq})} \right), \quad (244)$$

with $\delta T_{\mu\nu}^{(\psi)}$ determined by $\delta\psi$ and the metric perturbations.

Perturbed scalar equation. Linearize (208) about $\bar{\psi}$:

$$f^2 \delta(\Box\psi) - f^4 U''(\bar{\psi}) \delta\psi = -\epsilon \delta T_{(m)} + \delta J_{\text{neq}}, \quad (245)$$

where $\delta(\Box\psi)$ includes the usual metric-perturbation terms (e.g. Φ', Ψ' in Newtonian gauge).

Matter perturbations (Einstein frame) are not separately conserved. The covariant conservation law (217) implies at linear order:

$$\delta\left(\nabla_\mu T_{(m)}^{\mu\nu}\right) = \epsilon \delta(T_{(m)} \nabla^\nu \psi) + \delta Q_{\text{neq}}^\nu. \quad (246)$$

In practice: photons and neutrinos (traceless, $T = 0$) are unaffected by the conformal coupling at leading order, while CDM/baryons pick up coupling terms proportional to ϵ and $\delta\psi$. This is the standard “coupled quintessence” structure, except that here ψ is interpreted as an order field.

Implementation reality. A numerically correct Planck-era comparison therefore requires implementing (237)–(239) in a Boltzmann code (CLASS/CAMB-like), together with a recombination module that uses $\bar{H}(z)$ consistently. The background-only steps (Secs. E.3–E.7) are sufficient for deriving r_d and the geometric CMB scales, but not for C_ℓ precision constraints.

F Geometric completion (Model B): curvature–entropy scalar–tensor LTA

The chronometric model used in the main SN+BAO likelihood treats the order field $\psi(x)$ as a clock-bias field and, to leading approximation, neglects its backreaction on the large-scale geometry. Here we give a variational “geometric completion” in which the *same* entropy/order field is coupled into the field equations in scalar–tensor form, closely paralleling curvature–entropy constructions such as [?].

F.1 X.1 Action and conventions

We work with metric signature $(-, +, +, +)$ and $\square \equiv g^{\mu\nu} \nabla_\mu \nabla_\nu$. Let $M_{\text{Pl}}^2 \equiv (8\pi G)^{-1}$.

Entropy/order field normalization. We take ψ to be dimensionless, with an entropy-unit scalar $S_{\text{ord}}(x) \equiv S_* \psi(x)$ for some fixed $S_* > 0$.

Matter coupling and the derived chronometric factor. In the Jordan-first scalar–tensor completion used in this revision, matter couples minimally to the Jordan metric (the physical frame). The conformal/chronometric factor $A(\psi)$ appears only upon transforming to Einstein variables, where it is *derived* from the nonminimal coupling: $A(\psi) = F(\psi)^{-1/2}$ (Sec. 4.5). There is therefore no independent “clock coupling” parameter beyond the choice of $F(\psi)$ (and any additional sector couplings, if introduced).

Geometric completion (nonminimal curvature coupling). A minimal scalar–tensor completion is

$$\mathcal{S} = \int d^4x \sqrt{-g_J} \left[\frac{M_{\text{Pl}}^2}{2} F(\psi) R - \frac{f^2}{2} (\nabla_J \psi)^2 - f^4 U(\psi) \right] + \mathcal{S}_{\text{m}}[g^J, \Phi] + \mathcal{S}_{\text{neq}}[g, \psi, \Xi], \quad (247)$$

with dimensionless $F(\psi)$ and $U(\psi)$ and a stiffness scale f (mass dimension 1). For the linear coupling used in the main text discussion,

$$F(\psi) = 1 + 2\beta\psi, \quad \beta = \text{constant}. \quad (248)$$

The nonequilibrium sector $\mathcal{S}_{\text{neq}}$ is the covariant insertion point for sustained constraint-maintenance (dissipation, entropy throughput, etc.).

F.2 X.2 Field equations

Modified Einstein equation. Varying (??) with respect to $g_{\mu\nu}$ gives

$$M_{\text{Pl}}^2 F(\psi) G_{\mu\nu} = T_{\mu\nu}^{(m)} + T_{\mu\nu}^{(\psi)} + T_{\mu\nu}^{(\text{neq})} + M_{\text{Pl}}^2 (\nabla_\mu \nabla_\nu F(\psi) - g_{\mu\nu} \square F(\psi)), \quad (249)$$

where

$$T_{\mu\nu}^{(\psi)} = f^2 \nabla_\mu \psi \nabla_\nu \psi - g_{\mu\nu} \left[\frac{f^2}{2} (\nabla_J \psi)^2 + f^4 U(\psi) \right], \quad (250)$$

and $T_{\mu\nu}^{(m)}$ and $T_{\mu\nu}^{(\text{neq})}$ are defined by functional differentiation of $\mathcal{S}_m$ and $\mathcal{S}_{\text{neq}}$ with respect to $g_{\mu\nu}$.

Scalar equation of motion. Varying (??) with respect to ψ yields

$$f^2 \square \psi - f^4 U'(\psi) + \frac{M_{\text{Pl}}^2}{2} F'(\psi) R = - \frac{d \ln A}{d\psi} T_{(m)} + J_{\text{neq}}, \quad (251)$$

where $T_{(m)} \equiv g_{\mu\nu} T_{(m)}^{\mu\nu}$ and

$$J_{\text{neq}} \equiv - \frac{1}{\sqrt{-g_J}} \frac{\delta \mathcal{S}_{\text{neq}}}{\delta \psi}. \quad (252)$$

For the linear choice (??), $F'(\psi) = 2\beta$ so the curvature-driving term is $\beta M_{\text{Pl}}^2 R$.

Effective gravitational coupling. Equation (??) can be written as $G_{\mu\nu} = 8\pi G_{\text{eff}}(\psi) T_{\mu\nu}^{\text{eff}}$ with

$$G_{\text{eff}}(\psi) = \frac{G}{F(\psi)} = \frac{G}{1 + 2\beta\psi}, \quad (253)$$

so viability requires $F(\psi) > 0$ and (typically) small $|\beta\psi|$ and/or screening at Solar-System scales.

F.3 X.3 Curvature-sourced entropy production (Choice 1 link to [?])

Define the entropy-unit scalar $S_{\text{ord}} \equiv S_* \psi$ and the future-directed current $J_{\text{ord}}^\mu \equiv -\nabla^\mu S_{\text{ord}}$ (App. A.4). Then the production scalar is

$$\sigma_{\text{ord}} \equiv \nabla_\mu J_{\text{ord}}^\mu = -\square S_{\text{ord}} = -S_* \square \psi. \quad (254)$$

Using the scalar equation (??) gives the exact identity

$$\sigma_{\text{ord}} = \frac{S_*}{f^2} \left[\frac{M_{\text{Pl}}^2}{2} F'(\psi) R + \frac{d \ln A}{d\psi} T_{(m)} - J_{\text{neq}} - f^4 U'(\psi) \right]. \quad (255)$$

In the minimal decoupling limit $A(\psi) \rightarrow 1$, $U'(\psi) \rightarrow 0$, and $J_{\text{neq}} \rightarrow 0$, this reduces to a curvature-sourced production law of the form

$$\sigma_{\text{ord}} = \kappa_{\text{eff}} R, \quad \kappa_{\text{eff}} \equiv \frac{S_* \beta M_{\text{Pl}}^2}{f^2}, \quad (256)$$

which is the same structural statement as the curvature-entropy relations derived from the informational action approach in [?], up to normalization conventions and metric signature.

F.4 X.4 FLRW reduction and DE-like behavior

On a spatially-flat FLRW background with $\psi = \psi(t)$ and $H \equiv \dot{a}/a$, the 00 component of (??) gives

$$3M_{\text{Pl}}^2 F H^2 = \rho + \left(\frac{f^2}{2} \dot{\psi}^2 + f^4 U(\psi) \right) + \rho_{\text{neq}} - 3M_{\text{Pl}}^2 H \dot{F}, \quad (257)$$

where $\rho = \rho_m + \rho_r$ and ρ_{neq} is the 00 component of $T_{\mu\nu}^{(\text{neq})}$. For the linear coupling (??), $\dot{F} = 2\beta\dot{\psi}$. The scalar equation (??) reduces to

$$\ddot{\psi} + 3H\dot{\psi} + \frac{f^2}{f^2} \left(\frac{f^4}{f^2} U'(\psi) \right) - \frac{M_{\text{Pl}}^2}{f^2} \beta R = - \frac{1}{f^2} \left[\frac{d \ln A}{d\psi} T_{(m)} - J_{\text{neq}} \right], \quad (258)$$

with $R = 6(\dot{H} + 2H^2)$ in flat FLRW. In this completion, a homogeneous ψ can therefore generate DE-like modifications of $H(z)$ even with $\Lambda = 0$, while the observer-local chronometric limit corresponds to regimes where the observable imprint is dominated by $A(\psi)$ rather than by geometric backreaction.

G Pulsar timing in the chronometric/order-field theory: source term, propagation through space, and receiver term

This appendix derives the timing observable relevant for pulsar timing arrays (PTAs) in the chronometric/order-field framework, explicitly including:

1. the pulsar’s local clock sector (“their time hole”; **source term**),
2. propagation through interstellar space (“empty space”; **transfer**),
3. and the Earth’s local clock sector (“our time hole”; **receiver term**).

The resulting structure naturally yields a *pulsar term* and an *Earth term* analogous to the standard PTA response to gravitational waves, but here the underlying physics is a scalar chronometric potential $\mathcal{C}(x) = \ln A(\psi)$ rather than a tensor metric perturbation.

G.1 P.1 Chronometric potential and local clock times

Let the chronometric factor in Einstein variables be the standard derived factor

$$A(\psi) \equiv F(\psi)^{-1/2}, \quad \mathcal{C}(x) \equiv \ln A(\psi)(x) = -\frac{1}{2} \ln F(\psi)(x). \quad (259)$$

Along any worldline, the physical (Jordan) clock time is related to Einstein proper time by

$$d\tau_J = A(\psi) d\tau_E = e^{\mathcal{C}} d\tau_E. \quad (260)$$

A commonly used special case is $F(\psi) = e^{-2\epsilon\psi} \Rightarrow A(\psi) = e^{\epsilon\psi}$, but the PTA bookkeeping below does not require fixing a particular functional form.

We will need two worldlines:

- Pulsar worldline P : $\mathcal{C}_P(t) \equiv \mathcal{C}(x_P(t))$ and $d\tilde{\tau}_P = e^{\mathcal{C}_P} d\tau_P$.
- Earth/receiver worldline $:$ $\mathcal{C}(t) \equiv \mathcal{C}(x(t))$ and $d\tilde{\tau} = e^{\mathcal{C}} d\tau$.

In PTA practice one models times in a barycentric coordinate time; here we keep the derivation covariant and then specialize to a weak-field/slow-motion regime where coordinate time is an adequate affine label.

G.2 P.2 Pulse phase and the observable: arrival-time residuals

Let N be the pulse phase (integer pulse count). By definition,

$$\frac{dN}{d\tilde{\tau}_P} = \nu_P(\tilde{\tau}_P), \quad (261)$$

where ν_P is the pulsar spin frequency measured in the pulsar’s own clock time. In the simplest case $\nu_P \simeq \nu_0$ is constant plus known secular spindown that PTA models already fit.

A pulse emitted at event x_P propagates along a null ray to be received at x_E . Let t label receiver coordinate time and $t_P(t)$ the corresponding emission coordinate time (the standard retarded relation; in flat space $t_P = t - D/c$ up to Shapiro and kinematic corrections).

The key PTA observable is the *timing residual* $R(t)$: the difference between observed pulse arrival times and the best-fit timing model (which includes standard GR propagation terms, ephemeris, dispersion, spin-down, etc.). Any additional fractional frequency shift $z(t)$ contributes as

$$R(t) = \int_{t_P}^t z(t') dt', \quad (262)$$

up to the convention of which time coordinate is used in the integral (in practice one uses the same reference time variable as the timing model). The derivation below gives $z(t)$ induced by $\mathcal{C}$.

G.3 P.3 Instantaneous observed frequency including both source and receiver clocks

The arrival phase rate measured by the receiver clock is

$$\nu_{\text{obs}}(t) \equiv \frac{dN}{d\tilde{\tau}}. \quad (263)$$

Using the chain rule,

$$\frac{dN}{d\tilde{\tau}} = \frac{dN}{d\tilde{\tau}_P} \frac{d\tilde{\tau}_P}{dt_P} \frac{dt_P}{dt} \frac{dt}{d\tilde{\tau}}. \quad (264)$$

Now apply the clock relations $d\tilde{\tau} = e^{\mathcal{C}} d\tau$. In a weak-field slow-motion regime (appropriate for PTA baselines), we can take $dt \simeq d\tau$ along each worldline to leading order in standard gravitational potentials; all such GR terms are already part of the timing model. Then

$$\frac{d\tilde{\tau}_P}{dt_P} \simeq e^{\mathcal{C}_P(t_P)}, \quad \frac{dt}{d\tilde{\tau}} \simeq e^{-\mathcal{C}(t)}. \quad (265)$$

Also $dt_P/dt \simeq 1$ up to conventional kinematic/gravitational propagation corrections (again part of the baseline model). Therefore,

$$\nu_{\text{obs}}(t) \simeq \nu_P(\tilde{\tau}_P) e^{\mathcal{C}_P(t_P) - \mathcal{C}(t)}. \quad (266)$$

Interpretation in the “three-stage” picture you requested. Equation (247) factorizes cleanly:

1. **Flickering at a rate based on their time (source).** Any variability in $\nu_P(\tilde{\tau}_P)$ and/or $\mathcal{C}_P$ is defined in pulsar clock time $\tilde{\tau}_P$.
2. **Exiting their time hole $\rightarrow$ traveling across empty space (transfer).** If the interstellar path has negligible $\mathcal{C}$ (empty space: $A \simeq 1$), then the only chronometric contribution from the pulsar environment is the *endpoint value* $\mathcal{C}_P(t_P)$. This is exactly the “climb out of their local time hole” factor $e^{+\mathcal{C}_P}$.
3. **Entering our time hole (receiver).** Upon reception, the measured frequency is divided by our local clock factor $e^{\mathcal{C}_E(t)}$. This is the receiver “entering our time hole” factor $e^{-\mathcal{C}_E}$.

G.4 P.4 Fractional frequency shift: Earth term + pulsar term

Define the fractional frequency shift relative to a baseline in which $\mathcal{C} \equiv 0$:

$$z_{\mathcal{C}}(t) \equiv \frac{\nu_{\text{obs}}(t) - \nu_{\text{obs}}^{(0)}(t)}{\nu_{\text{obs}}^{(0)}(t)}. \quad (267)$$

If ν_P is treated as already modeled (i.e. we isolate the additional chronometric effect), then from (247) we have

$$1 + z_{\mathcal{C}}(t) \simeq e^{\mathcal{C}_P(t_P) - \mathcal{C}(t)}. \quad (268)$$

For small perturbations (the PTA-relevant regime),

$$z_{\mathcal{C}}(t) \simeq \mathcal{C}_P(t_P) - \mathcal{C}(t) = \epsilon [\psi_P(t_P) - \psi(t)]. \quad (269)$$

This is the PTA analogue of the standard “pulsar term minus Earth term” structure:

PTA chronometric redshift : $z_{\mathcal{C}}(t) \propto \psi_P(\text{retarded}) - \psi_E(\text{now})$.

(270)

The corresponding timing residual is

$$R_{\mathcal{C}}(t) = \int^t [\mathcal{C}_P(t_P(t')) - \mathcal{C}(t')] dt' \quad (\text{to leading order}). \quad (271)$$

G.5 P.5 “Flickering based on their time”: mapping pulsar-time variability into our time

Your explicit requirement is that any PTA equations must correctly include the fact that a process that is periodic in *pulsar clock time* will be observed through the combined (source clock $\rightarrow$ propagation delay $\rightarrow$ receiver clock) mapping.

Let a pulsar-local chronometric modulation be specified as a function of pulsar clock time:

$$\mathcal{C}_P = \widehat{\mathcal{C}}_P(\tilde{\tau}_P), \quad d\tilde{\tau}_P = e^{\mathcal{C}_P(t)} dt \quad (\text{weak-field } dt \simeq d\tau_P). \quad (272)$$

Then the pulsar term in (250) is evaluated at the retarded emission time:

$$\mathcal{C}_P(t_P) = \widehat{\mathcal{C}}_P(\tilde{\tau}_P(t_P)), \quad \tilde{\tau}_P(t_P) = \int^{t_P} e^{\mathcal{C}_P(t')} dt'. \quad (273)$$

To express the observed modulation in *our* clock time $\tilde{\tau}$, use $d\tilde{\tau}_E = e^{\mathcal{C}_E(t)} dt$, so

$$\frac{d\tilde{\tau}_P}{d\tilde{\tau}} = \frac{e^{\mathcal{C}_P(t_P)}}{e^{\mathcal{C}(t)}} \frac{dt_P}{dt}. \quad (274)$$

To leading order $dt_P/dt \simeq 1$, therefore

$$\frac{d\tilde{\tau}_P}{d\tilde{\tau}} \simeq e^{\mathcal{C}_P(t_P) - \mathcal{C}(t)}. \quad (275)$$

Observed flicker rate. If the pulsar modulation is sinusoidal in pulsar time,

$$\widehat{\mathcal{C}}_P(\tilde{\tau}_P) = \mathcal{C}_{P0} + \delta\mathcal{C}_P \cos(\omega_P \tilde{\tau}_P), \quad (276)$$

then the instantaneous angular frequency measured in receiver clock time is

$$\omega_{\text{obs}}(\tilde{\tau}) = \omega_P \frac{d\tilde{\tau}_P}{d\tilde{\tau}} \simeq \omega_P e^{\mathcal{C}_P(t_P) - \mathcal{C}(t)}. \quad (277)$$

This equation is the mathematically explicit version of your statement:

“Flickering at a given rate based on their time (needs to exit their local time hole to reach us), traveling across empty space, reaching us and then entering our local time hole.”

The pulsar-time frequency ω_P is warped by the ratio of clock factors between emission and reception. If the path is empty space ($\mathcal{C} \approx 0$ along the transfer), the only warp is the endpoint difference.

G.6 P.6 Allowing an extended “time-hole” profile and propagation domains

If the pulsar (or Earth) time-hole is spatially extended, then $\mathcal{C}(x)$ varies with radius. In the conformal-chronometric theory, the net chronometric contribution to a frequency comparison is still an endpoint difference in $\mathcal{C}$:

$$\ln \left(\frac{\nu_{\text{meas}}}{\nu_{\text{meas}}^{(0)}} \right) = -[\mathcal{C} - \mathcal{C}_P], \quad (278)$$

because $\mathcal{C}$ is a scalar and $\int_{\gamma} d\mathcal{C} = \mathcal{C}_{\text{obs}} - \mathcal{C}_{\text{emit}}$. Operationally, “exiting the pulsar time hole” is precisely the factor $e^{+\mathcal{C}_P}$ when the asymptotic region has $\mathcal{C} \rightarrow 0$.

If, however, one wants to *model* $\mathcal{C}(x)$ as a causal field sourced by the local nonequilibrium throughput, then $\mathcal{C}$ itself must be computed from a retarded Green’s function (Appendix D generalization):

$$\psi(x) = \psi_{\text{hom}}(x) + \kappa \int d^4x' \sqrt{-g} G_{\text{ret}}(x; x') \sigma(x'), \quad \mathcal{C}(x) = \epsilon\psi(x). \quad (279)$$

In that representation, both Earth and pulsar have their own retarded source histories, and the PTA observable $z_{\mathcal{C}}(t)$ samples *two* distinct causal solutions at two distinct worldlines:

$$z_{\mathcal{C}}(t) \simeq \epsilon [\psi_P(t_P(t)) - \psi(t)]. \quad (280)$$

G.7 P.7 Practical PTA implication: monopole Earth-term degeneracy and the role of the pulsar term

Equation (250) has an important structural consequence: the Earth term $-\mathcal{C}(t)$ is a *common* contribution to all pulsars, i.e. a monopole in the PTA. This is (partially) degenerate with terrestrial clock errors unless it is separated by: (i) independent time standards, (ii) pulsar-term structure, or (iii) sky-dependent propagation/anisotropy.

The pulsar term $\mathcal{C}_P(t_P)$ is different for each pulsar and thus offers a handle to break degeneracy, but it enters at a retarded time (set by the pulsar distance and standard propagation effects), exactly as in the gravitational-wave PTA response.

H Cross-validation under strong covariance coupling

This appendix reports cross-validation (CV) diagnostics for transparency, but it also explains why CV is *not* used as the primary evidentiary statistic in this analysis.

The key technical point is that the Pantheon+SH0ES release used here is not even approximately an i.i.d. dataset: the published covariance contains long-range, survey-spanning correlated modes (including calibrator–Hubble-flow cross-blocks and broad systematics). In that regime, the statistically correct object for model comparison is the *global* likelihood with the full covariance, and the statistically correct calibration for a bounded, nested extension is the null distribution of the global likelihood-ratio statistic obtained by parametric bootstrap (Sec. 10.6 and Sec. 12). Cross-validation answers a different question that depends sensitively on how one treats train–test cross-covariance; it is therefore best viewed as a *diagnostic of covariance-leveraging and subset sensitivity*, not as a replacement for the global full-covariance comparison.

H.1 Why CV is not the primary statistic here

For a Gaussian data vector with known covariance, the full-data log-likelihood is (up to an additive constant)

$$-2 \ln \mathcal{L}(\boldsymbol{\theta}) \equiv \chi^2(\boldsymbol{\theta}) = \mathbf{r}(\boldsymbol{\theta})^\top \mathbf{C}^{-1} \mathbf{r}(\boldsymbol{\theta}), \quad (281)$$

with residual vector $\mathbf{r}$ and the published covariance $\mathbf{C}$. This is the canonical quantity used throughout precision cosmology: it is exactly the statistic that correctly weights and de-correlates the measured modes, including modes that couple widely separated objects and surveys.

Cross-validation, by contrast, is most naturally justified when one has many approximately independent draws from a common distribution and one is primarily concerned with “generalization” to new samples. Pantheon+SH0ES is not of that type. It is one correlated realization of a single Universe observed through a complex inference pipeline with shared calibration and selection systematics encoded in $\mathbf{C}$. A CV split does not manufacture new independent information; it partitions a single correlated draw. As a result, CV scores can be dominated by *how correlated modes are apportioned across folds* rather than by whether a model provides a better global description of the realized dataset.

More concretely: let the data be partitioned into train (T) and test/validation (V),

$$\mathbf{r} = \begin{bmatrix} \mathbf{r}_T \\ \mathbf{r}_V \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} \mathbf{C}_{TT} & \mathbf{C}_{TV} \\ \mathbf{C}_{VT} & \mathbf{C}_{VV} \end{bmatrix}. \quad (282)$$

The Gaussian *conditional* predictive distribution implied by the same covariance model is

$$\mathbf{r}_V | \mathbf{r}_T \sim \mathcal{N}(\mathbf{C}_{VT} \mathbf{C}_{TT}^{-1} \mathbf{r}_T, \mathbf{C}_{V|T}), \quad \mathbf{C}_{V|T} \equiv \mathbf{C}_{VV} - \mathbf{C}_{VT} \mathbf{C}_{TT}^{-1} \mathbf{C}_{TV}, \quad (283)$$

and the corresponding conditional held-out score is

$$\chi_{V|T}^2(\boldsymbol{\theta}) = (\mathbf{r}_V - \mathbf{C}_{VT} \mathbf{C}_{TT}^{-1} \mathbf{r}_T)^\top \mathbf{C}_{V|T}^{-1} (\mathbf{r}_V - \mathbf{C}_{VT} \mathbf{C}_{TT}^{-1} \mathbf{r}_T). \quad (284)$$

Many “standard” CV implementations in the literature instead evaluate a *marginal* held-out score that ignores train–test cross-covariance,

$$\chi_{V,\text{marg}}^2(\boldsymbol{\theta}) \equiv \mathbf{r}_V^\top \mathbf{C}_{VV}^{-1} \mathbf{r}_V. \quad (285)$$

When $\mathbf{C}_{VT} \neq \mathbf{0}$ (as is emphatically true here), these are different statistics with different meanings, and they can rank models differently even when computed on the same split.

The full-data likelihood ratio $\Delta\chi^2$ used in the main text is not “in conflict” with CV: it is answering the physically and statistically relevant question for this problem—*which model better explains the single correlated dataset under the published covariance model?*—and it is calibrated by null injections that preserve that same covariance structure. CV is included for robustness and transparency, but it is not a more rigorous substitute for the global statistic in the strong-correlation regime.

H.2 CV protocol used in this work

We perform blocked CV over the *Hubble-flow* subset only, while always retaining calibrators in training. This choice is forced by the ladder-anchored likelihood construction: calibrators provide the absolute-distance anchor and define the intercept posterior used to marginalize M (Sec. 10.2). Leaving calibrators out of training would replace a physics-motivated anchored likelihood with a different statistical problem and would not test the same model.

For the representative run reported in the main text (run-tag 20260112-013808), the split is:

- Calibrators: $N_{\text{cal}} = 77$ kept in train for all folds.
- Hubble-flow pool: $N_{\text{HF}} = 277$ partitioned into $k = 15$ folds.
- Blocking: leave-one-survey-out folds over `IDSURVEY` (15 surveys), i.e. each fold holds out one survey block (sizes: min 1, median 14, max 72).
- Seed: `cv_seed=100`.
- Anchor normalization: t_{anchor} is recomputed from the training Hubble-flow subset only in each fold to avoid leakage of held-out information into the LTA amplitude normalization.

For each fold we: (i) refit baseline ΛCDM and $\Lambda\text{CDM}+\text{LTA}(\text{powerlaw})$ on the training set using the same anchored likelihood construction and numerical safeguards as in the full-data analysis; and (ii) evaluate held-out scores on the test block using both the marginal score (Eq. (266)) and the Gaussian conditional score (Eq. (265)), with the *full* train–test cross-covariance retained (`cond_cross=full` in the pipeline).

H.3 Why we report both “marginal” and “conditional” held-out scores

In a strongly correlated Gaussian model, there is no single CV number that is universally appropriate unless one states explicitly what is being conditioned on.

The marginal score $\chi^2_{V,\text{marg}}$ asks: *given the parameters fit on training, how large are the held-out residuals under the test-block covariance alone?* This resembles the classical out-of-sample score used in weakly dependent problems.

The conditional score $\chi^2_{V|T}$ asks a sharper question: *given the parameters fit on training and given the realized training residual field, how surprising is the held-out residual once we regress out the part linearly predictable from the training residuals under the published cross-covariance?* This is the correct predictive statistic if one treats the covariance model as the generative model and one genuinely wants $p(\mathbf{y}_V | \mathbf{y}_T)$.

Because Pantheon+SH0ES includes large-amplitude survey-spanning modes, conditioning can remove a substantial fraction of the variance in $\mathbf{r}_V$; in that regime, $\chi^2_{V|T}$ can become a measure of the model’s performance on *residual structure orthogonal to the correlated modes already realized in training*. That is a valid diagnostic, but it is not the same question as global model preference under the full likelihood, and it should not be used to dismiss a full-data likelihood-ratio detection that is itself calibrated under the same covariance assumptions.

H.4 CV results for run-tag 20260112-013808

We report fold-by-fold and summed held-out improvements in the convention

$$\Delta\chi^2 \equiv \chi^2_{\Lambda\text{CDM}} - \chi^2_{\Lambda\text{CDM}+\text{LTA}}, \quad (286)$$

so $\Delta\chi^2 > 0$ favors LTA.

For the leave-one-survey-out blocked CV described above, summing held-out contributions over all folds gives:

$$\sum_{\text{folds}} \Delta\chi_{\text{test},\text{marg}}^2 = +113.724, \quad \sum_{\text{folds}} \Delta\chi_{\text{test}|\text{train}}^2 = -7.783, \quad (287)$$

with the marginal score favoring LTA in 15/15 folds and the conditional score favoring LTA in 8/15 folds. The corresponding per-fold diagnostics are shown in Fig. 9.

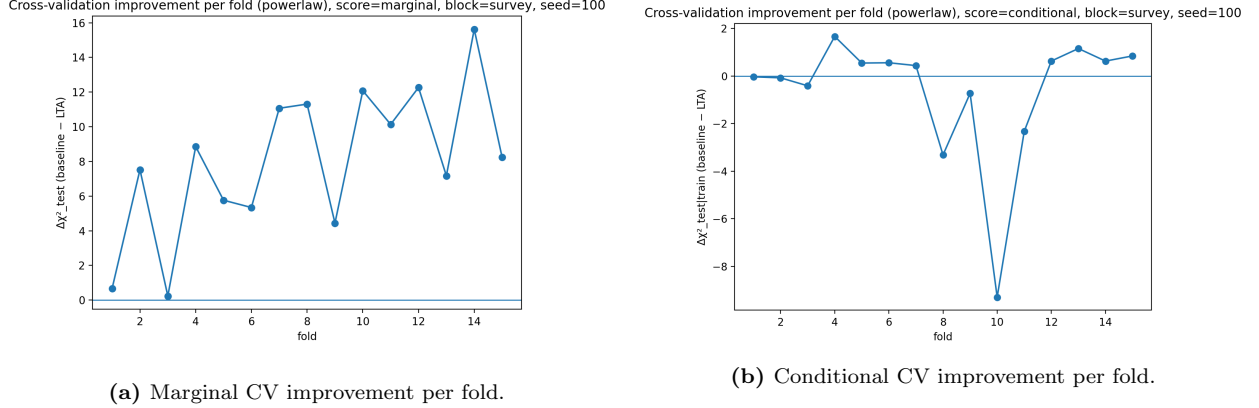


Figure 9: Blocked-by-survey cross-validation diagnostics for the Hubble-flow subset ($k = 15$, seed 100). Top: marginal held-out $\Delta\chi^2$ (Eq. (266)). Bottom: conditional held-out $\Delta\chi^2$ with full train-test cross-covariance retained (Eq. (265)).

We stress the interpretation: the marginal statistic indicates that the LTA mapping captures structure that persists across survey-held-out splits (consistent with the signal being distributed across multiple surveys rather than being a single-survey artifact), while the conditional statistic indicates that once one conditions on the realized correlated residual field in training, the remaining orthogonal held-out structure does not favor LTA in this configuration. These two statements are not mutually exclusive; they reflect different conditioning choices in a regime where $\mathbf{C}_{VT}$ is large.

For completeness, we also ran an alternative but closely related grouping-based CV diagnostic (`sn.group`, $k = 15$) which yields similar qualitative behavior:

$$\sum_{\text{folds}} \Delta\chi_{\text{test},\text{marg}}^2 = +60.266, \quad \sum_{\text{folds}} \Delta\chi_{\text{test}|\text{train}}^2 = -42.824, \quad (288)$$

again showing marginal preference but conditional reversal.

H.5 CV-null injections

We additionally perform CV-null injections as a sanity check on the CV statistics themselves: mock datasets are generated under the best-fit baseline model using the same published covariance model and the same anchored-likelihood machinery, and the entire blocked-CV pipeline is rerun for each mock. For the representative run, we performed $N = 50$ CV-null draws (seed 100). The observed CV totals were

$$\Delta\chi_{\text{CV},\text{marg},\text{obs}}^2 = 113.724, \quad \Delta\chi_{\text{CV},\text{cond},\text{obs}}^2 = -7.783, \quad (289)$$

while the null draws yielded:

$$\Delta\chi_{\text{CV},\text{marg},\text{null}}^2 : \text{mean} = 1.179420, \text{std} = 3.846321, \quad p_{\text{MC}} < \frac{1}{50+1} \approx 0.0196, \quad (290)$$

$$\Delta\chi_{\text{CV},\text{cond},\text{null}}^2 : \text{mean} = -0.242446, \text{std} = 1.004216, \quad p_{\text{MC}} \approx 1. \quad (291)$$

Here the reported p_{MC} is the Monte Carlo tail probability $P(\Delta\chi^2_{\text{null}} \geq \Delta\chi^2_{\text{obs}})$ estimated from the finite null sample. With $N = 50$ CV-null draws, none exceeded the observed marginal CV statistic, giving $p_{\text{MC}} \approx 1/(50 + 1) = 0.0196$ at this Monte Carlo resolution.

The CV-null exercise is included to show, explicitly, that the CV statistics behave as expected under the assumed covariance model: the marginal CV statistic can display broad dispersion under correlated systematics and can, in finite samples, register large positive totals, while the conditional CV statistic is anchored close to its null mean once the predictable correlated modes are regressed out. This is precisely why CV is not elevated to the primary evidentiary role in this paper: it is not measuring the same object as the global full-covariance likelihood ratio, and in the strong-coupling regime its interpretation depends on conditioning choices that are orthogonal to the main cosmological question addressed here.

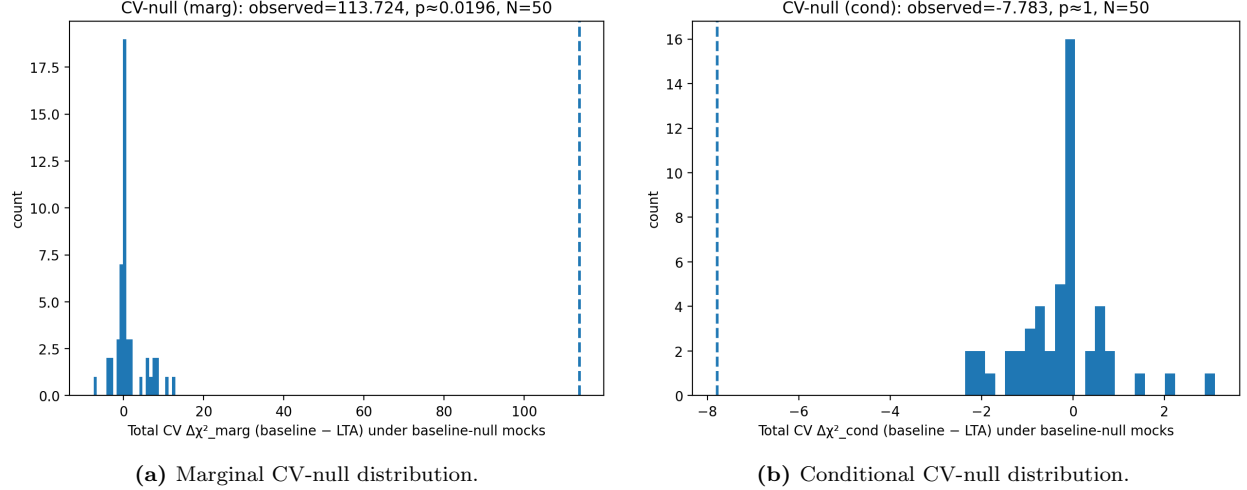


Figure 10: CV-null injections ($N = 50$) for the blocked-by-survey CV statistic, run-tag 20260112-013808.

H.6 Connection back to the main result

The central evidentiary quantity in this work is the *global* improvement in the full-data Gaussian objective, $\Delta\chi^2 = \chi^2_{\text{ACDM}} - \chi^2_{\text{ACDM+LTA}}$, evaluated using the anchored SN likelihood that respects the published Pantheon+SH0ES covariance (including calibrator–Hubble-flow cross-covariances) and combined with BAO and the optional Planck prior as stated in the main text. Because the LTA extension is nested with a boundary null ($s_{\text{anchor}} \geq 0$), the interpretation of $\Delta\chi^2$ is based on an empirical null distribution obtained by parametric bootstrap (null injections) that mirrors the same covariance structure and refitting protocol.

The cross-validation calculations reported in this appendix serve a different purpose. In a dataset with strong survey-spanning covariance, train–test partitions do not produce independent draws; they redistribute correlated modes between folds. As a result, held-out scores depend sensitively on whether one evaluates test residuals marginally (using $\mathbf{C}_{VV}$ alone) or conditionally (with the full train–test cross-covariance retained). Reporting both scores is therefore primarily a transparency exercise: it demonstrates how correlation structure propagates through blocked splits, and it provides a diagnostic check that the observed improvement is not confined to a single survey block under the marginal scoring convention.

Accordingly, the CV results should be read as *diagnostics of subset stability and covariance coupling*, not as an alternative evidence metric that supersedes the full-covariance likelihood ratio. The main conclusion of the paper remains anchored to the global $\Delta\chi^2$ computed on the complete dataset under the published covariance model and to its Monte Carlo calibration under baseline-only mocks.